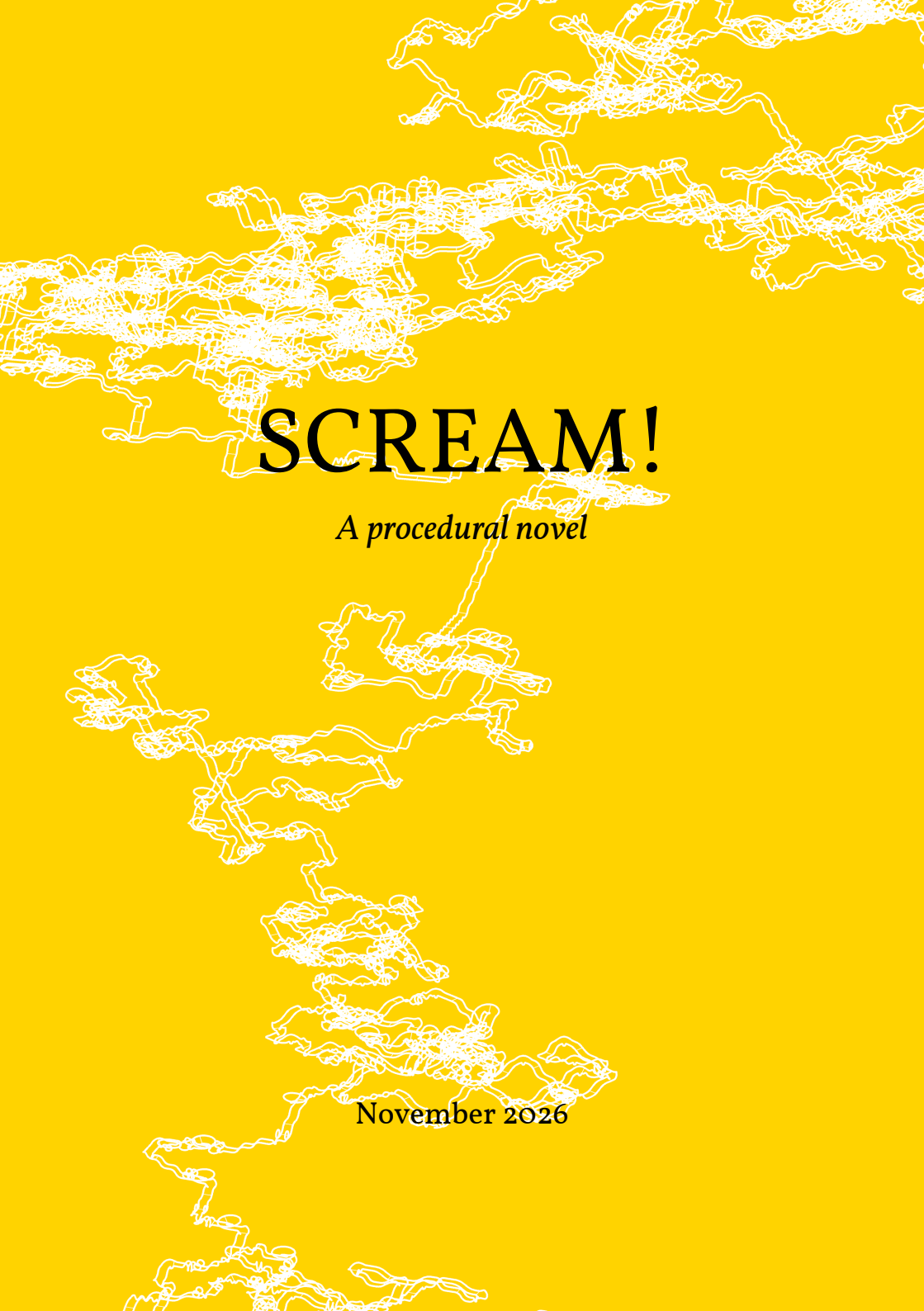


SCREAM!

A procedural novel

November 2026

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This novel is assembled from the ride narratives of 451 roller coasters from around the world. Passages have been shuffled and stitched together, with substitutions to improve continuity. Wikipedia text is licensed under CC BY-SA 4.0. The cover and chapter illustrations are derived from a model of the novel's continuous ten-hour coaster.

CONTENTS

Downward Helix 1

Tentacles of Devil's Snare 10

Surrounding Croatian-Themed Landscapes 24

Underground Tunnel 38

Cloud of Fog 49

Tiki Trail Treehouse Attraction 63

Shanghai Disneyland 76

V-Like Formation 88

Different Direction 102

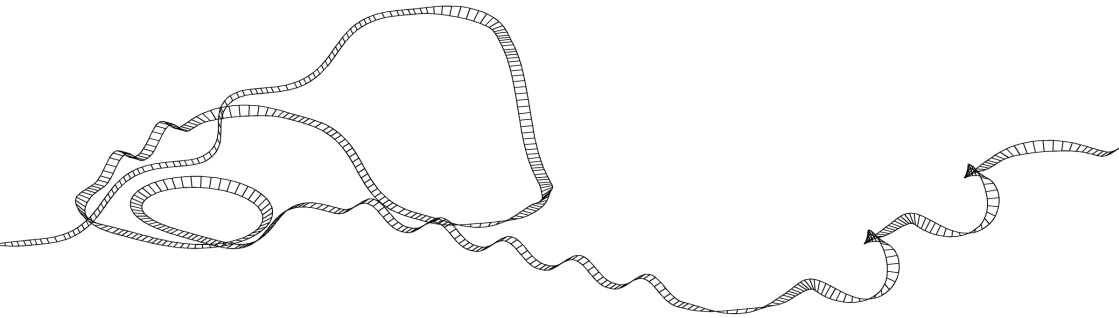
Chain Lift Hill 112

Chinese Tradition 124

Cavern Filled with Destroyed Bobsleds 136

Corkscrew 145

Distance and Soon Becomes Visible 154



Downward Helix

The train leaves the station.

The train then ascends into a roll over, in which the train goes through an Immelmann immediately followed by a dive loop. This element inverts riders twice and is shaped like a heart. The train travels through a banked hill and into a sidewinder, followed by a 270 degree downward helix into a double inline twist that features multiple footchopper effects. The train curves again, dips, and rises up into the trim brakes.

After a show scene in the building, riders are launched from 68 km/h (0 to 42 mph) out of the shed into the first of two vertical loops. They then careen into an airtime hill and a fast-paced corkscrew, before hitting a turnaround and jumping into the next launch, which is situated over an airtime hill. The launch boosts the speed from 56 to 80 km/h (35 to 50 mph), and sends riders into an elevated cutback.

The train then ascends a small hill and goes down a turn towards the double corkscrew element.

The train then makes a slight left turn into the first of five camelback hills before dropping again and entering a figure-8-like helix, also known as an ampersand turn. On the return trip, the train goes over a very small hill (speed hill); at the same time, passing through a trim brake which isn't activated. Immediately after, the train enters the second camelback hill, followed by a slight left turn, then crossing under this section's first drop to enter a splashdown element. Unlike other coasters featuring a splashdown (the ride, Sheikra, the ride, etc.), this section's train never hits the water.

The train continues and makes a downward right turn immediately followed by an upward right turn which leads directly to the 45-degree inclined chain lift hill. Once the train reaches the top of the 62-metre (205 ft) lift, it makes a right turn into a holding brake where the train slowly moves over the first drop, stops for six seconds, and then is released down the 90 degree drop. By the time the train makes it to the bottom of the drop, it reaches its maximum speed of 114 km/h (71 mph). The train enters a 44-metre (146 ft) Immelmann loop before dropping back to the ground and making a banked upward left turn into the mid course brake run.

Both tracks met back up to ascend the chain lift hill 28 m (93 ft).

After descending a drop from the brake section, riders crest a pair of back-to-back airtime bunny-hills. They then turn to the right slightly, then left.

This section's layout is akin to a large vertical U shape, that is loaded at the base and uses a series of LIMs to accelerate the train up each side of the track.

The track leads directly to the 56-metre-tall (183 ft) chain lift hill. Followed by a fairly level 180-degree turn to the right above the

quarry wall. This leads into a 51-metre (168 ft) drop off the quarry wall directly into the 44-metre-tall (145 ft) vertical loop. With the quarry wall on the track's right, the train climbs to the top of the wall and passes through a 360-degree clockwise helix.

After ascending the first lift hill, the train travels around the front facade of the enclosure to ascend the second lift hill. The first drop is this section's only steep drop, while the rest of this section contains three horizontal helices. The building is punctuated by strobe lights and sound effects play over a tribal or heavy metal music score. Occasionally, ride operators will operate the ride with the interior lights illuminated.

This move is done via a typical tyre drive system.

The train climbs up a steep slope into the mid-course brake run. After the momentary reduction in speed, the train turns to the right and descends into a downward 540-degree helix. Immediately following the first helix, the train climbs into an upward 270-degree helix to the left. Following the climb is a small descent and a hill. The train climbs up from the hill into the terminal brake run.

The ride started by climbing to a height of 23 m (75 ft).

The first segment includes a swooping drop and a climb into the second segment, which is a series of S-shaped turns. The third segment is a heavily banked figure-8; the fourth is a series of hills that form a semicircle; the fifth is a pair of small hills; and the sixth is a helix into the seventh segment, which is a long loop that leads to a massive hill and then into the brake run. For most of the ride, the cart is spinning around rapidly.

The train then makes a right turn followed by a slight upward left turn into a corkscrew, before turning right into the mid-course brake run. The train then drops to a point just above a body of water, at the same time making a 270-degree right turn, where water

jets spray up near the train to simulate the effect of the train actually hitting the water. After exiting the turn, the train passes a waterfall, traverses the second inline twist, and makes a left turn into the trim brakes.

The launch sends the train into a steep climb up to its tallest point, a 47-metre-tall (155 ft) top hat. The train then plummets down 43 m (140 ft) at an 80-degree angle, turning to the right and snapping back to the left to complete the 30-metre (100 ft) long zero-g stall. An incline follows taking the train into a 125-degree overbanked turn, leading into a banked airtime hill, a 133-degree overbanked turn, and a speed hill. Riders then reach this section's signature element, a heartline roll dubbed the "Mosasaurus Roll", which inverts riders over water at 85 km/h (53 mph).

The train then travels through a banked turn to the right, which is enclosed within a tunnel. The train then rises up into a hill where it banks to the left before travelling through a second corkscrew. The train then turns left and travels over a small hill before entering the trim brakes.

The track leads to the 27-metre-tall (90 ft) chain lift hill. Once at the top, the train traverses a small dip along a 180 degree turn to the left. The first drop of 23 m (74 ft) leads into a small bunny hop and a near-ground level 70-degree banked turn to the left. A series of small camelback hills are followed by a turnaround.

Next is the zero-g roll followed by the cobra roll which inverts riders twice. After the cobra roll, the train climbs through an upward spiral and enters a straight section. The train dives down to the right transitioning into another brief straight section of track. Riders then enter the first of two corkscrews which rotates the train 360 degrees to the left.

The train turns 180 degrees to the right, then begins to climb the chain lift hill, travelling at a speed of 15.5 kilometres per hour (9.6

mph). Once at the top of the 93.3-metre (306 ft) lift, it drops to the ground at an 80-degree angle, reaching a top speed of 148 kilometres per hour (92 mph).

Riders then take a right turn, travel through an enlarged cranberry sauce can, then turn left. Riders traverse a small hill adjacent to an 18-foot-tall rolling pin, then take a right turn followed by a left turn past larger-than-life recreations of a box of stuffing mix, a kitchen timer, a whisk, and a bottle of milk. The train then ascends a spike before rolling back and traversing the track in reverse.

The train then proceeds to go through a banked turn to the left, and goes through a "high-five" element. After this, the blue tracked train drops down and goes under the train on the green track while it does its top gun stall. The blue track goes through an airtime hill, taking riders close to the train in the green track's stall.

The track travels over an airtime hill above the lift hill, making a slight descending lefthand turn before traveling under the lift hill, where an on-ride photo is taken. It then traverses another airtime hill before entering a 90° banked right turn and 18 m (60 ft) drop into the ravine. The track continues turning to the right, rising up out of the ravine and through the lift hill in another right-turning airtime hill. It traverses two airtime hills and makes a righthand turn.

As the train exits the lift hill, it is sent on a double helix to the left.

According to the park, riders can experience between 2 and 6 g's during these loops. Riders fly through a tunnel containing red lights at 72 km/h (45 mph), before climbing a small hill, where their speed significantly decreases. At the top of the hill is the midcourse brake run and another pre-drop followed by a turn,

this time with a slightly greater angle than 180°. This leads into another drop, this one through the mouth of a stone structure. At the bottom of this drop, the track sharply banks to the right, and the train begins a double corkscrew. These corkscrews have a diameter of 11 m (35 ft).

The train traverses a tall, overbanked turn, hugging the side of the lake, before coasting over an airtime hill. After the hill, the track turns around in a hammerhead turn. Traveling over another camelback hill, the train goes over a series of airtime hills as the track travels back along the lake before entering the mid-course brake run. Dropping out of the mid-course brake run, the track makes a small left turn under the lift hill and goes into another airtime hill. It makes a banked turn to the right and goes into another banked turn to the left over a part of the park's lagoon.

The ride begins as the train continues and immediately enters the 29-metre (95 ft) lift hill. After reaching its peak, the train drops 28 m (92 ft) at a 45-degree angle and banks slightly left as it elevates into a short hill, providing riders significant lateral g-force.

Cresting the hill, the train turns right before descending into the second loop. After completing the loop, the train passes by a statue of the ride appearing to lunge at riders as it rises up into the trim brakes.

Riders begin by climbing the chain lift hill beside the highway to this section's highest point of 24.4 m (80 ft) before it heads down the first drop. Immediately after the drop, the train powers through a 21.9 m (72 ft) tall vertical loop before heading along an elevated section nearly level section of track with a slight dip to the right. The train turns downwards and left and then back up and left towards the second 15.8 m (52 ft) tall vertical loop. After the second loop, the train heads down and through a 4 m (13 ft) deep, 61 m (200 ft) long tunnel before climbing back to the brake

run. This section's famous double loops do not occur in a row, but are lined up to appear that way.

Some writers described the ride as being bumpy while the coaster banks sharply around the tower.

An eerie sound effect is played, then the train then ascends the lift hill. A pre-drop is followed by a nearly 180° turn just before the first drop, which leads directly into two vertical loops.

The track then turns a sharp 180° turn to the right before going back out of the building and onto the vertical chain lift hill. This hill takes riders up to a height of 43 metres (141 ft). Once at the top, the car slowly inches towards the 121° beyond-vertical drop.

The train continues into two smaller air time hills before traveling upward and leveling out into a set of trim brakes at the far end of the out-and-back layout. Here, the trains are slowed as they enter a large 560-degree helix where they circle downward over a small marsh, gradually picking up speed along the way. The helix circles to the left, allowing the red train to pull away, since it is located on the inside track. The tracks then separate upon exiting the helix at a high velocity.

The second inversion of the ride, an Immelmann loop, is followed by an overbanked turn around. An air-time hill, where riders experience the feeling of weightlessness, and a right turn leads into the mid-course brake run.

The car launches at 71.9 km/h (44.7 mph) into the outdoor segment of the coaster, navigating an airtime hill, turnaround, and corkscrew inversion before reentering the building. Riders encounter this section's forces on a bridge; one of the three helicopters in pursuit is shot down by the car, crashing into a the ride vehicle. The resulting explosion allows riders to escape, launching the car through an S-bend below and into the domed tilt track scene.

The train begins to climb the 28-metre-tall (92 ft) chain lift hill. Once at the top, the train makes an approximate 180-degree downward left turn followed by an upward right turn. Following the turn, the train enters the main 20-metre (65.6 ft) drop reaching a top speed of 77 kilometres per hour (48 mph) leading directly into a vertical loop. The train then immediately goes through an Immelmann loop.

An airtime hill carries the train back over the pathway before plummeting into a trench and two successive inversions; a Banana Roll and elongated vertical loop, offering a brief stall while upside down. The car follows up into a sharply ascending right turn only to exit into a dive loop/drop. Riders navigate a counterclockwise helix and airtime hill before making a turnaround to the lift in the brake run.

The ride is a combination of dark ride and launched roller coaster.

Meanwhile, the blue train turns through the wooden supports and traverses only one hill. Both trains then rise into the massive wooden structure of the lift hill, encountering another set of trim brakes to slow them down.

The train is launched forwards over the small hill, going partially up a top hat. The train is then launched backwards once more, back over the small hill and up the 28-metre (93 ft), 100° beyond-vertical spike. The train is then launched for the time, traveling forwards over the top hat. After cresting the top hat, the train turns to the right before traversing two airtime hills.

After dropping from the quarry wall again, the train enters the Sprialing Camelback (zero-g roll), where riders experience a feeling of weightlessness, then through a 24-metre-tall (78 ft) cobra roll. Exiting the cobra roll, the train makes a left hand ascent, then veers right as it approaches the quarry wall and

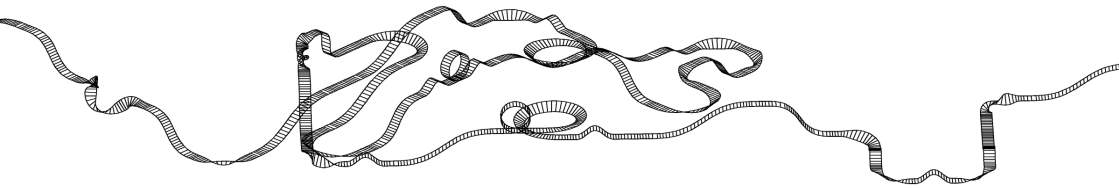
enters block brakes. the train dives to the right and enters a set of interlocking Flatspins (corkscrews). After passing through the second Flatspins, the train makes a right hand turn, and rises into the trim brakes.

Once the orientation is over, the train is launched up a camelback hump. The train then glides across the surface of the park's lake simulating a jet ski ride. Halfway through the ride, the train is launched one more time up the tallest point of the coaster (61 feet) and crosses the only section of the track that is on land (a small artificial island in the lake).

Trains on the ride climb the 55-metre-tall (180 ft) lift hill. At the top, riders take a small left turn and stop, while the train is tilted to peer over the edge. After a brief pause, riders plunge 52 m (171 ft) at a 96-degree beyond-vertical angle, reaching speeds up to 108 km/h (67 mph). Afterward, riders ascend into an Immelmann loop, followed by a dive loop.

Trains enter a 180-degree turn before ascending the chain lift hill. At the top of the lift hill, trains navigate a left turn into this section's 112-degree first drop. They then ascend a hill into another left turn before entering a mid-course brake run. Trains traverse an outer-banked turn into a dive loop, then ascend into a second mid-course brake run. Afterwards, trains make a left turn into a downwards in-line twist, then enter an elevated turn before the trim brakes.

The train launches up the attraction's signature 38.0-metre (124.7 ft) tall wave turn, but ultimately does not gain enough speed to crest it, thus rolling back into the launch. The train is blasted backwards into a 39-metre (128 ft) curved spike overtop the Dreamworld Express train ride; its one distinguishing feature from the standard the ride layout.



Tentacles of Devil's Snare

From the highest point, trains drop 55 m (180 ft) at an angle of 85°, 4.6 m (15 ft) below ground level, reaching their top speed of 116 km/h (72 mph). After this, cars ascend to a height of 38 m (125 ft) for an overbanked turnaround, followed by a descent back to the ground and a small air-time hill.

Traveling over a camelback hill, the ride enters its S-curve and into the 540-degree helix. After the mid course brake run, the ride travels over three camelback hills, followed by a trim brakes.

The train ascends a right-banked turn, completes a hill, then ascends into a left-banked series of dips. The train drops and banks right before going upwards into the trim brakes.

The train begins accelerating down its first 78 m (255 ft) drop at a 61-degree angle, into an underground tunnel, reaching a top speed of 137 km/h (85 mph). Upon exiting the tunnel, the train heads upwards into a banked right turn that towers above the ride. After

completing the turn, the train heads down another drop, flattening out to pass by the onride camera.

The train exits from the rear of the mountain and enters a large helix before being lifted back into the mountain a time. The train drops through a cave, where an audio animatronic of the yeti is reaching down toward it.

The train then banks to the left at high speed before travelling over a small hill. This hill is immediately followed by a much larger hill.

The last part consists of two hills. After those, the ride continues with a brake run and a corner left.

The ride continues with a chain lift hill ascending 28 m (93 ft), leading to a right hand near-vertical curved first drop, followed by two consecutive vertical loops. The ride then turns into a left-hand helix, up and through the center of the second loop.

The track makes a right hand turn, then immediately dives down the first drop, threading under this section's first drop and over its splashdown, and entering a 36 metres (118 ft) vertical loop. This is immediately followed by a dive loop. Right after this, the track goes through a zero-g roll which threads through the second vertical loop, and a cobra roll. The track then rises up into the midcourse brake run.

Immediately, a launch causes acceleration from 9.7 km/h (6 mph) to 72 km/h (45 mph) in 2 seconds, entering an Immelman loop. After this first inversion, the train enters a horseshoe turn, followed by an airtime hill. The train maneuvers through a second inversion, a combination of a half-corkscrew and half of a heartline roll. The train then goes through a 135° overbanked turn, and then enters the brake run.

The train immediately begins to climb the 53.3 m (174.8 ft) lift hill. Once at the top, the train drops back down 52.0 m (170.6 ft) at a

70-degree angle. The train then makes a banked right turn leading into the first of three consecutive camelback hills; each at a height of 37.0 m (121.4 ft), 29.0 m (95.2 ft) and 24.0 m (78.8 ft) (every hill is smaller than the previous one).

A pair of off-axis bunny hills and switch track lead into the main swing launch, where the train builds up most of its speed through three launches – the first isn't powerful enough to help the train clear this section's signature top hat, the second propels the train backwards and up a spike, and the third launch fully sends riders over the hill. Plunging down the top hat's 101° drop, riders make a quick turn into a large zero-g stall, exiting through a left hand turnaround. Riders complete the return trip through a trio of airtime hills – one straight, one twisted, and one fully banked to the side – before leading to a figure-8 turn overhead of the queue and underneath the top hat, which chiefly contains a heartline roll inversion. Riders make a right hand turn and traverse a pair of airtime moments into the brake run.

The train is then launched backwards out of the tunnel and partially up the dive loop. Once it loses its backward momentum, the train returns forward into the tunnel and is launched out of the tunnel and straight into a high-G turn to the left. Riders then go up the "top hat" that is located on top of the loop. Riders drop from the hill and reach the brake run.

A cartoon bomb is shown bouncing along the walls, and when it detonates, riders are launched into the layout. The first lap is navigated forwards and entirely in the dark.

The train then ascends a hill, before descending into a 510° downward helix. Upon exiting the helix, the train enters a decline into a slight left turn, and ascends into the mid-course brake run, before entering a series of three consecutive airtime hills and a double-up. The train passes through the second tunnel before the

second of these hills, and a camera takes on-ride photos at the bottom of this hill before the third hill.

The train then travels into a straight hill before travelling through a banked right-hand turn. The train then travels over a series of small twisting hills before entering a banked left turn. Following this turn, the train travels over three small hills before turning left into the brake run.

On the return run, the coaster slices through the support structure of the lift hill then dives into a surprise curved tunnel.

The track then passes into a trench and into an Immelmann of 22 metres (72 ft), and then a second overbanked turn 15 metres (49 ft). This section's track then descends to an 6 metres (20 ft) deep underwater tunnel.

It makes a sharp right hand turn into the 57-metre (187 ft) chain lift hill. Once at the crest, it enters a right hand banked turn before passing over another crest and into the first drop.

The train then traverses a small high speed hill, before entering a double up followed by a double down.

The train then travels over an airtime hill and down onto a 270-degree left turn which circles the Kisses Fountain. The train then dips slightly and then rises into trim brakes before curving to the right, downward, and up into the trim brakes.

DarKoaster's physical layout primarily contains two Drive Tire Launches and a handful of twists.

The train continues and immediately begins to ascend the 45-metre-tall (148 ft) lift hill. Once the train has crested the top of the lift hill, it banks to the right and drops at 100 kilometres per hour (62 mph) into a vertical loop. Exiting the vertical loop, the train then enters a zero-g roll, and then another vertical loop. Coming out of the second vertical loop, the train makes a high speed banked turn to the left and travels up into a cobra roll.

A small right hand turn leads the train onto the launch track. The trains are then launched at a speed of 105 km/h (65 mph) in 4 seconds, before transitioning into a 44 m (145 ft) "wingover turn".

An off-axis airtime hill is the last element before the train reaches the brake run. Owen thanks riders as they return but is abruptly summoned to tend to a situation over at River Adventure.

The train exits the tunnel and rises up a 20-metre-tall (65 ft) vertical spike before losing power and dropping backward through a sixth launch, taking riders through a tunnel and helix, then slowing down at the darkest part of the Forbidden Forest as the Muggles pass a centaur. The train eventually stops in the tentacles of Devil's Snare.

The Kings Island location has an on-ride camera that takes photos of the riders during the launch. The ride emerges into a "spaghetti bowl" of track inside of the 34-metre (110 ft) tall ride building. Following the launch, the train immediately enters a the ride (a element similar to a cobra roll), which contains two of this section's four inversions.

The track straightens briefly again before turning to the right and entering a short dip before taking riders into the second corkscrew. The ride continues with a 1.5 revolution flat helix where riders encounter strong positive G-forces before making one last left turn into the trim brakes. The ride is identical in layout to the ride at Cedar Point, the only deviation being the absence of a midcourse brake run.

The train then ascends into the first high-speed U-turn to the left, descending again beneath the lift hill and rising to the second 21-metre-tall (70 ft) U-turn to the right. It descends parallel to the lift hill, enters a camelback hill and rises to a smaller banked right U-turn, where it dives under the first high-speed curve. After the third U-turn, the train enters a second camelback hill with

a fan turn and a smaller airtime section as it approaches a fourth U-turn to the right. The train hops several times more, paralleling the second drop, before entering a right curve. It drops slightly, ascends into a tunnel with a small left fan turn.

The design is a reverse replica of the ride roller coaster at sister park Kings Dominion. The ride is the same ride experience and layout, as Legoland's the ride roller coaster.

It traverses two more airtime hills, separated by right and left turns, followed by a drop and a series of smaller airtime hills. After a left turn, the train passes through several additional drops and rises before entering the brake run, which runs parallel to the lift hill.

The train then dives into a cobra roll. Upon exiting the cobra roll, the train twists through two consecutive corkscrews before a short left turn into the trim brakes.

The train then passes under the lift hill through an over banked turn at 92 degrees. It then turns to the right into another over-banked turn followed by an airtime hill. The train then turns to the right into the brake run.

Kicker wheels propel the vehicle along a straight track segment toward the chain lift hill. The train performs a slight dip before ascending the 37-metre-tall (120 ft) lift. At the apex, the train executes a gradual descending left-hand turn, leading into the first inversion, a vertical loop. Immediately following the loop, the train enters a zero-g roll. The layout then transitions into a slight leftward curve onto a straight section where the first of two on-ride photo cameras is positioned on the left side of the track. The course continues into a double left-hand corkscrew, representing the two inversions of the circuit.

Trains on the ride make a right turn before climbing the 45-degree chain lift hill to a height of 55 metres (179 ft). Trains then

make a right turn before approaching the drop. Trains are held at the top of the drop for three seconds before being released down the 87.5-degree drop and under a bridge into the tunnel, where riders experience 5 g. After exiting the tunnel, riders traverse an overbanked turn to the right, then rise into the trim brakes.

The ride continues with an ascent up a 45-degree lift hill, after which the car disengages from the chain and enters a series of 180-degree turns along a gradual descent. This is followed by an elongated 270-degree right-hand turn leading into a straight section and a minor incline before the first set of trim brakes. The layout then continues through a 180-degree turn into a significant drop and rise, followed by another 180-degree right-hand turn and a series of airtime hills.

Riders then go through two negative-G hills and pass under a waterslide before climbing up a larger hill and making a turnaround to the right. The ride then drops down and rushes over three smaller hills before entering a banked turn to the left.

It climbs the second lift and turns left into the "mini horseshoe", which leads into a right turn downward spiral. From the spiral, it rides over the "c" dip, turns right and climbs the third lift hill. After clearing the third lift, it turns left into the "big horseshoe" and enters an elevated barn enclosure. When the train nears the end of the enclosure, it descends into a tunnel, reaching its actual top speed of 35 mph and completes an elongated S-shaped element, starting with a left turn (left right left).

The lift hill is completely enclosed in a 79 metres (259 ft) high tower. The roller coaster also has a non-inverting sea serpent roll, a vertical descent with a twist, airtime hill and a heartline roll at the end of the ride. After the first drop and the heart element, the layout of the roller coaster is kept closer to the ground in large parts of the route.

The train then traverses a wave turn circling the queue, followed by a corkscrew. After a left turn, the train reaches the trim brakes.

As the train reaches the top of the lift hill, the train goes through a pre-drop before making a sharp turn to the left towards the first drop. When the train reaches the bottom of the drop, it goes through a vertical loop followed by a slight turn to the left afterwards goes through the second out of four elements, a dive loop. After, the trains goes over a small airtime/bunny hill and goes through the zero-g roll. Immediately after, the train approaches the inversion of the coaster, an immelmann loop. The train then enters a twisted airtime followed by another right turn before hitting the trim brakes. After passing the brake run, the train makes a right turn going through another set of brakes.

As soon as the train crests the chain lift hill, it nears the first drop, a twisted drop then rockets the train to speeds of 45 miles per hour. After the drop, the ride soars over the ride and into a turnaround the then heads towards an airtime hill. The ride then circles the Tig'rr Coaster's area into more airtime hills.

After turning left out of the inversion, the train goes up a hill into a 180-degree stall featuring several head chopper beams. A wave turn element and several air time hills follow, along with an over-banked turn and an Asian camelback hill. The track winds sharply through another over-banked turn before entering the inversion, a zero-g-roll. Then, the train undergoes another banked turn and camelback hill.

A series of close-to-the-ground airtime hills take the ride under the lift hill of the nearby Hades 360 roller coaster, and out to the back of the park. A turnaround places the track parallel to the first set of airtime hills.

The coaster twists through an off-axis hill, a right-hand turnaround, and a zero-g roll.

The train continues and makes a right turn into the first launch. Riders hit a left turn and go over a smaller airtime hill. The train races through Andy's yard, going through a frame built out of blocks, with Jessie clinging to it. The train then moves through a left-handed helix.

The train is driven in reverse up a drive tyre lift hill. Once reaching its maximum height, the train is held for three seconds before being released. The train drops back down the incline passing through the covered section and over a small air time hump, banking and dropping right. It passes back under itself, banking upwards and left onto a parallel incline to the lift hill.

The train drops and immediately traverses a cutback followed by a corkscrew. The train continues through a quick dip into an over-banked turn and continues with a small ascent into the brake run.

After riders are secured, the train continues backward and ascends the first chain lift hill, backing into the themed pit stop covering at the top. Following a brief narration, it descends forward and travels through the covered section at approximately 48 km/h (30 mph). The train then navigates a banked left turn, an airtime hill, a headchopper element, and a helix that spirals clockwise.

The train takes a short drop and then does a 180° turn to the right. After passing the train storage area, it takes a short turn to the left. Then, up the first lift hill. At the top, a short dip followed by a slight right turn is taken. The train then dives 180° to the left to the ground.

The train turns right and is taken up a friction wheel lift hill to a height of 13 metres (43 ft). A small drop follows the lift hill that then leads into two downward helices.

The ride continues by climbing up a 120-foot vertical rise. Then zooms over hills while pivoting forward multiple times, before

plummeting into a steep raven drop.

The train then travels down a hill toward the ground and into the loop. Upon exiting, the train passes through a tunnel and over a hill into another descending U-turn. The train ascends slightly as it completes the U-turn and enters the trim brakes.

It immediately begins to climb the vertical (90 degree) lift hill. Upon reaching the top of the lift, the train turns 45 degrees backwards putting the train and its riders in an upside down position. Then, as the train exits the lift, it goes through a heartline roll before going down the drop in the layout of the roller coaster. The train speeds through the covered section going back up the lift.

The surfboard-themed train launches 97 km/h (60 mph) into a 34 m (110 ft) overbanked turn to the right.

The train then navigates a small airtime hill before rising into a large immelman loop.

Within the tunnel, the train traverses a helix, making 2.75 rotations. When the train exits the tunnel, it climbs a bunny hill, hits a short brake run, and then climbs a second lift hill.

The initial drop is followed by a left-banked curve and into a few more smaller hills until then reaching a right-banked curved leading into several more small hills. The train then curves right onto a section of bunny hills where it continues until it reaches the tunnel which contains the brake run.

The ride begins when the train slides onto the main launch. Soon afterwards, an alarm is sounded and riders are boosted through a 58-metre (190 ft) tunnel and up into a 46-metre (150 ft)-tall inside top hat that flips riders completely upside-down. This element is followed by a 34-metre (113 ft) overbanked turn that banks riders at a 140-degree angle and a vertical spike. As the train climbs this spike and slows, it is gently pushed up almost all the way to the top by another set of LIMs. The train then freefalls

down the tower back through the course.

It then drops 54 m (176 ft) at a 76-degree angle, reaching a top speed of 110 km/h (70 mph). As the train reaches the bottom of the drop, it comes close to the track above, creating a headchopper effect. It then travels up a 34-metre (112 ft) camelback hill followed by a second camelback hill at 30 m (100 ft). It then rises and then travels through a 180-degree downward-banked turn to the right, and up another banked turn to the left.

The train then goes over a third camelback hill, immediately followed by a camelback hill. The train makes a left hand turn into the mid-course brake run which slows the train down. After the train exits the brake run, it goes over the fifth camelback and makes a sharp right hand turn into an element known as a Diving Spiral.

The train proceeds over a 55-metre (182 ft) parabolic hill, which provides a moment of zero gravity as it passes over a lagoon and down onto Adventure Island. It completes an 27-metre (87 ft), 360-degree right-handed helix, followed by a 30-metre (100 ft) 122-degree left overbanked turn. This is followed by a small right-hand turn and another hill that leaves the island. The train then travels left through a second tunnel where the on-ride photo is taken, followed by a left turn and a small hill, passing by the queue. Then, the train travels 21 m (68 ft) high through another right overbanked turn over the queue and is stopped by magnetic brakes.

The car then accelerates backward over a small hill before traveling through a straight section of track. This straight section of the track goes through this section's splashdown pool, which at this point in the ride is low enough to allow the vehicle to pass over it. The car then travels part of the way up a vertical spike of the track before traveling forward over the hill. The car then launches

forward into a second vertical spike and comes back down before entering a third backward launch.

A straight uphill climb leads to a curving dive to the right as the coaster assumes a more typical GCI layout. According to the designers, the soaring curves mimic the action of a large ray swimming in the ocean. Multiple curves and dips are combined with airtime pops.

The train then crests a hill and curves left over the covered section, followed by a second major descent into an enclosed tunnel. Within the tunnel, the train navigates a turn before traversing two successive camelback hills, after which it enters a manual brake run to conclude the circuit.

It ascends a 55-metre-tall (179 ft) chain lift hill, before dropping 52 m (171 ft) at an angle of 81°. The track then ascends to the top of the quarry wall where it enters a 110° over-banked turn. A 95° over-banked turn leads to the left, back off the quarry wall. The ride then rises into the zero-g roll which sits atop the quarry wall. A camelback hill is followed by two more over-banked turns measuring 98° and 93°, respectively.

The train then drops 91 m (300 ft) at an 80-degree angle and reaches a maximum speed of 150 km/h (93 mph) at the bottom of the hill. This is followed by a climb of 52 m (169 ft) through a right overbanked turn at 122 degrees from the horizontal axis, of which the train then travels through a tunnel as it passes over the Frontier Trail.

The train enters a zero-gravity roll followed by a right turn leading into interlocking corkscrews. After exiting the first corkscrew, the train makes a left turn before going through the second corkscrew.

Then, the train traverses a helix and a turn before reaching the trim brakes. This section's 885.0-metre (2,903.5 ft) layout features

two inversions and reaches a height of 40 m (131 ft) and a speed of 85.0 km/h (52.8 mph).

The train makes a slightly banked right turn into a zero-gravity roll. After an upward left turn, then another right turn, the train rises up leading into the trim brakes.

The train then falls back through the covered section, where it is accelerated backwards into a twist.

Riders are taken up 14.8 metres (49 ft) by a wheeled lift hill, and go through a tight helix, followed by a series of small turns and drops. The ride is slowed by a magnetic brake run.

It then proceeds up a 30-metre (100 ft) vertical lift hill. After peaking the hill, the ride enters into a beyond vertical drop at 95 degrees.

The train then travels through a large overbanked turn to the left, passing underneath the lift hill structure. This turn is followed by a series of small hills and banked turns before the trim brakes. There is a water feature consisting of fountains on either side of the track immediately before the trim brakes.

The train then rises up into a turnaround, wrapping around the outer side of the initial turnaround's structure.

Trains climb a 49 m (160 ft) lift hill before making a slight left turn onto the tilt track. After the train locks in place, the track tilts downwards to a 90-degree angle before being released. Following the drop, trains ascend a hill before turning to the right and entering a corkscrew. Trains descend a triple down into a tunnel. Trains then traverse two hills, a second corkscrew, a wave turn, and a set of hills before entering the trim brakes.

Riders instantly start climbing the lift hill. After a little dip in the track, riders go through 2 helixes and go into 2 quick bunny hops. Riders then go through 2 more helixes.

Riders immediately enter the record-breaking 49-metre-tall

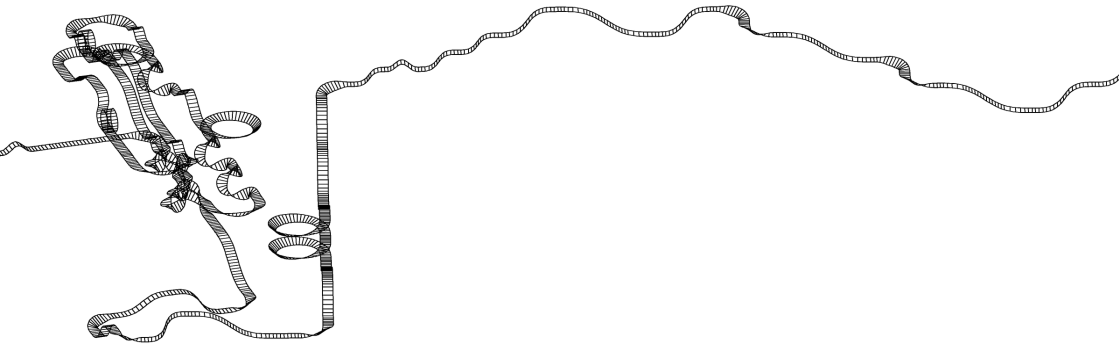
(160 ft) vertical loop, which is followed by a high-banked curve to the right and a to the left as the train approaches the second inversion, a dive loop.

The train then traverses over seven airtime hills and a third tunnel into a mid-course brake run. The train turns left and goes back over the walkway to Cedar Point Shores.

The train rises 18 m (60 ft), and then plunges down the first hill, and up the next and down again around the west turn, into several more series of hills which gradually get smaller. It goes through two more turns.

The car ascends into a double-up and encounters an exterminator animatronic as it takes a left turn into a hallway. At the end of the hall is a large mirror that creates an illusion of colliding with another car.

A miner's evil laugh echoes down the first small drop as the car rushes toward the first curve. A wall of cawing crows and a caged canary watch the car as it passes. After turning the corner, passengers come face to face with a giant spinning rock crusher.



Surrounding Croatian-Themed Landscapes

Cars are painted to look like cats or mice to fit the theme of the ride.

As the train continues, a recorded message, "Clear for Takeoff," is played. The ride then climbs its chain lift hill, reaching a peak height of 34 m (113 ft). After a small pre-drop, the roller coaster train then drops to the right before entering its first inversion, a 27 m (90 ft) vertical loop. Riders then travel through a small ravine that dips below ground level and enter an Immelmann loop which sends the train in the opposite direction.

A high speed S-curve follows, as the train traverses a bunny hop and a turnaround into the trim brakes.

The train dips to the right before ascending the 25.4-degree and 24-metre (80 ft) tall lift hill. Cresting the hill, the train descends the 52-degree and 23-metre (75 ft) drop, banking to the right and reaching maximum speeds of 76 km/h (47 mph). It then traverses a series of left-banked hills before descending into a drop. The train

ascends to the left before dropping into a succession of curved hills and entering a right-banked hill.

Both hills are built side by side and head in opposite directions. The ride turns to the right. After ascending the first lift, which travels partially underneath this section's lift hill and first drop, the ride descends down first drop and then rises into a wide left turn. This leads into a right turn and a near-miss with this section's lift hill, followed by a slight turn to the left and another turn to the right.

The recorded sound of a wolf howling can be heard before the train dives down a sweeping left-hand 34 m (113 ft) drop at 95 km/h (59 mph) into a covered tunnel. The track then rises to the left, uphill in preparation for the spiral drop. The train then descends into a 23 m (77 ft) spiral drop to the right before traversing an airtime hill. After heading uphill again, the train enters a 20 m (64 ft) drop through the second underground tunnel. After exiting the tunnel, the train makes a turn to the right, then to the left, before dipping down another drop and rising into the double helix.

The ride continues with a 35-metre (115 ft) lift hill. Followed by that is a 31.5-metre (103.5 ft) drop, a rollover, sidewinder, turn and back to back inline twists, before slamming to a stop, or trimming into the brakes.

The train climbs a 31.0-metre (101.7 ft) tall vertical Lift hill before plunging down a 73° curved drop.

The coaster continues into a left-hand turn before ascending the 21.0-metre (68.9 ft) tall lift hill. Upon cresting the top, the car traverses an initial short drop and a trick-track over the pathway, and makes an uphill right hand turn into the first block brake section.

It is then propelled by linear motors from a standstill to 99.9 km/h (62.1 mph) in two seconds. The train ascends from the launched track section into a 38-metre (125 ft) tall corkscrew. Im-

mediately afterward, the coaster enters a banana roll inversion, a second corkscrew, and an airtime hill. The train ascends onto a set of block brakes, slowing the train down before it makes a U-turn to the right and ascends the 43-metre (141 ft) vertical chain lift hill. Once at the top, the car is held by a brake before entering the 121.5°, 43-metre (141 ft) beyond-vertical drop. Once the car is released from the top of the hill, it traverses a dive loop, an inline loop, and an Immelmann loop, before hitting the trim brakes.

The ride sends riders into its first LSM launch and into a sharp 180° right hand turn, where the train meanders through a series of low-to-the-ground twists and airtime hills. Upon entering the second launch, the train reaches a top speed of 99.9 km/h (62.1 mph) and ascends a 38.5-metre (126.3 ft) tall twisted top hat. Plunging down the drop, riders enter the first of four inversions on the coaster - a vertical loop - and proceed through a twisted airtime hill, which in turn dives under the covered section and leads the train into a batwing double inversion. The coaster dives underneath the covered section once again, navigating a series of low-ground twists and turns throughout the established footprint, taking riders through a handful of airtime hills, 270° helix, and a corkscrew inversion on the way.

It then goes over a 23-metre (74 ft) airtime hill, followed by a turn to the right into the Twisted Horseshoe Roll that features two 360-degree corkscrew rolls, back-to-back, with a 180-degree banked curve between the two corkscrews. The train then makes a turn to the right into the tunnel, during which riders slow down and encounter multiple railroad crossing signs and lanterns. From the tunnel, which is underneath the covered section, the train is launched at 110 km/h (70 mph), traveling for 120 m (400 ft). At the end of the tunnel, it goes up a hill with trim brakes, followed by a high-banked curve. The train then goes through a

turn to the right into a canyon with an S-curve.

The train begins climbing the 57-metre (188 ft) chain lift hill. At the top, the train curves into the 52-metre (171 ft) drop and enters a 44-metre (144 ft) vertical loop.

Making an U-turn, the car rolls toward a large electrical transformer, which the car swerves away from with a hairpin turn. After navigating another hairpin turn, the car passes under a sign reading "Danger - Boiler Room". The car then makes a sharp left turn into this section's first drop into a large boiler room, passing by a giant furnace. The car rises again, making two more left turns into its biggest drop.

Rolling forwards once again, the train hits the launch track for the third time, where it reaches speeds of approximately 104.9 km/h (65.2 mph) and tops the wave turn, leading into the first of four inversions on the ride; a 32 m (105 ft) vertical loop. Exiting such, it twists around and up into a mid-course brake run, which then leads into a pair of back-to-back twisted horseshoe rolls and an airtime hill through the centre of the loop. Riders enter another slow turnaround and ride through the inversion on the ride, a heartline roll, before making a right-hand turn into the brake run.

After speeding through a turnaround encompassing the ride queue, the train enters the second lift hill, which is powered by drive tires, with reasonable speed. Riders plunge into a curved drop and speed through the brake run towards the water, navigating a trick-track section before rolling out onto the pond and up the spike element, similar to those seen on Schwarzkopf the ride coasters. The train loses speed, as intended, before rolling back through the switchback and into the brake run.

The train immediately climbs the 32-metre (105 ft) chain lift hill. At the top, the train makes a sharp 24-metre (80 ft) downward

right turn into this section's first element, a vertical loop, reaching a top speed of approximately 80 km/h (50 mph). After exiting the loop, the train enters a zero-g roll inversion before entering a second vertical loop.

The train makes a large right turn and then travels up a chain lift hill. It gently trundles around a curve at the top of the hill. The train then violently dips downwards. A ratchet that is in place to prevent roll backs precedes a short straight section. After this straight section, there is another smaller dip.

It then makes its way through a 17-metre (56 ft) airtime hill. This is followed by a 45-metre (147 ft) airtime hill with a trim brake. The train zooms through a 38-metre (125 ft) spiral before passing the on-ride camera. The train goes through a left turn into two more airtime hills measuring 27 m (90 ft) and 25 m (83 ft). Following this, the train slows down on the brake run, makes a left turn.

The ride continues and ascends a chain lift hill. The train enters the first drop, dips to the right, and rises back up through this section's centrepiece theming element – a portal – which correlates to this section's space theme. Immediately following this, the track twists, turning riders from the prone position onto their backs. The train then embarks on a large upward left turn before twisting again, returning riders to the prone position. After exiting from this lie-to-fly element, the train passes underneath a small ravine before pitching up into a tight turn over the plaza area.

The ride will have an elevation change of 24 m (80 ft) and a largest drop of 18 m (60 ft), as well as top speeds of 74 km/h (46 mph). The ride will be 574 m (1,885 ft) long.

Guests begin by entering the queue, which is adorned with numerous gothic-themed elements.

The train then enters an Immelmann loop followed by a dive loop. A short stretch of high speed "S" curves lead the train into

a cutback followed by an inclined loop. As the train completes the inclined loop, it reaches the brake run.

There then follows various coaster elements, including a booster launch, a turntable, a reverse launch, Dive Loop, Top Hat, Immelmann Jr. The ride Tail, Zero-G Stall, two Corkscrews, ejector airtime, multi-pass/swing launch and more. Throughout the ride, the trains pass through the surrounding Croatian-themed landscapes, which contain ruins, rocks and a church tower.

When it does, the train makes a 45-degree, right-hand turn toward the lift hill. Upon reaching the top of the 13-metre (43 ft) hill, the train makes a small dip and a 180-degree turn to the left, followed by a small drop into a right-handed corkscrew.

After a third airtime hill that soars over the lake, the fourth hill takes riders into this section's structure and mimics the "S-bend" turn of the Hurricane. The ride then soars over the Kiddieland section and goes into a helix. Following the helix, riders are treated to two more bunny hills of airtime before turning into the brake run, ending a ride on the ride.

The attraction (queue and launch) also features an original 30 minute electric/orchestral hybrid score by composer Rick McKee.

Riders on both tracks climb adjacent chain lift hills, reaching a maximum height of 26 metres (84 ft). Riders make a right turn followed by a drop at a 52-degree angle, reaching their maximum speed of 74 kilometres per hour (46 mph), then climb a slope before making an ascending right turn. Riders then drop back to ground level before traversing a double up, a left turn, and a double down. Entering a tunnel, the tracks split; the red track drops while twisting to the right before making an approximately 270-degree curve in the same direction, while the blue track traverses an enclosed elevated right turn prior to a nearly 360-degree curve to the left. This creates a near miss effect of the trains approaching each other

head-on. The tracks then join together once more for a hill and right turn into the brake run.

Riders continue in a standing position, then ascend a small 27-metre (90 ft) lift. A pre-drop, characteristic of early Bolliger & Mabillard coasters, follows the lift hill, preceding the curved right drop. After the drop it reaches its full speed of 50 mph. A vertical loop follows the drop and is followed-up by an upward right and downward banked turn.

The ride is a suspended coaster with 5 inversions. The ride continues with a LSM launch to its max speed of 82.1 km/h (51.0 mph) before diving into a cobra roll.

The train makes a slight turn to the right followed by a turn to the right, then the train turns left onto the 33-metre (108 ft) lift hill. After the train crests the hill, it enters a tunnel then drops 50 degrees to the right, followed by another slight drop, reaching a top speed of approximately 87 km/h (54 mph). This section's first drop wraps around the park's signature Star Tower attraction.

The second corkscrew passes directly under the Sea World Monorail System before curving up and back over it.

Riders are dropped 33 m (108 ft) down, fly back through the covered section and into a cobra roll element. The riders then are taken through a 360-degree vertical loop and are sent up a second 35.5-metre (116.5 ft) hill. The riders pause, and are sent down to do the full circuit again backwards.

The train then falls back through the covered section, where it is accelerated backwards into a twist.

The train climbs a 87-metre (287 ft) chain lift hill. Upon reaching the peak, it drops 91 m (300 ft) at an 85-degree angle, reaching 146 km/h (91 mph). The train speeds through a 53-metre (174 ft) banked airtime hill (or "wave turn") and a 62-metre (202 ft) horseshoe.

It climbs up a 56.02 m (183.8 ft) lift hill using a cable lift system, then curves to the right. It then drops to the ground at a 77-degree angle. It then goes back up to a 46 m (151 ft) tall hill and then drops into an intense right-hand turn drop.

After some time, it goes into a corkscrew. It then plunges into a 5-meter deep pit filled with artificial fog and goes straight into a vertical loop that goes around the launch (which is the most intense part of the ride) and dives down into the pit again. Lastly, it has a zero-g roll over the launch and goes into a helix before reaching the brake run.

As the train climbs the chain lift hill and enters the drop, it falls into a large vertical loop, and then into multiple 180-degree helixes. Then ride goes around a large 180-degree helix and into a brake run.

The train then goes for its second lap, navigating the launch and layout once again, although with the riders facing backwards. The turntable once again rotates the train into a forwards-facing position, and riders then unboard.

After climbing halfway up, the track leaves the first lift hill, and makes a left turn out of the structure, then slams into a 180 degree turn passing under the exit from the double helix and climbs the upper section of the lift hill. At the top of the lift hill, the track passes through the swoop curve before diving down an 89.6-foot first drop, then rising up a second hill for the first turnaround.

This section contains about 790 m (2,600 ft) of track and three inversions.

The train makes a left hand turn, entering the 50-metre (164 ft) lift hill. Once at the top, the train drops back down and enters a diving loop. The train makes a short left hand turn before making an upward right turn. The train then drops back down and goes through an artificial splashdown.

The train makes a 90 degree turn to the right, then begins to climb the 76-metre (249 ft) tall chain lift hill. Once the train reaches the top of the lift, the train drops back down to the ground at a 77.4 degree angle reaching speeds of up to 134 kilometres per hour (83 mph) as it passes through a tunnel and races past the drop off this section's midcourse brakes.

The train begins its ascent up the first of two chain lift hills. Once reaching the peak, the train disengages from the chain, and gravity takes over as it navigates a series of helixes and turns through a dense forest. After losing momentum, the train ascends a second lift hill. It then continues through more banked turns and twists, soaring through the forest canopy. The coaster reaches its most intense moment as the train dives toward the ground, passes through a tunnel, and executes a sharp right turn to exit the tunnel.

The train soars up and rolls out to the left as it exits the Norwegian loop. Upon its exit, the train dives down and to the left into a cobra roll. Following the cobra roll, the train rolls into two consecutive corkscrews and a banked turn to the right through the lift hill. Exiting the turn, the train flies over a small airtime hill and through a wide low to the ground left-banked turn up into a slanted downward trim brakes. The train makes a left and right turn.

Riders instantly start climbing the lift hill. The first drop curves towards the right before going through the vertical loop, which is the inversion on the ride. After the loop, riders go through a right over-banked turn before going down a small drop followed by a smaller left over-banked turn. After going through another small drop, the ride does a banked turn to the left before dropping once more and going through a right banked turn through a tunnel.

After a slight right turn, it climbs the second lift hill. The track

does the same thing as the first lift hill, but continues with a 360° turn instead. After a bunny hop, a slight climbing right 180* turn, the train climbs a small lift hill. The train goes slight right, then slight left, with another slight right.

The ride passes directly overhead Timmy's Halfpipe Havoc, navigates a few smaller airtime bumps.

Trains pass over a trimmed airtime hill and make a right turn inside the structure of the second hill. The train descends another drop, traveling within the structure, and traverses a banked right turn into an underground tunnel, the entrance into which being where the on-ride photo is taken.

After passing through a tunnel and reaching the maximum speed of 95 kilometres per hour (59 mph) at the bottom of the drop, riders enter their first inversion: a reverse sidewinder. Right after, the train turns to the right into a twisted airtime hill, which twists riders to the left.

The train continues into a small airtime hill with a trim brake and a whip around an Immelmann turn. The track levels out before sending riders up into a zero-g roll, followed by a short descent into a shallow turn. The train descends into a double halfpipe, twists right then left, and proceeds into a downward helix that twists back to the left.

A cable lift carries the train to the top of the lift hill. The train then drops at an 85-degree angle, traveling through a large right hand turn and into a large airtime hill. The train next traverses a 270-degree helix before entering another airtime hill. After this, the train enters a right-hand turn that transitions into an overbanked turn.

The ride takes off and goes through a loop. It then goes up a hill, loses momentum, and goes backwards through the loop. It goes past the covered section to another hill.

While this section's only elements are several small hills, it uses the park's topography to its advantage. Built over steep terrain, this section features many tight twists and turns, which intensify the ride and its sense of speed. It is located next to the carousel and the ride.

The song "Turbulence" by Steve Aoki and Laidback Luke is played from this section's PA systems on the lift hill, although it is all but drowned out by the sounds of the lift mechanism. Once at the top, free spinning mode is enabled and the ride makes a left turn into this section's main drop, achieving a top speed of 64 km/h (40 mph) and pulling up into a large turnaround. Following an airtime hill, the coaster twists through a low-to-the-ground series of track. A cross-layout figure-8, airtime hill, and helix compose much of this section, with a tight turn leading the trains to hit the brake run with plenty of momentum to spare.

The coaster continues and goes up a lift hill. Riders then drop 30 metres (98 ft) and reach a speed of 80 kilometres per hour (50 mph) before entering the first loop, followed by the second. The coaster then goes through a corkscrew, which is one of the inversions.

The ride momentarily flattens out before turning right through a wide and heavily banked flat turn. The train traverses a series of airtime hills and another wide, right hand banked turn. Following the second large turn, the train traverses another series of airtime hills, followed by an elevated right hand banked turn. Following this is another smaller airtime hill, another large, right hand banked turn, and one airtime hill into the trim brakes.

Riders begin to ascend the 63-metre (207 ft) cable lift hill at an angle of 51 degrees and a speed of 30 kilometres per hour (19 mph). The train then descends the main drop, reaching a maximum speed of 120 kilometres per hour (75 mph). Immediately after the drop, the train then turns right and enters a 52-metre (171

ft) non-inverting loop. The train then travels through several banked turns and hills before entering a heartline roll near the end of the ride. Shortly after the heartline roll, the train enters the trim brakes.

An audio buildup featuring a voiceover repeating the word "icon" marks the LSM launch, which accelerates the train to 80 km/h (50 mph) through a tunnel into a 25 metre (82 ft) tall top hat. The top hat then passes under this section's lift hill. This is followed by a shallow non-inverting inclined loop before a series of banked turns, intertwining with the ride and Steeplechase, before turning over Steeplechase's brake run and navigating through a downward barrel roll. After diving into another tunnel, a second LSM launch propels the train to 85 km/h (52.8 mph) into a non-inverting 27 metre (88.5 ft) junior Immelmann loop, the tallest part of the layout. A series of highly banked turns, airtime hills, a helix and two s-curves follow, before the train hits the trim brakes.

Upon latching on to the lift hill chain, a pre-recorded voice chastises the riders saying, "Hey! Hey! I see you up there! What on Earth do you think you're doing? This is security. Seriously, guys, you need to take that truck, go home to your video games. This place is not safe." Cresting the 109 foot lift hill, the chain slows again so that a pre-recorded voice can tell the riders over loudspeakers, "Whatever you do, don't go in the shed!" At the top, the train makes a swooping 98 foot drop to the left, before rising over a hill. At the crest of this second hill, the train makes another left turn and crosses over the lift hill, before diving close to the ground and traveling over a series of airtime hills. The track briefly enters a covered tunnel as it makes a left-hand turn to change directions.

Continuing down the valley, the train enters the third inversion, a zero-gravity roll, where riders experience the feeling of

weightlessness. The train makes an upward left-hand turn into a "pretzel knot" (an element where the train enters a dive loop, followed by an Immelmann loop). Upon exiting the pretzel knot, the train enters a second vertical loop before making a right turn into an inline twist.

A roll over (also known as a Sea serpent roll) inverts riders twice by featuring a half loop followed by a twist, then a twist and a half loop.

The train continues heading north and immediately turns right at an almost 180-degree angle, which leads to an 23-metre (75 ft) lift hill. It then moves over the first 58.1-degree drop; as the train reaches the bottom of the drop, it comes close to the track above for a headchopper effect.

The train continues to go left before going through an upward helix. After, the train makes a downward right s-bend leading into the second inversion, a corkscrew. The train makes a left turn back up before going through another s-bend, small over-banked turn which leads to the brake run.

It drives onto the lift hill, which ends in a dive drop at the top. There follows a right turn with 360 °, at the end of which is the looping. After the loop, the train goes into zero-g-roll.

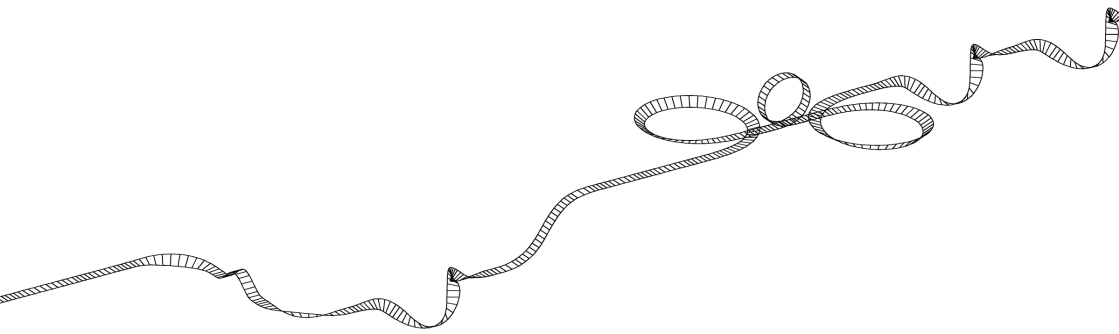
This section's layout consists of two camelback hills and several banked turns, some of which the park marketed as "inclined loops", though they differ from true inclined loops as they do not invert. The ride additionally features a 63-metre (208 ft), 65-degree drop from its 62-metre (202 ft) chain lift hill, which leads into a 1.8-metre-deep (6 ft), 38.7-metre-long (127 ft) tunnel, where the coaster reaches its top speed of 117 km/h (73 mph). The train takes a left-hand U-turn and begins to ascend the chain lift hill. At the peak, it traverses a pre-drop before descending into its main drop and passing through a tunnel.

For the majority of the ride, the vehicles remain vertical with only slight banking occurring on various turns. Riders therefore do not experience traditional roller coaster inversions. On-ride photos can be purchased following the ride.

This section is themed around the effects of a Category 5 Tropical Cyclone at a coastal shipping port. Theming surrounding the ride depicts a path of destruction from the storm, with boats, cars, shipping containers and other debris strewn throughout. However, the port is in the eye of the storm and everyone must evacuate via Coast Guard rescue boats.

To ride alone, guests must be 1,200-millimetre (48 in). The height limit of the ride is 12.6 m (41.5 ft).

Due to this section's placement, it can be seen from anywhere inside the park, as well as from nearby hotels and the Reus airport.



Underground Tunnel

The train then traverses a zero-g roll, turns down and to the right into a corkscrew above a barrel roll separated by a 270-degree turn. This is followed by a 180-degree turn to the left, then a 360-degree helix into the trim brakes.

A helicopter rises to the train's left, shooting simulated gunfire.

As the train reaches the bottom of the first drop it reaches its maximum speed of 110 km/h (70 mph). The train enters a 44-metre (145 ft) Immelmann loop before making a banked, upward, 3.5 g left turn into the mid course brake run. The train slows down and descends a second 42-metre (138 ft), 90-degree drop into an underground tunnel. The train makes a right overbanked turn leading into a splashdown that sprays two 18-metre (60 ft) lines of water in the air and takes approximately 1.9 seconds to pass through.

The coaster layout for the ride is particularly close to two other

Euro-Fighter (model 320+) roller coasters: Rage at Adventure Island and the ride at Duinrell.

The train continues into a 58-metre-high (190 ft) barrel turn, into a high speed s-curve that crosses the North Carolina and South Carolina state line. After passing over the park's north entrance, the train makes a banked turn to the left that leads into a 91-degree overbanked horseshoe turn 48 m (157 ft) off the ground. The train then dips down, passing under the entrance pathway and through several illuminated hexagons – an element the park refers to as the "hive dive". This is followed by a second banked turn to the left at a height of 31 m (101 ft), which crosses the state line once again. After a straight section of track, the train travels over a 34-metre (111 ft) camelback hill.

Riders then go through three more small hills and enter the trim brakes.

The actual layout is quite simple; after a brief vertical lift hill, the coaster descends into a mostly downhill track course of flip-inducing hills and raven turns. Each side of this section's train is nicknamed Knight and Squire; the Knight side offers a more thrilling ride and uncontrolled spinning experience while the Squire side offers a milder ride.

This section is themed around flying on the back of an eagle through a landscape inspired by the Rocky Mountains. Riders ascend to the top of a rock formation known as Ørneklippen (Eagle Rock), where an eagle nest with a mother and her chicks can be seen, before descending into a series of turns passing LEGO animal figures and scenery. The attraction simulates a flight through a western-themed environment, incorporating rock formations, lakes, and prairie animals as part of the experience.

It then climbs the first camelback hill before descending this section's tallest drop of 53 m (175 ft). A banked left turn leads to

a second camelback hill with a 39-metre (129 ft) drop. The train then climbs into a 540-degree descending helix, passes through a set of trim brakes, and traverses a third camelback hill with a 36-metre (118 ft) drop. The train then enters a horseshoe turn, followed by three progressively smaller camelback hills with drops of 24 m (79 ft), 17 m (56 ft), and 15 m (48 ft). Following a banked left turn, the train makes a short drop.

The ride is a custom Gerstlauer Euro-Fighter roller coaster. The 1,000-metre (3,300 ft) ride continues with a sudden drop into pitch black darkness before entering a slow heartline roll. In just two seconds, the car is launched by linear motors down a 63-metre (207 ft) long tunnel to a speed of 100 kilometres per hour (62 mph). It then continues building and directly into a large corkscrew.

The train continues into a left-handed curve. The train approaches a right turn, heading towards an on-ride camera, which takes photos of the riders. After a 180-degree right turn, riders go through a fly-through the covered section element while traveling 64 km/h (40 mph).

The train then banks right making a 180 degree turn and entering a 30 m (100 ft) tall 90 degree drop, diving and reaching a maximum speed of 84 km/h (52 mph) before entering by a dive loop. The train then rises up to the left into an airtime hill and then drop again, entering a hill followed by several turns. After the turns it drops and turns into a cutback, followed by a corkscrew. The train then goes through an over-banked turn to the right before hitting the brake run.

The train then rises up into a banked turn to the left. Riders then go through two hills next to the lift hill followed by a turnaround that is positioned over the tunnel at the beginning of the ride.

The train then makes a sharp left, entering two additional verti-

cal loops and climbing into the mid-course brake run. This is followed by a zig-zag into a the ride element, where the riders experience a half-corkscrew followed by a half loop. The train then proceeds into another half loop and half corkscrew, sending it in the opposite direction. An on-ride camera photographs riders in the middle of this element.

The train turns left in a slight decline before leveling out. Once leveled, it enters the diving helix into the ravine. This is achieved by the left rail lowering much more steeply than the right rail as the train descends into a left-turning helix. The train begins to climb and level out after reaching the bottom of the ravine, producing significant positive G-forces. After exiting the top of the helix, the train enters a hill followed by a U-turn and a second set of trim brakes as the track levels out.

Riders ascend an 25-metre (83 ft) lift hill, followed by a small right hand turn before reaching the first drop. During the drop, an on-ride camera takes photographs of passengers.

The train continues making a left U-turn and ascends the 46-metre (150 ft) chain lift hill. Cresting the top, the train descends the first 46-metre (150 ft) drop reaching its maximum speed of 105 km/h (65 mph). At the bottom of the drop, the train passes the on-ride camera and curves slightly to the left, going over a camelback hill before descending 20 m (65 ft), and ascending into another one.

Then quickly travels down the third large drop. This is followed by a climb into a hill leading into a small left turn. Which leads into a long bunny hop over the creek. The ride then transitions into a 200° right turn leading into the brake run. The ride passes over Johnson Creek a total of four times.

It is the tallest roller coaster in the state and the longest ride in the park.

The train continues into a sharp left turn into a 19-metre-tall (63 ft) airtime hill with a 22-metre (72 ft) drop at a 32-degree angle. The train climbs upward, makes a right turn, and speeds into a covered brake shed.

The train ascends a hill and completes a large left turn where a brake run is located. Riders then drop into a 270° helix to the right which closely hugs the terrain and has strong banking. Afterwards, the train climbs another hill into the midcourse brake run. Riders then turn about 200° to the right, directly around (and within a few feet of) the spiral lift hill. Directly after this, the track banks left and dips down over a small pond, then rises once more, turning to the left to cross over the initial drop, then turning to the right to cross a pedestrian path below.

A dive loop is next, where riders are inverted before descending a half loop. The track then enters the second inversion, a zero gravity stall, where the train is suspended upside-down as it crests over a hill underneath the lift hill.

The train dives down a steep drop into an airtime hill and then proceeds counterclockwise through a 360-degree helix that ends at the trim brakes.

Trains climb the 33.3-metre-tall (109.3 ft) chain lift hill. Trains then descend the 28-metre-tall (91 ft) drop reaching a top speed of 80.0 kilometres per hour (49.7 mph). The train then navigates a rollover (2 half loops connected by two half heartline rolls). Once the train exits the rollover, it traverses a banked turn, sidewinder, helix, and double in-line twist. The train then performs a few turns before entering the brake run.

Due to the length of the lift hill, it utilizes two chains with separate motors. At the crest of the lift hill, the track plummets down a 94 metres (308 ft) drop to the ground, before passing over a 77 metres (253 ft) tall airtime hill. After this hill, the track rises

over a 64 metres (210 ft) tall hill before dropping to the right into a pair of helixes, the first one being clockwise and the second being counterclockwise. Following the second helix, the track maneuvers through the supports of the first helix and third hill and makes a left turn into the midcourse brakes, which start the return trip. The return trip consists of a series of airtime hills, running parallel to the outbound track, before hitting the trim brakes next to the base of the lift hill.

Soon after the second encounter, riders plunge into the alpine lake. No on-ride photo is available for this ride.

It is common for riders on each train to give high fives through curves. This section's 38-metre (125 ft) lift hill sends riders down a 36-metre (118 ft) drop at a 55-degree angle up to 97 km/h (60 mph).

The train turns up and to the right, then drops down and twists to the left. After this, the train traverses two banked turns and an airtime hill.

Built over a hillside, the ride is an out and back coaster. It uses skid brakes operated manually with a lever. The train consists of four cars. The first car has two rows, and the other cars have one row. Each row has two seats.

The five Premier LIM catapult coasters share a similar layout and have the same technical specifications. This section uses a LIM launching system instead of a traditional lift hill to propel riders into its first inversions.

The train continues into an upward right turn into the first inversion, a 31-metre-tall (103 ft) corkscrew. This leads into a downward left turn immediately followed by an upward left turn, and then the train enters a 29-metre (96 ft) zero-gravity roll. Riders then experience a sharp drop followed by an 24-metre (80 ft) horseshoe element. The train makes a left turn and enters a record-breaking 38-metre (124 ft) pretzel loop.

As the ride continues, it turns to the right, dropping slightly before a right turn to start up the 37-metre-tall (122 ft) lift hill. Upon cresting the lift, the roller coaster dives downward to the right before swooping up into the first major element, a two-inversion butterfly. Exiting the butterfly, the train enters a wide-radius 270-degree curve to the left, setting up the third inversion, a reverse sidewinder. The ride climbs a gentle slope before making a U-turn to the left and entering its element, a double corkscrew.

The train travels into a right hand helix which goes through the center of the second vertical loop. Pulling out of the helix, the train enters the mid-course brake run. Following the mid-course brake run, the train drops and enters a corkscrew which leads into a wide banked turn to the left. The train then travels over a small hill before making a right turn that leads into the trim brakes.

The front train is launched while the rear train sits at the bottom of the lift hill. Once the front train has reached the tunnel, the rear train is launched. The ride starts out with a LSM launch up a 32-metre (105 ft) hill at 20 degrees. At the top of the hill, the train drops 30 m (100 ft) at a 95-degree angle. The train reaches a speed of 92 km/h (57 mph) then goes through a series of banked turns, twisting right, left, and then right again.

The train enters a 34-metre (110 ft) vertical loop followed by a slight left, then into a zero-gravity roll where riders experience the feeling of weightlessness. Upon exiting the roll, the train immediately enters an immelmann loop. The train then goes through a trim brake, before entering a corkscrew, then a camelback hill which is a common way of achieving air-time on roller coasters.

After a medium-sized curved hill, it negotiates a 380° upwards helix and a bend to take the track parallel to the lift hill. Before entering the brake run, riders experience considerable air-time on

three bunny hops.

An airtime hill and banked left turn (rather than a helix on "Titan") follows into the mid-course brake run. The train is decelerated quickly and makes a hard left turn out of the brake section. Another 180 degree banked turn directs the train into a 585 degree, descending helix. The track turns upwards and banks left. After an ascending right turn, the train reaches the trim brakes.

The ride is a multi-lift hill roller coaster that features short dips and turns throughout a lightly wooded area and over a lagoon. The ride continues with the train entering a covered building resembling a mine shaft, where additional unused trains are stored, and then proceeds up the first lift hill. After reaching the top, the train dips slightly onto a straight section of track, followed by a short first drop that winds 270 degrees to the left during its gradual descent. As the train exits the turn, it ascends onto a straight section of track with a set of trim brakes that slow the train. This is followed by this section's climax, its largest drop that send riders over a lagoon close to water level.

The train veers up into a 30-metre (98 ft) tall, sharply outwards banked airtime hill, and into the signature non-inverting cobra roll.

During the journey back, there are more airtime and direction-changing hills before the trim brakes.

The blue train dips low close to ground level, picking up speed and making up for lost ground, while the red train rises up and passes over the blue train. Both trains turn right 20 degrees until they are running parallel with each other once more, though separated by the outbound tracks. The red train dips down after passing over the blue train, passing over two more airtime hills.

The train continues into an airtime hill and a second drop through a tunnel, which features a banked right turn that quickly

transitions into a banked left turn. The track then leads into a large, swooping left-hand turn toward the ride effigy. As the train passes through the figure, flames flare from its shoulders while the track rises into a right-hand banked turn. The train passes through the figure a second time before executing a right-hand turn followed by a rising left-hand turn. After a 180-degree right-hand lateral turn, the train drops once more and passes through the ride statue for a third time. The experience continues with a twisted airtime hill.

The train goes over a small inverted hill, drops to the ground, and makes a left turn into a corkscrew. The train makes a right turn, which leads to the brake run.

The ride begins in an indoor Antarctic-themed section with glaciers, snow, and an LED screen display of penguins before launching into the outdoors, utilizing a linear synchronous motor (LSM) launch in a wide right turn. A series of twists and turns including a banked turn lead to another LSM launch. This second launch leads to a overbanked turn. The ride takes a left turn over an airtime hill and then a wide left turn. Riders have compared the ride experience to that of sledding and bobsledding due to this section's use of airtime hills and turns.

The train moves through a corkscrew and a cutback. Following the cutback, the train goes through a small camelback hill. The train then enters a cobra roll located in front of the negative-g stall loop. After exiting the cobra roll, the train passes the on-ride camera, which takes photos of the riders.

On-ride speakers allow riders to hear the flaps of a phoenix's wings. The ride at Busch Gardens Tampa Bay features no full inversions, consistent with the family-friendly coaster theme.

Riders continue and make a right-hand turn, proceeding into the 30-metre (98 ft) tall chain lift hill. Upon reaching the top, yet

another right hand turn sends the train down its initial drop and into a below ground tunnel, reaching a top speed of 90.0 km/h (55.9 mph). Following a turnaround, the coaster proceeds into a signature 90-degree banked airtime hill. A speed bump underneath the lift hill sends the train through a banked turnaround and then through a series of straight hills spanning the coaster site, before being greeted by yet another lower-ground turnaround.

The ride is one of over 50 the ride coasters installed by Vekoma around the world, but it is the first roller coaster to feature the company's re-designed MK-1212 trains. The train is pulled backwards up the 35.5-metre (116.5 ft) lift hill.

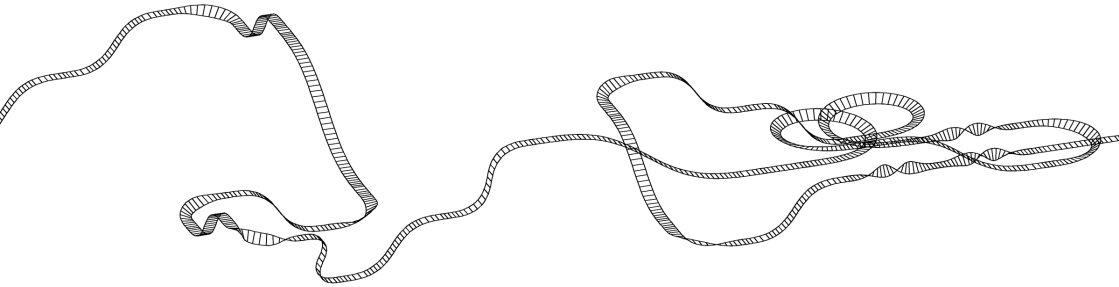
The ride descends as it continues right and climbs upward, entering a descending 270-degree helix as it passes Top Thrill 2's brake run. The train then enters a brake run and second lift hill, traveling in the opposite direction as the first.

The train drops down into the next two inversions, two consecutive dive loops before travelling over a trimmed airtime hill into this section's largest element, a the ride (this element consists of a sidewinder and reverse sidewinder). The train then travels through another corkscrew before reaching the second set of block brakes, after a brief pause the train ascends the second lift hill, this time at a 90° vertical angle. The train then enters another drop, 180-degrees to the left, banking into a downward corkscrew. Riders then navigate through a sea serpent roll, followed by a short drop into another trimmed airtime hill where the on-ride photo is taken.

The train crosses over the valley between The Voyage's second and third hills, traverses an overbanked turn, and then goes through an overbanked elevated spiral. The train then goes through a zero-g roll. After several curves next to The Voyage's triple-down drop and "spaghetti bowl", the train dips down into

a barn, where it then curves left back into the barn via an in-line twist. Once the in-line twist ends, the train enters the brake run.

The train continues to climb a series of left-curved hills, transitioning into a zero-g stall. Completing the stall, the train traverses a small outer-banked hill and continues into a series of upward-curved hills to the right. The ride continues with a turnaround into a hill and a left turn into the trim brakes. Upon completion, the train makes a left turn passing the car barn, and a last right turn.



Cloud of Fog

The drop at the top of the hill begins with a sharp left turn. After the initial drop of 28 m (91 ft), the train reaches its maximum speed of 80 km/h (50 mph). The train enters a vertical loop, then performs a right-banked 270-degree turn before dipping down and up into a zero-gravity roll.

The train continues into a large left-hand turnaround before ascending the 46 m (150 ft) lift hill. At the top, the train traverses a pre-drop to alleviate tension to the chain, followed by the drop, which descends into a below-ground trench. The train then enters a 39 m (128 ft) vertical loop, followed by a dive loop to the left, and a zero-g roll. Afterwards, the ride then enters a two-inversion sea serpent roll.

The car makes a sharp turn to the left and begins to ascend the lift hill. Nearing the top of the hill, a spotlight reveals another exterminator animatronic standing on the left side of the track, who

calls for backup into his radio. The car crests the top of the hill and makes a sharp turn to the left into a straightaway where another exterminator waits at the end of the hall and tries to blast the riders as the car makes a U-turn.

The SkyLoop experience begins with a vertical chain lift hill to a height of 46.2 metres (152 ft). At the top of the lift, the car is pulled slowly backwards over itself before being released into a full heartline roll.

The train turns 180 degrees. It then lines up with the launch motors and is accelerated from 80 km/h (0-50 mph) in 3 seconds. It goes up the 29 m (95 ft) top hat element.

A sharp left hand turn leads to the transfer track area and this section's main 21.0-metre (68.8 ft) tall chain lift hill. At the top, riders make a slight left turn before descending this section's main drop, proceeding to crest a flat camelback hill that crosses over the Romus et Rapidus rapids ride and a pair of twisted airtime hills with a tunnel in between. Another airtime hill leads into a shallow double up and into a turnaround, passing through a faux billboard in the process. The coaster maneuvers over and under itself in a series of twists and then ascends its second lift.

The on-ride photo is taken just before the drop into the corkscrew. The train then navigates a right-handed, 270-degree downward helix, which features 90-degree banking. This leads into an elevating second inversion, a zero-g roll, where riders experience a feeling of weightlessness in an air time moment.

The ride continues with a slight left-turn and then the train begins to climb up the 44-metre (145 ft) chain lift hill. Once the train reaches the top and passes through the pre-drop, the track makes a 180-degree right turn, leading into the first drop. Riders then drop 42 m (137 ft) at a 52-degree angle, reaching a top speed of 97 km/h (60 mph) and enter a 36-metre (119 ft) vertical loop.

Riders experience a 121° overbanked left turn that leads into a zero-g roll. After the zero-g roll, riders make another left turn up into the first block brake. After exiting the block brake, riders make another left hand turn and navigate through two back-to-back corkscrews then make a right turn up into the second block brake. After the second block brake, riders experience a tight right hand downward helix and a small hill before hitting the trim brakes.

The chain mechanism activates, and the coaster car climbs up the 22-metre (72 ft) tall vertical lift hill. Upon cresting the top, the ride is released into a 97° drop and into a vertical loop—a homage to the old Looping Thunder coaster—and a cutback inversion. The cars rocket up into a high-profile turnaround, followed by a heartline roll.

Exiting the helix, the return journey followed a 19-metre (62 ft) drop and two more 11-metre (35 ft) airtime hills.

The track straightens as the train makes a second descent into a short hill which produces an "air-time" experience. The third hill is followed by a steep climb into this section's 180-degree turnaround that features brakes to slow the train. On its way back, the train descends from the turnaround into a series of two shorter hills, the second of which banks to the right at its crest. The following hill is larger and sits directly beside this section's lift hill, with wooden supports surrounding riders as they pass through. The train continues through two very short hills and another 180-degree turnaround before the coaster meets the trim brakes.

The train begins to ascend the 33-metre (108 ft) lift hill. Following the main drop, the train makes a banked turn to the right before ascending into a twisting hill that passes underneath the lift hill.

From there, the train descends the first drop, reaching a maximum speed of 64 km/h (40 mph) and banking right. The train makes another banked turn over the mid-course brake run before circling back to it. From the mid-course brake run, the two separate cars on the train rotate backwards and King Venymyss can be heard saying, "Turn, turn, and face your fear". The train then descends through a minor helix, which carries it to a wheel lift hill.

At the base of the first drop, the train banks a few degrees to the right and climbs the first air-time hill and descends. Following this, the train climbs up to a hairpin 180-degree hammerhead turn to the left. The train descends from the height of the turn to travel up and over three more air-time hills.

After dropping off the brake run, the train passes the on-ride camera and crests this section's second camelback hill. This hill leads into a three-quarter turn to the right that wraps around the first drop. The ride continues with a descent to ground level, followed by a figure-eight turn. It then enters the trim brakes, which passes through this section's maintenance bay.

It begins its ascent up the 70 m (230 ft) chain lift hill. Once reaching the top of the hill, the coaster drops 66 m (215 ft) at 74° over the Rivertown midway reaching speeds up to 130 km/h (80 mph). It then curves to the right up a 59 m (193 ft) hill, past the ride' first drop, and then dives into the ravine, curving again to the left. The ride then ascends a 40 m (131 ft) camelback hill and drops, curving left into the hammerhead turn.

The ride then drops into a small helix before navigating through a corkscrew. A third helix follows before the train slows in the trim brakes.

The train ascends a 18-metre (60 ft) chain lift hill at a 45-degree angle, then levels out and turns towards the drop. The car reaches the drop and pauses, facing over the edge for 3 seconds.

The train immediately begins its ascent up the 47.0-metre (154.2 ft) tall lift hill. At the top, the train makes a 90° right-hand turn and engages with the holding chain at the drop. After halting for a few seconds, the train is released into its signature 50-metre (164 ft) drop into a tunnel, and immediately pulls up into an immelmann inversion. It crosses over the canal into a banked turn and a zero-g roll. A clockwise turnaround helix and a brief dip leads to the inversion, a lengthened heartline roll, whereas a right hand turn leads into the trim brakes.

They are launched forward into an immediate drop. The propulsion system consists of an electric motor and pulley system. The drop leads into a vertical loop as riders experience a maximum of 4 g.

The track makes a wide U-turn to the lift hill, which then pulls the train up 37 m (121 ft) above the ground. This section features a 3-metre (10 ft) pre-drop, which gives the train enough momentum to round a bend to the right, before falling down the first real drop of roughly 23 m (75 ft). At the bottom of the first drop, the train reaches its maximum speed of 80 km/h (50 mph), which is followed by a vertical loop. The train enters a batwing element (also known as a "boomerang" on Arrow Dynamics coasters), which is a heart-shaped series of two inversions, consisting of a reverse sidewinder followed by a sidewinder.

The ride makes a 90 degree left turn, is sent through a set of drive tires to boost the train towards the pre-lift section, and makes a left turn. After going through a series of small hills in the pre-lift, the train makes its way up the right side of the lift hill, catching up to the train on the green track. Immediately after the lift hill ends, riders are sent down an 80 degree, 128 foot drop. The trains then go through a small airtime hill and then up another tall hill, crossing under the ride.

After dropping out of the hammerhead, the train travels up a 39 m (129 ft) hill (which has a trim brake). After the fourth hill, the train turns down into a 287° counter-clockwise helix, containing the on-ride photo camera, and into the mid-course brake run. After the short brake run the train drops down into the ravine, raising into a pair of bunny hop hills before entering the clockwise helix of 323°.

As the name suggests, the ride roller coasters are themed to the DC Comics supervillain, the ride. At Six Flags Great Adventure, the supports are all black. At Six Flags Great America, Six Flags New England and Six Flags Over Texas, they have purple and green supports parallel to each other. At Six Flags Great America and Six Flags Over Texas, those coasters have 2 the ride signs facing back of each other, while the ones at Six Flags Great Adventure and Six Flags New England have only one the ride sign.

Speeding through soaring turns and near-misses with the surrounding buildings, the train navigates a helix situated directly overtop of the projection chamber. Traversing the part of the layout, riders enter a second corkscrew-esque inversion and complete a tight turn around the foundations of the hotel and into the brake run.

The train then climbs a 90 degree vertical lift hill with more special effects in complete darkness. The train crests the top of the lift hill and drops down a 97 degree, beyond vertical drop. The train then moves into a quick banked turn to the left, and onto a straight track (which is alongside the launch track), which has brakes to slow it down.

However, the amount of power required by the LIM infrastructure led to the park only launching one train at a time.

The train ascends up its first hill of 30 m (100 ft), followed by

a curved drop at 78 degrees – re-profiled as a steeper drop from the original Roar coaster. It accelerates to a maximum speed of 85 km/h (53 mph) before entering the "step-up under-flip inverted roll" element.

The train then enters a series of turns to the left and a second lift hill. After another short dip at the top onto a straight section of track, the track turns left, then right twice, followed by a set of trim brakes that once again slow the train.

Because of this, the entrance to the tunnel does not cross over the second climb.

A tunnel also wraps around the outside of the first drop. Upon exiting the Sector 3 Transport Tunnel, the train flies through the covered section. Riders go through the Sector 12 Transport Tunnel. After this, the train emerges on the other side of the ride and moves through a 360-degree right turn around the queue line.

It descends a 7.6 m (25 ft) first drop before immediately ascending up into a left turn. It drops again, into two airtime hills, and curves through a righthand turn over the covered section. It drops into a series of two bunny-hops and traverses a right curve directly underneath the first left curve.

This section features a vertical 41-metre (133 ft) lift hill and a first drop at a 101-degree angle. At the bottom of the hill the riders reach a top speed of 105 km/h (65 mph). The next element is a large overbanked turn, providing a significant amount of hang-time, after which the riders are redirected into a hill and then navigate the twisted drop that goes directly into a Finnish loop, before going into a dive loop that will take riders into an air time hill then an Immelmann loop. The car then goes into its only set of trim brakes, an overbanked turn and its inversion, a corkscrew.

Riders take an S-turn into the pre-launch sequence, where Owen warns the riders one last time. Four Velociraptors can be

seen in their cages on either side of the track; Blue and Delta are caged on the left, and Echo and Charlie are caged on the right. The boosters power up as the Velociraptors seemingly escape, and riders are launched from 0 to 80 km/h (0 to 50 mph) in 2 seconds. Immediately following the launch track, the train passes through an Immelmann loop and dives down and back up through a dive loop.

The train turns left toward the 61-metre (200 ft) chain lift hill. From the top, it drops 52 m (170 ft), reaching a maximum speed of 110 km/h (70 mph).

The train navigates a short left-hand turn before rolling into the first and only LSM launch, accelerating riders from 104.4 km/h (0 to 64.9 mph) in 1.9 seconds. The train ascends a 40.0 m (131.3 ft) top hat, before pulling up into a large airtime hill followed by this section's signature Finnish loop. A mid-course brake run follows, after which a series of left and right-handed bends lead into the second inversion, a dive loop. The coaster then dips into an underground trench before rising back up into the inversion, a corkscrew.

The train turns a big 180 degrees to the right and begins to climb the chain lift hill at a 40-degree angle. At the top of the 52-metre (170 ft) lift, it immediately enters the first element, an inversion that B&M calls the "Wing Over Drop". The train rotates 180 degrees before descending 50 m (164 ft) in a half loop. During this drop, the ride attains its maximum speed of 108 km/h (67 mph), and riders experience approximately 4 Gs. After exiting the first drop, the train enters an Immelmann loop situated underneath the lift hill. After the loop, the train turns right into its only camelback hill, where riders experience a feeling of weightlessness.

The ride continues with several large hills, which give a sensation of air-time, especially in the rear of the trains, and a number

of overbanked turns, before becoming more twisted as this section runs through woods and over a lake. Just before entering the brake run, the trains negotiate a series of bunny hops, again giving riders considerable amounts of air-time.

The train enters a camelback hill producing a moment of air-time. Upon exiting the hill, the train immediately enters an Immelmann loop. The train goes through a corkscrew inversion, followed by a left-handed 270-degree downward helix.

The train then enters a cobra roll element that transitions into a vertical loop. Upon exiting the loop, the train climbs the second lift hill making it most of the way up on its own momentum. Once towed to the top, the train is quickly released sending riders through the ride in reverse before being caught by the first lift hill.

It dives into a sharp turn invoking a near miss element. Each train rises back up and proceeds to the second launch, where this section's top speed of 100 km/h (62 mph) is reached. This leads to this section's Celestial Spin element, a 41-metre-tall (133 ft) inversion where the ride side traverses a zero-g roll while the Photon side traverses a downward barrel roll.

The train then navigates a small dip and turn before engaging the lift hill. After climbing to the top of the lift hill, the train begins turning to the left with the help of drive tires (as opposed to the ride at Alton Towers, which turns with the help of a chain lift) before being suspended over a near-vertical drop on a holding brake. After a few seconds, the train is released and plummets down the vertical drop into a subterranean tunnel. After leveling out and ascending back out from the tunnel, the train flips through an Immelmann and over an airtime hill, before navigating a 270° helix. Proceeding from the helix, the train passes through a heartline roll.

Each train makes a 180-degree turn inward.

The train then runs back parallel with the lift hill before entering two separate helices and entering the brake run.

The track leads directly to a 47-metre-tall (153 ft) chain lift hill. Once at the top, riders drop 45 m (147 ft) at an angle of 79°. A double up leads into a 90° overbanked turn, a dip and a 95° overbanked turn. Another dip and rise leads the train into a 115° overbanked turn.

Most of the track is painted red, but the highest parts of the track are painted dark green due to a rule which states that no roller coaster should be visible above the tree line.

The train continues and makes a small dip and into a left hand turn into a covered building resembling a mineshaft. The train makes another dip into a right hand turn and begins to ascend the 50-metre-tall (163 ft) chain lift hill. The train make a descending banked right turn, before descending to its longest drop of 39-metre-tall (128 ft) and through a tunnel.

It then flips through a Heartline Spin (zero-G roll), followed by a vertical loop. The train then travels upward around a tight spiral to the left, then through a wider turn to the right, drops slightly and quickly turns through the first corkscrew (referred to as a "flatspin" by the manufacturer). Following this is a tight right turn and another flatspin, then a tight left turnaround before the train enters the trim brakes.

And the train ascends a 44-metre-tall (144 ft) half-enclosed chain lift hill. As the train descends, it turns to the right and then proceeds over the first of many camelback hills. This hill navigates over the roof structure covering the park's Hollywood area.

After dropping and turning slightly to the left, the train enters a series of four camelback hills. The train then traverses an s-curve into the trim brakes.

The train ascends the 45-degree lift hill at 21.2 km/h (13.2 mph) to a maximum height of 93 m (305 ft). Once the train crests the top of the lift, the train descends down the 91-metre (300 ft), 85-degree drop, reaching speeds up to 145 km/h (90 mph).

The train continues into an Immelmann loop, a vertical loop, and a cobra roll – the latter of which inverts riders twice. The train heads up a banked curve into a corkscrew and a 230-degree turn into the trim brakes.

The ride continues by passing through the transfer table and up the 35-metre (116 ft) lift hill. A slight dip into a right-hand U-turn leads into the first drop. The ride then negotiates two back-to-back vertical loops, and then travels up a hill into a small mid-course brake run. At this point, the ride turns right into a second U-Turn and dives down the second drop.

The train then enters a zero-g-roll before going through a cloud of fog. The train then makes a slight left turn before entering an Immelmann loop followed by a right turn over a pond. The train then leads into the second zero-g-roll over the main entrance of the ride.

The train continues and makes a right turn and climbs the 37-metre-tall (121 ft) vertical lift hill. Cresting the hill, the train falls into the 97-degree drop. At the bottom of the drop is an on-ride camera. The train ascends into a "Norwegian loop", rolling to the left and dives through a half loop down to the ground.

It turns right and travels alongside of the lift hill and then turns right 180 degrees. At this point the train is very close to the ground. It turns 180 degrees to the left into the activating brakes.

Further down the launch track, an on-ride photograph is taken. Immediately following the launch, the train climbs vertically through a 150 foot top hat. The train then drops 180 feet, reaching top speeds of 75 mph, before climbing into a 135 foot tall cobra

loop. After another drop, the train traverses the flying snake dive (a heartline roll followed by a dive drop). Following these inversions, the train descends and crosses over Spring Creek and a section of the ride, while passing under a section of the monorail.

The train reaches a slight decline that allows the car to roll out and around a 180-degree turnaround and ascends the 30-degree and 26-metre (85 ft) chain lift hill, operating at a speed of 6.4 km/h (4 mph). The train then descends 20 m (65 ft) at a 45-degree angle at a top speed of 77 km/h (48 mph). The train enters a bunny hop, drops lower than the main drop, and enters a vertical loop. The train goes up to a short straightaway before descending a banked 180-degree right turn into the two consecutive corkscrews over the midway of the park, traveling at 61 km/h (38 mph). In its stretch, the train enters a slight ascending right turn followed by a shallow left turn.

It then travels up to a height of 46 m (150 ft) where it completes a half-loop and a heartline roll before exiting in a second non-inverting half-loop. It then enters a full non-inverting loop.

The ride continues with the train going into a tunnel, then using an linear synchronous motor for its initial large overbanked turn to the right. It does a vertical loop and a small overbanked turn the left. It goes up slightly and stays for a while, then goes down, performs a twisted horseshoe roll followed by a small camelback then tilts to the right.

Guests go through a series of bunny hills and a left turn. Passing by some blocks, the train makes a right turn. Riders hit the trim brakes and spot a Wheezy animatronic on the right singing "You've Got a Friend in Me".

The train dips to the right before ascending the 37-metre (120 ft) chain lift hill. Cresting the lift hill, the train enters a left turn pre-

drop before descending the 52-degree, 31-metre (102 ft) drop. At the bottom of the drop, the train banks slightly left and completes a camelback hill before ascending into a left-banked hill. Continuing the turn, the train dips, before flattening into a drop where the train completes a hill, then rises to the right into a continuous right banked turn. Continuing the turn, the train dips, slightly ascends, then exits into a drop where it completes another hill.

The train goes through a small second hill that speeds past the covered section and the lakeside. The train then makes another turn and up a smaller hill where riders experience -2 g forces on an ejector airtime hill, crossing over the former the ride track. After coming down the drop, the train snakes through twists and turns. After coming out of the twister section, the train slows down as it moves through small "S" curve camelback hills and into the brake run.

The ride ascends the 12-metre (40 ft) chain lift hill and then travels through a series of camelback hills and slightly banked turns.

The ride drops riders into a tunnel through Caddo Lake, emerging just outside the trim brakes and queue house. Runaway is prone to flooding from Johnson Creek after heavy rainfall amounts.

Riders continue begins to climb a 27-metre (90 ft) tall lift hill. After cresting the lift, the train begins a sweeping helix to the left. Following this turn, riders plunge into a 38-metre (124 ft) drop into The Hollow.

The train then maneuvers 3 switchbacks before entering the high-speed turn. The train then climbs another airtime hill with brakes, followed by another airtime hill before entering a twist and then climbing a small, twisty bunny hop into the magnetic brakes.

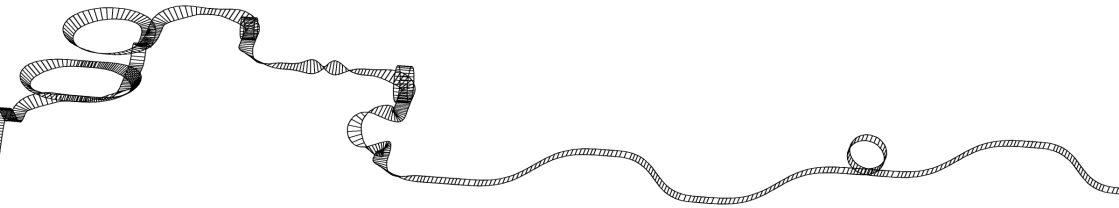
This time isn't usually followed, as the train often passes over

the sensor anywhere between 7 to 10 seconds after launch. This is because the purple train is slightly heavier than the red train, causing both trains to run at noticeably different speeds when running two trains. After launching, the train climbs a vertical 62 m (205 ft) top hat element, then soars through two banked turns of 34 m (110 ft) and 29 m (95 ft), respectively. It then reaches the brake run. The magnetic braking system consists of mounted magnetic clippers on the trains and copper alloy fins mounted onto the track. The alloy fins on the launch section retract during the launch procedure so as to not interfere with the train.

The train continues and enters the lift hill. At the top of the lift hill, the ride is released onto the main downhill track, which the ride is not physically held onto. It travels through various bobsled track style curves, picking up speed, before reaching the lowest curve and curving back up again.

The train enters a short drop and lefthand turn to the chain lift hill. After climbing the 26 m (85 ft) lift hill, the train immediately descends the 37 m (120 ft) first drop while turning sharply to the right through a tunnel, and travels over a 50 m (165 ft) long airtime hill as it crosses PA 832 in an enclosed bridge. The track curves to the left, then right, while ascending the turnaround to the top of the second major drop. The train descends this 32 m (105 ft) drop into the second crossing of PA 832 over another airtime hill in the enclosed bridge. This is immediately followed by a pair of tunnels enclosing two airtime hills, and an ascending turn to the left.

It then navigates a non-inverting Immelmann, followed by an outer-banked barrel roll down drop and a stalled dive loop. A short brake trim section moderates the train's speed before it ascends an outer-banked airtime hill. The train continues through a turn near the lake and rises over a concluding airtime hill.



Tiki Trail Treehouse Attraction

After dropping back down to the ground, the train goes over a three camelback hills; each at a height of 21.0 m (68.9 ft), 18.0 m (59.1 ft) and 16.0 m (52.5 ft). The train makes an upward 14.0 m (45.9 ft) right banked turn, immediately followed by a downward left banked turn. After another 13.0 m (42.7 ft) left banked turn, the train rises back up and goes over a small bump before entering the trim brakes.

Riders are pulled up by a 9,000 lb (40 kN) chain lift. At the top, the train dips slightly into the pre-drop before turning 90 degrees to the left as it drops 36 m (119 ft) down the first hill. From the bottom the train immediately enters a 30 m (100 ft) vertical loop.

The train then makes a sharp left turn before making a sharp right turn which leads into the brake run. The train then makes a left turn into another set of brakes.

Riders continue and immediately ascend the 14-metre (47 ft)

tall lift hill, before turning slightly to the right and plunging down the 13-metre (44 ft) drop at a 52° angle.

The train then goes around a right U-turn and into the second inversion – a downwards inline twist. This leads into a cutback-style turnaround.

Riders on the ride continues by climbing along a 120-foot vertical rise. Then zoom over hills while pivoting forward multiple times, before plummeting into a steep raven drop.

The train goes through an Immelmann loop. After going over an airtime hill, the train makes a banked right turn into the trim brakes. The train makes a left turn into the brake run.

The track leads into the second launch, accelerating the train to its top speed of 66.9 km/h (41.6 mph) and re-entering the top hat, this time using a track section above the first element. The train traverses two airtime hills – a camelback and off-axis twist – before navigating a couple of low-to-the-ground twists and a wave turn to the right. A heartline roll takes place over a pond before the train hits the brake run.

It ascends a chain lift hill before going down a small dip and into an s-bend followed by a U-turn. After a slight ascent, the car enters a 120.5° drop – the steepest in the southern hemisphere. The track then returns to approximately two-thirds of its original height before entering the first set of block brakes.

Modern coasters can be made out of wood or steel, and may incorporate different launch mechanisms or train configurations to enhance the ride experience.

The train ascends the 33-metre (109 ft)-tall lift hill before dropping into a tight right-hand turn. After leveling out it goes into a roll over element followed by a left hand turn. After the turn, it navigates into a sidewinder element and a 270 degree right turn, which immediately leads to a double inline twist. After a small

bunny hop, the train enters the trim brakes.

This section features three lift hills. The ride continues with the trip up the first and highest lift. From there it travels around the track to the second lift. The second lift is housed in a building designed to look like a rock crusher. The lift leads into the "Ace Hotel and Saloon".

Riders sit face-to-face, staring at fellow passengers through the course of the ride.

This section features a double out-and-back layout.

Riders descend low to the ground once more into a small airtime hill heading back toward the first drop, entering a sea serpent element with two more inversions.

The track turns right while dropping down a 52 m (170 ft) spiral drop, going past The Land of The Dragons, with the train hitting 108 km/h (67 mph). Following the drop, the train passes through a 37 m (120 ft) Immelmann loop, followed by a 32 m (106 ft) vertical loop. The track then races through a wooden tunnel before passing through a cobra roll over the Rhine River, adjacent to the ride. Out of the cobra roll, the track crosses over the entrance to the cobra roll, passes by the ride, then rises into the midcourse brake run.

On the return trip, the second launch area decelerated the train from 113–56 km/h (70–35 mph) before traveling back through the vertical loop and the underground tunnel.

In the double helix, the train turns to the right, making two complete circles. At the conclusion of the double helix the train goes downhill, then makes a lefthand turn through the double-down drop and enters a 90° right turn, followed by a drop and a banked 90° turn to the left. The train then makes a 180° turn to the right. Once the train exits this element, it immediately enters the trim brakes.

The train begins to ascend the 77-metre (253 ft) lift hill. After

reaching the top of the lift hill, the train drops 82 metres (269 ft) at an angle of 85 degrees into a tunnel, reaching a speed of 142 kilometres per hour (88 mph). The train then exits the tunnel and travels over a large hill. Following this hill, the train enters an inversion that consists of a large turnaround similar to a dive drop.

The train speeds up into a helix, following a jump over the covered section building and yet a helix.

The drop off of the second lift is not very steep, with the train navigating a series of gradual curves. After passing under Top Thrill 2's back spike, the train then pulls up into a sharp right turn. The next drop curves left slightly, followed by a right-hand turn and drop into a clockwise helix over the lagoon.

The train emerges from the show building and makes a slight left towards the 30-metre (98 ft) chain lift hill. Once at the top of the hill, riders drop down 30.5 metres (100 ft) at an angle of 100°.

The car turns slightly to the left, and ascends the 30.0-metre (98.4 ft) vertical lift hill. The car then descends a 97-degree beyond-vertical drop, reaching top speeds of 85.0 km/h (52.8 mph). The car traverses an Immelmann loop before traveling through a curved drop to the right. It then enters a cutback, followed by an inclined loop, before pulling into the brake run.

Sections of the outdoor track have been fitted with speed-reducing devices due to early problems with the ride entering its second lift hill too quickly.

After the ascent to the top of the 63.1 m (207 ft) lift hill, the train plunges down the 59.7 m (196 ft) drop, reaching the maximum speed of 119.1 km/h (74 mph). The train then enters a 31.4 m (103 ft) parabolic hill. After the hill, the train climbs a larger 39.6 m (130 ft) hill and turns left followed by a large figure-eight turnaround. The train then makes a s-turn into a mid-course brake run. followed by several airtime hills, two of which are inside a tunnel

where riders are photographed via on-ride camera.

Following, the train enters a 480-degree-helix, then exits into a series of S-bend curves.

The track makes a dive to the right into a trench where it hits a 18-metre-tall (60 ft) vertical loop. The track then exits the trench, making a three-quarter clockwise turn. After passing under the zero-g roll, the track makes a left-hand turn over the first vertical loop's exit and dives into another trench to hit a corkscrew, before making another right turn onto the trim brakes.

Riders ascend the first lift, then are held until the lift disengages. After disengaging the lift mechanism, riders drop down the lift hill then fly through the covered section and enter a cobra roll. Immediately after, riders enter a vertical loop before engaging the second lift halfway up the incline. Riders climb up to the top and are released with no warning, traversing the course in the opposite direction.

At the peak of the lift, the ride descends into a pre-drop gradual left hand turn over the Sky Ride before hitting the main drop. Riders immediately navigate a turnaround over a pond and speed through a gradual S-bend, before hitting this section's helix. The train dives into a tunnel under the Wonderland Special Narrow-gauge railway, hits a right-hand turnaround.

The train then drops 49-metre (160 ft), reaching an 83-degree angle and its maximum speed of 114 km/h (71 mph); before banking up into its first element, an inverted zero-g stall. From here the train twists through two airtime hills and two high banked turns before encountering its next inversion, a heartline roll. Following this, the track descends to ground level as it passes through several more strong airtime hills, prior to a second heartline roll. The ride continues with a ski-slalom style right to left section before hitting the brakes.

The train immediately ascends the 64-metre (210 ft) lift hill. After reaching the top, it enters a 64-metre (210 ft) drop, in which the train reaches at a top speed of 122 km/h (76 mph). The train ascends and descends a 52-metre (169 ft) airtime hill over Spring Creek. After the hill, the track reverses direction via a 123-degree Hammerhead turn. After climbing over another camelback hill, a series of airtime hills follows as the track travels back above Spring Creek before making an upward helix to the right and a leftward bank.

It enters a second 40-metre (130 ft) near 90 degree drop into an Immelmann loop. Following a small airtime hill, the train goes through a splashdown where the car's rear scoops spray two 15-metre (50 ft) plumes of water into the air. Then, the train makes a banked turn to the left leading into the trim brakes.

The train rises up into a mid-course brake run which is located next to the lift hill. Similar to the first drop, the train drops to the left and enters a 270-degree helix, before passing through two Interlocking corkscrews.

After a 27 m (90 ft) S-turn over the launch track, the train turns back around going through a "wing slide" over the pathway. As it reaches the back of the layout, the train goes through a corkscrew and airtime hill.

The train then enters a 315-degree left hand turn leading into the trim brakes.

The train then falls back through the covered section, where it is accelerated backwards into a twist.

The trains continue to the right to engage with the chain lift hill, elevating them to the maximum height of 19m (62ft). From there the trains travel through a slow, wide right-hand turn into this section's drop, ascending again into a brake. Afterwards, the trains navigate three tight turns in alternating directions, gather

speed and emerge into two helices.

The ride starts as riders climb the curved lift hill, followed by a quick pre-drop and immediately go into a sharp half loop, then through a half loop and then a half loop before being slowed back down by near-vertical magnetic brakes; the coaster continues with a quarter loop.

The train makes a left turn to climb the 43-metre (142 ft) tall chain lift hill. Once riders reach the top, they go through a small pre-drop before dropping 40 m (132 ft) to the left at a 55-degree angle. The train then reaches a top speed of 98 km/h (61 mph) as it enters a 35-metre (114 ft) vertical loop, followed by a 29-metre (96 ft) diving loop.

The train continues into an upward right-handed helix and a downward left turn into a series of corkscrews separate by a sharp left turn. The train then veers right into a brake run.

It climbs the 47-metre-tall (155 ft) chain lift hill. It then drops 46 m (150 ft) before entering the first 37-metre-tall (120 ft) vertical loop. This is followed by a 29-metre-tall (95 ft) cobra roll and a 23-metre-tall (75 ft) zero-g roll. A second, smaller vertical loop leads the train into a large helix before the mid-course brake run.

On-ride photographs are taken as the car rises into a brake run on the left. A sudden drop then leads into a dive loop, after which the ride continues with a banked turn into the trim brakes.

Riders immediately ascend a 17.2-metre (56.3 ft) chain lift hill. Once at the top, riders turn over and drop under the existing Dawson Duel racing coasters, navigating a trip of elevated airtime hills in the direction of the Bellewaerde pond. Once at the water, the train makes an abrupt right-hand turn and speed through a bunny hill and left-hand turnaround, followed by a series of twists and turns, including another jump over a park pathway.

The train then drops towards the ground, climbs back up, and

spins through a zero-g roll, an inverting element where riders experience the feeling of weightlessness. Afterwards, the train dives back towards the ground and enters the double-inverting batwing element that crosses through a tunnel underneath the park's rear entrance. After the train exits the batwing, it climbs through a camelback hill over the covered section before entering the inversion, a corkscrew to the right. The train then begins a 200-degree turn while climbing to the left, after which it reaches this section's brake run.

It then launches backward along the same launch track before travelling backward up a twisted section of track. The train then launches forward and travels over the top hat. Following the top hat, the train makes a slight left turn before entering an Immelmann loop immediately followed by a heartline roll. The train then enters a banked turn to the right before entering a tunnel through a building. The train then travels over a series of small hills before turning right turn into the inversion, a corkscrew.

The train continues, makes a sharp left turn and ascends a 29-metre (95 ft) lift hill. A small pre-drop and turn follows, which leads into the track's 20-metre (65 ft) drop. This is followed by two 26-metre (85 ft) vertical loops and a small hill with a banked turn. As the train makes its way over the walkway, it flips riders twice in a double corkscrew.

The train then banks left into a 180-degree turn and enters the signature 87-degree drop, diving and reaching a maximum speed of 84 km/h (52 mph).

The ride is the first of more than fifty identical the ride roller coasters made by Vekoma.

The riders enter a decorated light tunnel, followed by a downward banked curve to the left. The train passes through the covered section, then completes the layout a second time. With

a height restriction that accommodates children as small as 102 centimetres (40 in) tall, this section's design is geared towards families. While most of the ride is experienced in the dark, riders also encounter visual effects and props, most notably the ride (nicknamed "Herman" by Wonderland employees). Other visual effects include lighted tunnels and strobe lights.

The train then heads from the unload section to the load section.

To give the ride operators extra time, the green lift can run slower than the blue lift, giving the blue side time to catch up.

Riders' cars will ascend the lift hill to this section's highest point before immediately entering a series of switchback turns. Following a sweeping downwards right-hand turn is a straight track, followed by two more right-hand turns which directly precede this section's first drop.

The layout of the track consists of a double-loop with two cable lift-hill sections. The drops off both of the lift hills are double-drops.

This circular shape differed from the teardrop-shaped loops which are used in modern roller coaster inversions—producing lower g-forces and less rider discomfort.

The train makes a 180 degree right turn before reaching the chain lift hill. After reaching the top, the train dives down towards the right-hand side. The train dashes horizontally before traversing through town buildings inside a German village. Once past the village, the train arrives at the second lift hill. After descending again and narrowly avoiding a wolf trap, the train turns around near an Oktoberfest celebration plaza featuring with tables and beer mugs.

The train then departs and immediately begins to climb the 35.9-metre (117.8 ft) lift hill. Once at the top, the train goes through a pre-drop before making a sharp downward left turn back to

the ground. When the train reaches the bottom, it then enters a 29-metre (95 ft) vertical loop. Upon exiting the loop, the train then rises back up and goes through a 26-metre (85 ft) dive loop. The train then makes a 180 degree left turn through the loop before going through a set of trim brakes.

On one of the lift hills, riders are able to interact with the surrounding screens by effectively collecting "stardust clouds" to help power the sun. Aside from interactive technologies, the ride also features a collection of standard projection screens, lasers, fiber optics. Throughout the 275-second ride, the individual four-seater cars perform both magnetic-controlled spins and uncontrolled spins; the latter relies on the riders' weight distribution coupled with the force of gravity.

Instead, it enters the first drop, with the lift hill being located in the middle of the ride after the second drop. After the lift hill, riders descend a 27 m (90 ft) long drop. After this drop is a tight bend which results in the rider on the right being pushed into the rider on the left due to the lack of a seat divider. After this, riders experience several turns and hills.

The vertical loop is followed by an elevated right turn passing over the entrance to the nearby Scenic Skyway, after which the car dives down into an upward counterclockwise helix which wraps around the lift hill. Upon completing the helix, the car passes through a heartline roll that threads through the vertical loop. This is followed by a clockwise 540 degree helix, after which the car hits the brake run.

The train starts ascending the 34 m (112 ft) lift hill. The train turns 180 degrees during the first part of the lift hill. After cresting the top, it enters the drop. At the bottom of the drop, it immediately enters its first inversion.

A hairpin turn leads into a drop over the pathways, followed

by a twisted airtime hill and a gradual right-hand turnaround composing of a brief outwards banked twist before hitting the second block brake. A twisting descent takes riders close to the ground and under the Tiki Trail treehouse attraction. The train then rises into another twisted airtime hill and dives under the Tiki Trail once more before popping up into the third block brake. Riders make a right-hand turn before entering three camelback hills and a 270 degree helix over the nearby pond.

Once the momentum of the train runs out on the 70° spike, the train begins to traverse the track backwards, returning through the loop.

The train makes a turn to the left, passing through this section's structure. It then begins to climb the 55-metre (181 ft) tall cable lift hill. Once the entire train is on the lift the cable increases its speed to around 13 mph. Once at the top of the lift the speed of the cable gently slows down, but it is barely noticeable on the ride. After cresting the top of the lift, the train briefly travels forward and makes a 180-degree turn to the left.

The drop is the longest on the ride, measuring 43 m (141 ft). The signature double helix features two long tunnels and turns riders counterclockwise twice at high speeds while ascending. After exiting, the train dips through another small hill into the trim brakes.

The train continues and immediately begins ascending the chain lift hill, reaching a maximum height of 30 m (98 ft). After disengaging from the lift, the train navigates a 180-degree left turn into its first drop, passing through a vertical loop. It then traverses a long airtime hill that passes beneath the ride exit and gift shop, creating a "head-chopper effect", before sharply ascending into a cobra roll positioned in a partially flooded pit, allowing guests to view the element and enhancing the attraction's visual spectacle.

The train makes a left turn leading directly to the beginning of the 45-metre (149 ft) lift hill. Once the train reaches the top of the lift hill, it makes a sharp, downward 39-metre (129 ft) drop at a sharp right turn, in which the train reaches a top speed of 97 km (60 mi) per hour, followed by an 27-metre (89 ft) Vertical loop. After exiting the loop and making a slight right turn, the train goes up, leading into a zero-gravity roll. The train drops back to the ground before going through an Immelmann loop. The train enters a banked turn.

From there, the train descends a banked drop, causing the individual cars to spin freely. The train goes through a series of overbanked turns, and changes in acceleration and velocity while spinning. The trains travels under the Serengeti Railway attraction.

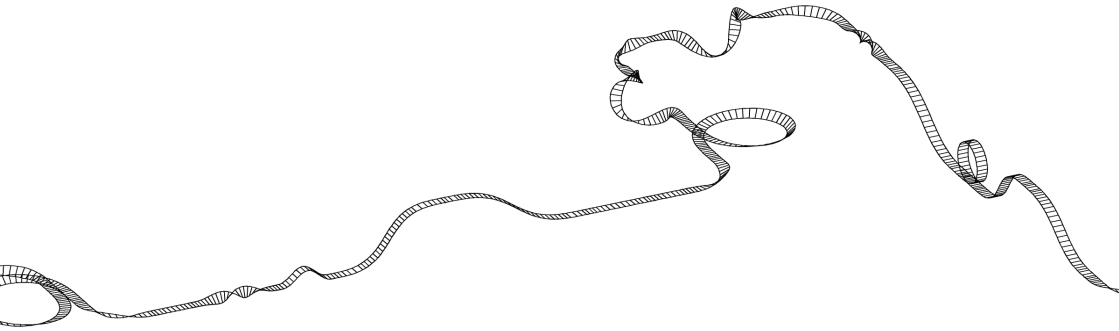
The physical track layout is fairly minimal; various twists and turns are navigated close to the room's floor, and the attraction's second launch is located halfway through the layout to keep the train moving.

The train climbs a 33 metre (109 ft) tall lift hill before arching into a steep curved incline. The train rises up into a butterfly loop (two half loops connected by a corkscrew) before pulling upwards into a banked apex. From here, the train drops sharply into a sidewinder (a loop that transitions into a corkscrew). A tight helix follows, providing a near-miss with the ride, before the train straightens out and pulls into a double inline twist. The train subsequently enters a 90 degree turn before pulling downwards and ascending into the brake run.

The train then rises through two banked turns, first to the right, then the left. After a brief moment of airtime, the train enters the trim brakes. Following the brake run, the train curves to the left.

This type of roller coaster features cars that hang beneath the

track by a chassis allowing them to swing side to side. Some parts of the ride are built on artificial lagoons, and the track takes riders into wooded areas and over water. The ride is one of two roller coasters at Cedar Point to utilize two separate lift hills, the other being the ride.



Shanghai Disneyland

The train then drops from the top of the helixes into a "street" lined with highway signs and three police cars, with sirens and lights flashing. Riders then swerve between the police cars. Riders then go through an overbanked turn at 88 degrees, followed by a dip, and then travel down a set of "subway stairs".

After being released, the train passes through the covered section, enters a Cobra roll element (referred to as a boomerang by the designers), then travels through a vertical loop. Upon exiting the loop, the train runs up a second spike, which is angled so that the two spikes meet at the top in a "V"-like formation. Once the train runs out of momentum, it is lifted to the top of the second spike via chain lift, and is held for several seconds before being released; travelling through the layout in reverse.

The train then ascends the chain lift hill, reaching a height of 72 m (235 ft). Ascent slows toward the top of the chain lift hill, a

safety feature that reduces stress on the chain; it is also an attempt to reduce positive g-forces and increases rider anticipation.

It then transitions into a left-hand turn and then climbs into a 28 m (92 ft) mid-course brake run, and then drops again. It then goes into an airtime hill, turns right, then climbs a larger airtime hill. It then turns right, going into two back-to-back airtime hills.

Immediately after, the trains follow upwards into a left-banked turn and subsequently head downward into a right-banked curve where it transitions into a zero-gravity roll. The trains after traverse into another left-banked turn upwards before dipping down into the second LSM launch going from 48 km/h (30 mph) to 72 km/h (45 mph) in 3.5 seconds.

The train begins its backwards climb up the first of this section's two 116-foot lift hills, both of which are placed diagonally towards each other. The train continues to slowly rise for thirty seconds before dropping at 47 miles per hour through the covered section and through this section's first inversion, a cobra roll, exerting as many as 5.2 Gs on riders throughout the two elements. The train then goes through a loop before entering the second lift section.

The train continues and makes a slight right turn before proceeding up the 28-degree incline lift hill. At the top of the lift hill, the train makes a left turn with a small dip and descends a long 23-metre (75 ft) drop, where the train reaches its maximum speed of 72 km/h (45 mph), then enters the 17-metre (57 ft) vertical loop. Exiting the loop, the train makes a long, ascending sweeping left turn that travels through the middle of the loop. The train traverses a gentle right curve which includes a short tunnel.

While the train is being loaded with passengers, the catch car for the cable lift descends the lift hill and latches onto the middle car underneath the train. Once the train is cleared, the cable lift immediately pulls the train up the 45-degree lift hill at 24 km/h

(15 mph) to a height of 94 m (310 ft).

The train plunges down a 34-metre (113 ft) drop into a ravine at 100 km/h (62 mph), then enters the mid-course brake run. Exiting the brake run, the train drops at a right banked 44-degree curve before ascending another hill.

The track hits a batwing, entering a trench at the valley between the two inversions. Leaving the batwing, the track rises up into the mid-course brake run.

The train is propelled up to a height of 13.0 m (42.5 ft) via a drive tire lift. A switch track hastily diverts the train into an indoor show scene, depicting the temple of the ride.

The ride is one of three wooden roller coasters at Worlds of Fun and Worlds of Fun's first wooden roller coaster. This section's highest point is 30 m (100 ft) and its largest drop is 29 m (95 ft), at which point it reaches speeds of 72 km/h (45 mph) and incurs g-forces of 2.8.

The track passes through a tunnel and zooms into a 27-metre-tall (90 ft) tall Immelmann loop, providing a footchopper element with the pylon. After completing the Immelmann, the train goes through a zero-g roll, where riders experience a feeling of weightlessness.

It makes a banked right turn back into the mountain, then a tarpaulin drops, blocking the entrance to prevent light from entering. The train slows and the cars rotate 90 degrees to face the screens. Stansein tells riders to prepare to battle the enemies. The first scene takes place in forest environment where riders fight Sneaky Spiders and Beasty Bats.

The ride is a combination of a dark ride and a launched roller coaster. The ride continues with a dark ride portion lasting roughly 45 seconds. The train is then accelerated to 100 km/h (62 mph) in 2.5 seconds using a linear synchronous motor launch,

along an 80-metre (262 ft 6 in)-long launch track. The launch propels the train over a turn that peaks at 38 metres (124 ft 8 in) high.

At the opposite end of the layout, it ascends an adjacent booster-wheel lift hill and re-enters the garage. Following a brief narration warning riders they are falling behind in the race, the train reverses direction, retracing the layout while traveling backward. It passes through the covered section as a checkered flag appears on a video screen, climbs the first lift hill.

After pausing for 3 seconds, riders plunge 75 metres (245 ft) through the middle of this section's helix and into an underwater tunnel, reaching a top speed of 130 km/h (81 mph). After exiting the tunnel, the train rises into an Immelmann loop. The Immelmann loop itself is 57 metres (187 ft) above the floor of the tunnel, 45 metres (148 ft) above the exit of the tunnel, and 50 metres (165 ft) above the water the ride sits on. Following the Immelmann loop, the train then enters a Zero-G Winder, a combination of a Zero-G Roll and a Sidewinder.

The ride continues by taking a small dip, turning 90 degrees to the right into the 40 metres (131 ft) tall lift hill. Upon reaching the top of the lift hill, riders plunge down a beyond-vertical drop (unknown angle), twisting 90 degrees to the right in the process.

This section includes five inversions after the lift hill that stands 40 metres (130 ft) in height. After the initial drop, a dive loop follows going 30 metres (98 ft) upside-down (the ride is the first inverted roller coaster to feature a dive loop).

Instead, small wheel motors drove it up the incline of a tight helix.

The train then descends down the first drop and immediately enters the first loop. Upon exiting the loop, the train climbs a hill, levels out and reaches the first set of trim brakes.

Riders rise into this section's first inversion, a dive loop (dubbed a Raven Truss Dive) that also serves as a turnaround. A speed hill directly proceeds, leading up into the second inversion, a zero-g stall underneath the lift hill. An outwards-banked airtime hill follows, redirecting the train 90° to the left.

The train heads through an overbanked turn and rises up into an off-axis airtime hill reaching a turnaround that rises through the rock work. This is followed by a downward S-bend, where the train passes Blue and Charlie, two of the four Velociraptors seen on the ride. The train goes through another overbanked turn and hits a sharp S-bend, passing Delta and Echo. Following a slow outward-banked hill, the ride turns right and enters a tunnel and its second launch, propelling riders to 110 km/h (70 mph) in 2.4 seconds.

The train then ascends a hill that curves left into the second, 144-foot drop, next to the New York-New York's entrance. The train ascends another hill and curves left again onto the roof of the casino, dropping slightly to enter a vertical loop. Immediately afterward, the train twists to the right and enters the dive loop. Exiting the dive loop, the train ascends into a hill with a mid-course brake run, then turns 180 degrees to the left, ascending an airtime hill. The train turns 180 degrees to the right, dips into two airtime hills, and traverses a 540-degree helix to the right, traveling through one last airtime hill before hitting the trim brakes. The ride then travels indoors above a replica of Little Italy, Manhattan, on a structure that resembles an old elevated railway.

Typically a single train is used while the other comes out during busier times.

The ride is a wild mouse roller coaster manufactured by Mack Rides.

The train turns right into a 270-degree turn before ascending

the 46-metre (150 ft) airtime hill. The train then descends into a high-speed bunny hop before entering another high-speed turn.

The train then proceeds to dive under the wing of a wrecked plane and into a zero-g roll 30.6 metres (100 ft) above the ground before passing narrowly through a themed billboard, immediately following which the on-ride photo is taken. It then enters an inclined loop 23 metres (75 ft) in height. The ride continues into a turnaround featuring a wrecked helicopter with slow-spinning rotor blades in the centre.

The train blows its whistle and continues to the right and climbs a small lift leading to a short drop, then circles around to the 36-metre (118 ft) lift hill, carrying the riders into the mountain. On the way up, it passes through a ransacked temple with murals of the yeti, warning the riders that the mountain is its territory. A noise can also be heard during this lift hill, foreshadowing the yeti. At the top of the mountain, the train curves around the main peak and goes through a cave.

After twisting through the ride, riders then go through a 31-metre (102 ft) tall vertical loop which crosses over the covered section and hit the second vertical tower of the ride. A catch car there pulls the train up the second vertical tower, this time with riders facing the sky. After the train reaches the top of the tower, it is released to cycle backward through the layout.

There are two on-ride photos; one at the start of the drop and the other at the end of the high-banked turn. This section's track is 372.5 metres (1,222 ft) long, and it has a ride time of one minute and fifteen seconds.

The train twists clockwise for one full turn; at the same time, the seats rotate forward one full turn. This is followed by an overbanked turn and a half-twist "fly-to-lie" maneuver, in which the train twists counterclockwise one half-turn as riders flip

backward one half-turn to return to the original position of laying on their backs. The train then enters an outside raven turn immediately followed by another half-twist and half-backward seat rotation.

The train descends a right-hand swooping drop into a tunnel, where riders are exposed to simulated fire effects. The train then makes a brief left turn before ascending the 29-metre-tall (95 ft) chain lift hill. At the top, it descends in a long left-hand swing into its first inversion, a vertical loop, the train then rises into the second inversion, a zero-G roll. The ride continues with a right-hand turn into the first of two interlocking corkscrews, followed by a right-handed overbanked turn, and then passes through the second interlocking corkscrew. After a large right-hand helix and a sharp left-hand upward turn, the train enters the trim brakes.

The second launch accelerates riders to their top speed of 97 km/h (60 mph) before climbing a 31-metre (102 ft) figure-eight element (also known as a Windcatcher Tower). Riders weave through a series of helixes before dropping 40 m (130 ft) into a trench, then proceed over a direction-changing airtime hill crossing over the Skyride, before completing this section's inversion, a heartline roll. Upon exiting the barrel-roll and its subsequent brake run, the ride remains low to the ground and goes through a series of short banked turns through a rock fixture simulating several "near misses".

This section is located in the Gotham City section of the park near Superman: the ride roller coaster.

Guests then ride through: a set of speed-retarders, several helix turns, an additional set of speed retarders, a series of large track-hills, a set of speed retarders, and a series of fast sharp bumps.

The train makes a right turn to the launch. A voice is then heard saying, "Initiate in 3, 2, 1!" (In Shanghai Disneyland, "3、2、1，

發射!") and then the train launches out of the building and underneath the canopy. This is where the on-ride photo is taken. The trains then enter the building again, then proceed to twist and turn along the grid, passing by the other lightcycles and capturing energy gates.

The train drops and immediately traverses a cutback followed by a corkscrew. The train continues through a quick dip into an over-banked turn and continues with a slight uphill popup into the brake run.

The train heads straight to the chain lift hill. Once the train reaches the top of the 71-metre (232 ft) lift, it drops 64 m (211 ft) at a 74-degree angle reaching speeds of up to 121 km/h (75 mph). Following the first drop, the train then goes over the first camelback hill before making a sharp right hand turn back to the ground followed by a left hand turn. After the turn, the train goes over the second camelback hill. Following the hill, the train enters a 37-metre (121 ft) hammerhead turn.

After the train stops for a few moments on a pitch black section of track, magnets suspending the train release, quickly dropping riders 5.5 m (18 ft) vertically while they remain facing horizontally. The train rests for a moment before descending a short dip into the second LSM launch, accelerating the train to its top speed of 85 km/h (53 mph) out of the event building and through an ascending reverse S-turn. This leads to another brake run on a decaying covered bridge accompanied by sound effects of the bridge creaking in the wind.

The boats travel through a short flume section before emerging from the building and ascending a 28-metre-tall (92 ft) chain lift hill. Once at the top of the hill, the track dips and turns 180 degrees to the right, entering a mid-course brake run. A sweeping 180 degree downward turn to the right is followed by an upward

turn into another brake run. The ride then drops below ground, passing under the queue path before emerging over an air-time hill, where riders experience a feeling of weightlessness. This hill drops into the hull of an upturned boat and is followed by this section's splashdown.

A hairpin turn then leads the train into the second Barrel Roll and turns again. The train is then slowed by the trim brakes and then twists through another set of hairpin turns and twists until the train slows by the trim brakes.

The train begins to climb the 41-metre (135 ft) lift hill. Once at the top, the train goes through a pre-drop before making a sharp 37-metre (120 ft) downward right turn. The train then enters a 30-metre (98 ft) tall vertical loop. After exiting the loop, the train goes through a zero-gravity roll before dropping back to the ground and entering an Immelmann loop. The train then makes a full 360-degree upward right turn followed by a left turn leading into another drop.

The train continues, entering a slight decline followed by a slight left turn leading into the chain lift hill, ascending 61 m (200 ft). At the top, the train drops down 62 m (205 ft) at a 61-degree angle. At the bottom of the first drop, the train passes through the first of the two tunnels, each 37 m (120 ft) in length. The train then enters a 49 m (161 ft) camelback hill, where riders experience airtime.

Following a zero-gravity roll, the train then makes a banked left turn and an upward right turn leading into the mid-course brake run. After passing through the brake run, the train drops down into a tunnel (under a path). Upon exiting the tunnel, the train makes a left turn, followed by a right which leads directly into the inversion, an inline twist. After making a right turn into a second tunnel, the train makes a left hand helix which leads into the trim

brakes.

The train continues and starts climbing the 46-metre (150 ft) vertical lift hill. Upon reaching the top, the train is held for five seconds before plunging at a 96-degree angle. The train reaches a max speed of 92 km/h (57 mph) and heads through a negative-g stall loop, a one-of-a-kind inversion. This inversion is similar to a Norwegian loop, except the train enters a non-inverted dive loop followed by a sidewinder before exiting in the opposite direction.

Guests then dip into an archway of blocks and turn left. Along the way, riders can spot more scenery, a tower of Jenga blocks with Rex on top.

The train then makes a sharp left turn leading into a 17.0-metre (55.8 ft) zero-gravity-roll. After a sharp left turn, and right turn, the train enters a 10.0-metre (32.8 ft) inline twist, going through the supports of the track, a fake tree and a steel rectangle.

The track leads directly to the 47-metre-tall (153 ft) chain lift hill. At the top, the train descends a small pre-drop and follows a level turn to the right before dropping 44 m (144 ft) towards the ground. The ride then enters the first 39-metre-tall (128 ft) vertical loop, followed by a dive loop. The train traverses a zero-g roll, which is followed by a cobra roll.

The train curves to the right. One cycle of this section is about 175 seconds long.

A banked turn to the left leads into the mid-course brake run. The exit from the mid-course brake run drops down directly into the second vertical loop.

The train then does a 180-degree turn and enters the 87-degree drop, diving and reaching a maximum speed of 84 km/h (52 mph), which when viewed from the angle of the park's entry courtyard, gives the illusion of it diving into Mura's volcano.

Riders then turn left into an over-banked turn, which then turns

right.

The train then drops down a small dip which leads onto the launch track. The train is then accelerated to 100 km/h in 1.4 seconds, immediately into a hard banked turn to the right, curving upwards, and out of the building. The train passes a top-hat which provides a moment of airtime, before dropping back down. The train then travels through a large overbanked turn to the left and into a pretzel-turn. Following the pretzel-turn, the train travels through a heartline-roll and onto a brake run which slows the train to a halt before entering the show building once more.

The wooden track is 800 m (2,640 ft) in length, and the height of the lift hill is 21 m (70 ft).

The track then goes through a small pre-drop, similar to that on Bolliger & Mabillard steel roller coasters, before dropping 49 m (162 ft) at an angle of 81°. The train then ascends the first element, which is an outside banked turn, where the track is banked over to 153° before rolling back out of the bank.

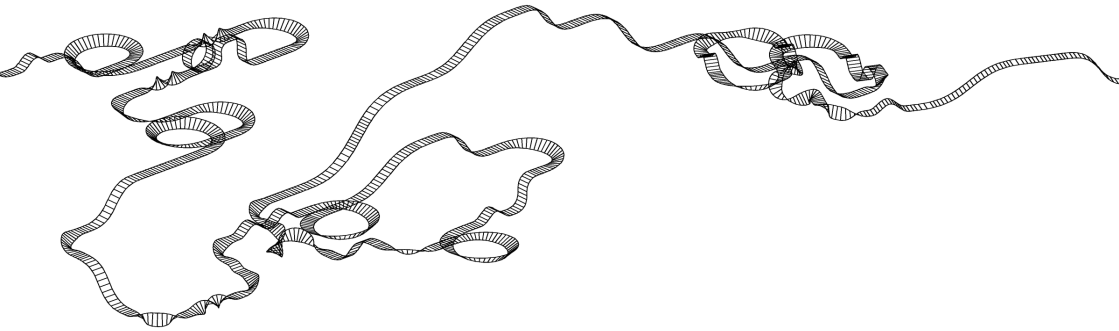
The ride continues with a small drop, before the trains make a sharp U-turn to the right. Riders immediately climb the lift hill . Trains again make a small drop, and another turn to the right, this is followed by a 27-metre drop.

Riders pass under the second tunnel and make a 360-degree left turn into the trim brakes. The train slowly makes a left turn past the maintenance shed and transfer track.

The train then enters the first drop and goes into a small camelback where riders experience a sensation of airtime. The train makes a sloped banked right turn and into a set of two consecutive corkscrews.

A few more curvy airtime hills sprint the train towards a low lying, left hand turnaround, up a two hills, and into the brake run. The ride is one of 8 wooden roller coasters with at least one inver-

sion as of its construction.



V-Like Formation

The train continues and makes a slight left turn. It then makes a sharp right turnaround, passing a boat shack with an old ride vehicle, and starts its climb up the 40 m (130 ft) first lift hill. Cresting the lift hill, the train makes a right turn, and descends 34.8 m (114.2 ft) toward the park's Rhine River at its maximum angle of 55 degrees and reaches its maximum speed of 97 km/h (60 mph). The train climbs a large upward hill and turns left before it briefly ascends, then drops into the first of the two interlocking vertical loops. This loop is followed by a right turn and uphill ascent into the mid-course brake run, followed by a descending, spiraling tunnel.

The train turns slightly to the left before entering the launch track, which is enclosed within a tunnel. The train briefly stops before accelerating from zero to 79.2 kilometres per hour (49.2 mph) in 2 seconds.

From the lift hill, riders can view other rides at the park, as well as the Downtown Denver skyline. Leaving the lift hill, trains snake around a swooping 3.0-metre (10 ft) drop, mimicking the drop on the original Mister the ride. This is followed by a 27-metre (90 ft) drop to the ground, and a rise up into a big turnaround, and another drop. After the second drop, the track goes through a double up type element wrapping around the first drop turn, before making another drop and hill and descending into the big helix. Upon leaving the helix, there is a slight straight segment before dropping to the left, entering a tunnel in the structure of the second turnaround.

It goes straight into a zero-G roll followed by a 180 degree upwards curve into the mid-course brake run.

A 90° vertical lift hill is followed by a 22-metre (72 ft), 97° beyond-vertical drop. After the drop the 8 person cars go through three inversions: a large vertical loop, a cutback and a heartline roll.

Riders are taken through a large turnaround to the right. Following the turnaround, riders encounter a large camel back that provides significant airtime. The train subsequently enters a 540 degree upwards helix which leads to the mid-course brake run. Off the mid-course brakes, the trains maneuver a left hand drop, leading into an over-banked turn.

The train then drops, reaching a top speed of 89 km/h (55 mph) before entering an overbanked turn. This is followed by a zero-g roll and "Fly-to-Lie" element that turns riders from a face-down prone position to a face-up lying position.

Unlike modern roller coasters, there are no brakes on the track at all.

As riders continue, they climb up the 4.6-metre (15 ft) lift hill. When riders reach the top of the hill, they do a left downward

bank turn, which takes their picture in the middle of the element. Riders climb up a small camelback hill, then a smaller hill, with a downward left helix, following a series of airtime bumps.

From the tunnel, a straight segment precedes a turnaround that leads into the brakes. Overall, the layout of the ride III has similarities to the Mister the ride, but a noticeable difference is the helix entry.

The train then ascends into the SLC signature roll-over inversion, then drops again back down to the surface of the lake. This element inverts riders twice and is shaped like a heart.

The train then banks left making a 180 degree turn and entering a 32 m (106 ft) tall 90 degree drop, diving into a tunnel and reaching a maximum speed of 84 km/h (52 mph) before entering a dive loop. The train then rises up to the right into an off-axis airtime hill, followed by a right-facing upwards helix. After the helix, the train makes a left turn and quickly drops, entering a right-facing cutback and a corkscrew. Then, riders go through an over-banked turn to the left before hitting the brake run.

The train climbs a 67-metre-tall (220 ft) lift hill, which is angled 50 degrees off the ground. The ascent up the hill is typically accompanied by the Styx song "Renegade". The lift hill uses a variable speed drive. The train climbs most of the lift hill relatively slowly, speeding up near the top after it has passed several sensors. After cresting the hill, it dips slightly and veers left into the United States' tallest inversion, a 60-metre (197 ft) "Drachen Fire Dive Drop". The train drops 62 m (205 ft) out of the inversion.

The last helix lead into a tunnel and the brake run.

The lift hill crosses over a portion of this section's track. After reaching a height of 65 metres (213 ft), the train drops 62 metres (205 ft) at 65 degrees, while entering a quarter right turn. As the drop levels out, riders experience up to 3.5G and rise up a large

hill with minimal airtime. The track curves slightly right into a large 180-degree, heavily banked left turn followed by three slightly banked airtime hills.

The train enters a sidewinder followed by a high-speed turn to the left. Following this turn, the train travels over a small hill before entering a corkscrew.

The exit from the overbanked turn leads into a small air-time hill, which delivers the greatest negative g-force on the ride, before ascending up into the mid-course brake run. The train drops from the brake run and continues through a series of low-to-the-ground air-time hills and directional changes.

Gotham City Gauntlet: Escape From Arkham Asylum follows the same layout as many other wild mouse roller coasters, with four people per individual car. The highest point is 15 m (49 ft) and this section includes 17 turns and numerous drops. This section is primarily intended for younger children and families, but it is not specifically a junior roller coaster as the minimum height requirement is 1.2 m (46 in).

The train passes through two consecutive horseshoe turns, first to the right and then to the left. As the name suggests, Horseshoe turns are highly banked horseshoe-shaped turns which feature track entering from roughly the same direction as where it exits.

Both keyholes are connected by a zero-g roll inversion. The train then dips low to the ground and enters an inclined dive loop, where the track reaches its turnaround point.

Ride vehicles then descend a drop, traverse a smaller camelback, followed by two more right-hand turns before the trim brakes.

The ride continues with an ascent up a 27-metre (90 ft) chain lift. At the top, the train dips slightly and turns roughly 225 de-

grees to the right. The train then drops 21 m (70 ft) into a valley banking right at the bottom as it begins to climb into the horseshoe element. The cars swing up and around to the left exiting the horseshoe parallel to same position during entry. Dropping back into the same valley, the train makes another banked turn to the right followed by a slight turn to the left as it passes by the observation area located near the exit. The last part of the ride takes riders through a series of sharp turns, each sending the train swinging quickly from one side to the other.

This section's acclaim is featured in the "History of Roller Coasters" in the ride at Frontier City in Oklahoma City.

The train then goes around a right U-turn and into the second inversion – a downwards inline twist. This leads into a cutback-style turnaround.

After being released, the train passes through the covered section, enters a Cobra roll element (referred to as a boomerang by the designers), then travels through a vertical loop. Upon exiting the loop, the train runs up a second lift hill, which is angled so that the two lifts meet at the top in a "V"-like formation. Once out of momentum, the train is towed to the top of the lift hill, and is held for several seconds before being released; traveling through the roller coaster in reverse.

The car curves to the right to enter a small tunnel, where it rises 33 metres. At the highest point it reaches 52 metres above sea level. It again curves to the right, and starts to fall. After this, it turns sharply left and travels for 50 seconds in the woods Collserola.

Almost immediately after, the train goes through a zero-gravity roll followed by a downward left helix. The train enters the first of two corkscrews which are separated by a banked left turn. The train enters an upward left helix (the beginning of the helix is close

to water) before entering the brake run.

The train is then launched again into a tunnel, twisting and dropping before coming out of the "broken billboard", splashing down in an L.A.

The train makes two left turns through this section's support structure before climbing a chain lift hill to a height of 30 m (100 ft). The train then descends an 24-metre-tall (80 ft) drop, reaching a top speed of 80 km/h (50 mph). After passing below a later section of track, the train rises into an elevated turnaround before dropping beneath this section's supports once more and ascending a camelback hill.

A series of direction and elevation changes lead into a second corkscrew, the inverted part of which is just above a portion of the queue line. The track then turns to the right.

The pre-drop serves to reduce the stress and pull of the chain. After the pre-drop, the train goes down a 64 m (210 ft) drop toward a water-filled ravine at a 65-degree angle and reaches a top speed of 117 km/h (73 mph). At the end of the ravine, the train enters a second airtime hill with a 40-metre (131 ft) drop. A short, narrow, above-ground tunnel is at the bottom of the second drop.

Minebuster will be familiar to most coaster riders as a modified out and back wooden coaster, specifically with the addition of an upward spiralling helix at the end as the most obvious modification to the traditional out and back layout.

Each train then dives opposite each other into a 360-degree helix. The blue train circles at a significantly smaller radius further making up ground it lost in the earlier helix. They exit the helix and ascend into the trim brakes to determine the winner.

The train then heads up the spiral lift hill, which slowly twirls the train upward. After the lift, the train then heads down a short, steep drop and then ascends slowly back up. A hairpin turn then

occurs and takes riders through the first Barrel Roll inversion. The train then hits another hairpin turn into another short drop.

The train continues into a slight left turn leading into an inline twist and a 135-degree downward right turn. The train rises back up slightly, leveling out and entering a mid-course brake run. The train makes a left turn and dips slightly, ending at the trim brakes.

The message before launch is "Keep your head back! Hold on tight! You MUST escape! Go, go, go!". According to Alton Towers, this section's maximum capacity is 1,150 riders per hour.

This section consists of two raised sections of track with a vertical loop in the centre.

The train turns left and passes through a zig-zag double down before reaching the second launch, which takes trains up to the top speed of 72 km/h (45 mph).

The train continues and dips to the left into a trench before hitting the lift hill. After climbing to a peak height of 40.0-metre (131.2 ft), the coaster plunges down an 80° drop into its signature "stall loop" inversion. The train exits through an S-bend and first airtime hill of many, leading into a right hand turnaround and heartline roll.

A heavily banked turnaround is followed by a camelback over the roof. After a turn to the left the ride track runs parallel to the park's Hollywood street, after which the train navigates over two more camelback hills, and is slowed as it runs through a set of block brakes at the top of another hill. After descending out of the brakes, the train enters a one-and-a-half turn upward-spiraling helix followed by another camelback hill.

After a slight upwards hill, the train straightens out, then curves right into a descending double helix. After exiting the helix, the train continues on an upward sloping right hand turn into the trim brakes.

It is accelerated by linear synchronous motors from 0 to 180 kilometres per hour (112 mph) in 5 seconds. The train then ascends a 112-metre (367 ft) tall top hat, twisting 90 degrees to the left. Once the train goes over the top hat, it descends back down, again twisting 90 degrees to the left. This sends the train travelling parallel to the launch track but in the opposite direction. The train then enters a flat brake run and then ascends a small airtime hill before entering the set of brakes.

After a moment's hesitation, the train is then launched up a steep incline to a peak height of 33.0 m (108.3 ft). Riders navigate an outwards-banked 90° left-hand turn before plunging into a vertical drop over the park pond, which leads to this section's signature pair of inversions; a banana roll and a vertical loop.

It travels over the resort gate walkway, down a slight decline into a 90-degree turn to the right and engages the chain, which operates at a speed of 9.7 km/h (6 mph), to begin the 62-metre (205 ft) climb to the top of the lift hill. The train descends 59.3 m (194.7 ft) at a 60-degree angle, reaching a top speed of 116 km/h (72 mph). The train then ascends a 48-metre (157 ft) camelback hill, curving to the left into the first tunnel. Coming out of the tunnel, the train ascends a third, 24-metre (80 ft)-tall hill. After the third hill, it enters a set of trim brakes and a pretzel-shaped turnaround where it curves to the left toward the beach and then left again into another tunnel.

Immediately as the train exits the bend, it passes through a cobra roll, then an immense upward helix to the right, and then the mid-course brake run. Following that, the train plunges into a corkscrew that travels into a tunnel after completing the mid-course brake run.

Hagrid offers congratulations for finding them, and the train exits the building into the launch, accelerating to this section's

maximum speed of 80 km/h (50 mph). Fog effects engage near the tracks, as the train hits a left banked turn and passes the on-ride camera. After this, the vehicles make a right turn before stopping at the abandoned castle, where a unicorn and her foal are spotted.

The train climbs up a left-turning, 35-metre (116 ft) outward-banked hill, dips slightly right, and passes over a double up element, which leads to the first inversion, a zero-g roll variant. The train dips right, passing underneath the lift hill into an overbanked turn to the right that incorporates an airtime hill, known as a wave turn, which leads into the second inversion, a zero-g stall. The train passes through the lift hill structure for a second time and dips down, entering another double up element into the mid-course brake run. This is followed by a sharp left turn into a short, steep drop that dips slightly left. The train then navigates an upward turn to the left, traverses another airtime hill, and rises once more into a slight left turn.

The pretzel knot element is different from the much more common pretzel loop element.

The train makes a winding 180-degree turn. It ascends 63 m (208 ft) up the lift hill before dropping 62 m (205 ft) at an angle of 68 degrees.

After passing through the storage transfer track, the train makes a slight curve to the left into the lift hill. After cresting the lift, the train momentarily flattens out, then begins a slight descending curve to the right, passing over part of the queue line for the ride. After passing under the lift, the train heads heads straight, then into a slight curve to the left, followed by a small hill into the mid-course brake run. The train then traverses down another drop, curving to the left and passing under the track of the Dry Gulch Railroad.

The coaster initially goes up a lift hill, then falls slightly to the

left. It then falls down another time, goes right, then left and right again. This section has two trains with ten cars per train, with two riders across per row, and thus can have 20 riders per each run. This section's tracks are 1030.2 ft long and made of steel, and this section runs for 1.02 min. This section is themed after The Tortoise and the Hare, an Aesop fable, due to its location in Aesop's Village.

The ride begins as the train turns to the right and begins to climb the 32-metre-tall (106 ft) chain lift hill. After cresting the top of the lift hill, the train drops 30 m (100 ft) to the right at a 50-degree-angle, reaching a top speed of 82 km/h (51 mph) and preparing to enter the pretzel loop. In a pretzel loop, a train swoops up to a height of 24 m (78 ft) before diving toward the ground, looping back under the starting point. At the bottom of the loop, riders face upward and experience strong, positive g forces.

This section is located entirely inside the structure of American Dream. The train immediately enters a slow heartline roll.

With a sudden left turn, the trains enter the re-entry tunnel with many colorful stars flying towards them and two flashes of light as the onride photo is taken.

The track leads directly into a 23.2 metres (76 ft) drop to ground level. Levelling out, the track has a short straight segment as it passes through back to back vertical loops and crosses under this section's brake run. After the vertical loops, the track makes a left turn, passing a backstage area, and enters into a counter-clockwise double corkscrew as it crosses back under the ride. This is followed by a set of safety brakes, then a clockwise upward helix to end the ride.

This section has a similar layout to Rage, a Gerstlauer Euro-Fighter roller coaster at Adventure Island in Southend-On-Sea, Essex, UK, and has the steeper-than-vertical 97-degree drop characteristic of all Gerstlauer Euro-Fighters. The coaster is

reported in some sources as having 2 inversions and in other sources as having 3 inversions.

The train makes a small dip before making a 180-degree right turn to climb the 48-metre (157 ft) tall lift hill. Once the train apexes the top of the lift, riders drop 45 m (148 ft) to the right at a 57-degree angle, reaching a maximum speed of 105 km/h (65 mph), into a 41-metre (135 ft) tall vertical loop.

Initially the ride turns sharply right, then into a left-hand downward helix. Rising up from the helix it transitions into a downward right-hand helix.

The ride drops and enters a quick transition into a highly-banked right turn, which transitions into a twisted airtime hill. The train enters another airtime hill and a highly-banked turn to the left, crossing over the ride. The train enters the trim brakes, then makes a right hand turn.

The track crosses over the exit from the Immelmann loop and then into a tunnel and down a drop before entering a zero-g roll, alongside the Le Scoot log flume. After a short section of straight track close to ground level, the track inverts into a corkscrew followed by a clockwise upward helix, before making a left turn to the trim brakes.

The ride will feature four linear induction motor launches to reach a top speed of 72 km/h (45 mph), including a multi-pass launch following a backward element of the ride. This section's 1,260-metre-long (4,120 ft) track will provide riders 32 moments of floater airtime during the two-minute thirty-six-second ride. Riders will experience a 12-metre-tall (40 ft) drop and a 26-metre (85 ft) total elevation change during the ride. The ride will be set in an environment with quarry walls and pine trees. Part of the ride will be themed to an escape from the quarry's werewolf curse, featuring themed elements including werewolf figures and

a crumbling quarry scene to end the ride.

The sound of a bike bell goes off, cuing the entrance of the professors assistant, a monkey which cycles over the left hand side of the lab on a skybike. A video is shown revealing a machine which is supposed to collect grapes for wine, and the professor is seen climbing down a ladder and removing a large spanner out of his lab coat. The monkey appears to the left holding a lever whilst the professor adds the finishing touches to his machine.

Up to 14 riders at a time are electro-magnetically accelerated to 160.9 kilometres per hour (100.0 mph) in seven seconds along the extended, 206-metre (676 ft) launch tunnel. The track then pitches up 90 degrees to the vertical, with the passengers pulling 4.5 g. Riders are almost weightless during the entire vertical section of the ride for about 6.5 seconds, 3.25 seconds going up and 3.25 seconds falling back down. The car then pitches back down to horizontal and enters the tunnel where an on-ride camera takes photographs of riders.

The car usually sits for a few moments.

It goes into a low over-banked turn under the cutback, followed by a corkscrew over a piece of track, creating a head-chopper effect. It then goes into another over-banked turn, followed by a series of S-curves. It does one more 180 degree turn into a series of bunny hops under the corkscrew. It traverses one last twist into the trim brakes.

Billy appears on screen, accompanied by Jigsaw's voice delivering the line "Game over." The screens then switch off, Billy's laughter is heard, and a bell sounds as the lift hill engages and the car begins its ascent, gradually increasing in speed. At the summit, the car plunges into a 100-degree freefall drop, passing beneath large rotating blades styled to appear bloodstained. This is followed by an Immelmann loop, an overbanked turn, and an air-

time hill.

The vehicles then make a sharp turn at the west-end of the building. The coaster proceeds to make two fast turns and runs to the opposite end of the building, with a few track-bumps along the way. The ride then slows to a crawl as it travels around the park's Log Chute attraction.

Riders ascend a chain lift hill before going down a small dip and into an s-bend followed by a U-turn. After a slight ascent, the car enters a beyond-vertical drop.

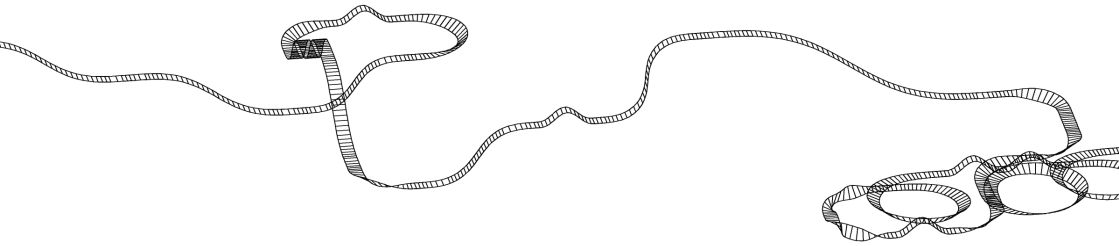
The ride then travels through the second vertical loop and navigates several low-to-the-ground turns. The train ascends one last airtime hill before hitting the trim brakes.

The ride layout is Out and Back, characterized by the long, stretching launch and the hills parallel to the launch. The ride continues with an S-curve, after which the train will be stopped before the launch. When the train is launched to 75 km/h (46.6 mph), it takes a stretched hill, which is followed by a turn to the right and the horseshoe. Then the horizontal loop and a left turn follow.

Then the blue tracked train does a zero-g roll. After a double up and a quick turn to the left, the blue track turns green and the train hits a brake run, before returning to the left side of the lift hill. The green track is identical to the blue tracked segment up through the high-five, which is banked in the opposite direction as to create the "high-five" illusion with the blue track. After the high-five, the green track does a double down and does a top gun stall crossing over the blue track.

The ride continues with a 90° turn into the chain lift hill. After climbing this lift hill and descending the twisted first drop, riders experience a pretzel loop. The track then turns to the left, passing through the pretzel loop. After turning to the right, riders dip

down towards a pond. After another right turn, the track crosses over the trim brakes into an inline twist.



Different Direction

Then a blue streak is seen followed by Superman saying "Don't worry folks, there's only one way out of this mess. Heads back. I'll push you out of here. Fast. Superman fast." This line is followed by the main 100 kilometres per hour (62 mph) launch. The train launches up this section's top hat, followed by a heavily banked turn to the right.

The train enters a small drop into a high-speed overbanked left turn, which leads into the third inversion, a zero-g roll. It exits into a double-up and another high-speed overbanked left turn, followed by one last zero-g roll as the fourth inversion.

An upward helix follows the downward turn and is followed by a corkscrew to the right. After another upward helix and a wide turnaround, the train enters the trim brakes.

This section's hydraulic catapult motor accelerates the train from 132 km/h (0 to 82 mph) in 2.3 seconds. It only uses 48 m (157

ft) of track to launch the train. Each of the twin hydraulic catapult motors achieves a maximum of 10,500 horsepower (7,800 kW) during the launch of the train. This section's hydraulic motor system has the mechanical capability of accelerating vehicles to a speed of 188 km/h (117 mph) or more, as tested for Top Thrill Dragster, although less power is needed for the train to crest the hill. In order for a train to launch, the catch car must first roll back slightly into position.

The ride hit a straight piece of track, then headed towards a helix through a wooded area adjacent to the village.

Navigating a tight dive loop, the train breezes past the covered section and through the switch-track into this section's endgame portion; a 90° banked airtime hill similar to the wall stall elements on various RMC coasters, and a twisted rollback spike. This particular element twists riders upside down over park pathways and ends in a dead end of mangled track, forcing the train to roll backwards through the banked airtime hill once again.

After traversing the 360° helix, the train immediately turns into a second overbanked turn. Traveling into a banked airtime hill, the train turns right and then whips to the left into a second 270° helix. Upon exiting the helix, the train turns left before making a right turn into the brake run. A ride cycle is approximately one minute and 50 seconds.

Riders continue and move to depart the base, but first pass by a lab display where a holographic scientist demonstrates the prototype withstanding bullets, bombs, and flamethrowers. Passengers then travel through a village before being accosted by this section's strike team.

Continuing straight down a dark hallway, an animatronic exterminator dressed in a bright orange hazmat suit appears, aiming a poison gas gun at the riders.

Additionally, the ride bears several similarities to the SkyLoop model including design, height and speed. On speakers outside the attraction entrance, John Williams' score is played in a loop.

The train takes a right turn into the 61.6-metre (202 ft) lift hill. Upon reaching its maximum height, riders take a twist down the 59.4-metre (195 ft) 85-degree-drop. At the bottom, the ride reaches its maximum speed of 115.1 kilometres per hour (71.5 mph). The train ascends this section's tallest camel back hill before descending left to follow into the 40-metre (130 ft) non-inverting loop.

There is a small left turn and initial descent into a ravine, which is followed by a gradual sweep to the right. The train passes through the transfer track (which leads to the storage track under the covered section) and climbs the lift hill. Riders descend 33 m (108 ft) into a drop covered by a metal canopy. The train then turns left and rises over an airtime hill before making a sweeping left-hand turnaround. After the turnaround, riders descend again and rise into a gradual right-hand climb before descending into the structure of the lift hill and making a left-turn chicane.

The roller ride continues by entering a cable lift hill. It then reaches its highest point, travelling above Luna Park's entrance before the first drop. This section's layout circles the perimeter of the park twice, the first circumnavigation featuring large drops and tunnels. The second pass around the park features significantly smaller drop heights, and lower speeds.

The train turns right and begins to slowly climb the lift hill through the use of a chain lift, taking riders to this section's maximum height of 70 metres (230 ft). From the top, a passenger can clearly see the iconic CN Tower to the left and overlook the entire park to the right. The train drops from its peak at a 75-degree angle and reaches a maximum speed of 125 km/h (78 mph) in 3.9 seconds.

The train makes a turn to the right before being launched upward at 39 km/h (24 mph) before dropping down a 55-metre-tall (180 ft) twisted drop into a set of airtime hills, turns, and a wave turn. The train is then launched up the Tuwaiq cliffs to a speed of 160.0 km/h (99.4 mph). After the launch, the train traverses a series of turns and hills at the top of the cliffs, including an outer-banked turn over the edge. Afterwards, the train is slowed down by brakes at the edge of the cliff before dropping at an angle of 90 degrees into a tunnel. Exiting the tunnel, riders reach this section's third launch, accelerating to a top speed of 249 km/h (155 mph).

According to the Gerstlauer website, there are two different versions of its spinning roller coaster model, as well as a customization option for a park to fit its desires. The ride is the only 400 model in operation, as the rest are the 420 model. The 400 model is shorter at 15 m (48 ft), more compact and has a lower capacity. The layout consists of several "segments" separated by brakes.

The train pulls forward into the launch area. The catch wagon then attaches to the front of the train prior to launch. During the launch, the train accelerates from 126 km/h (0 to 78 mph) in 1.8 seconds, followed by a "dog tongue" double inversion at 53 m (175 ft) off the ground.

The train makes two right turns and climbs the 18-metre (60 ft) chain lift hill. Upon reaching the top, it makes a left turn, goes through a tunnel and drops 23 m (74 ft) at 77 km/h (48 mph). The train makes a sharp right turn and ascends several airtime hills, before making a right and then left turn into a helix. Afterwards, the train enters the trim brakes.

The train (the vehicle mode for an Autobot-named Coaster) launches from 0 to 64 km/h (0 to 40 mph) in 2 seconds at an upward angle through the belly of a Decepticon Driller (a tunnel

with special effects and lighting). The train then enters a zero-g roll and plunges a 32-metre (105 ft) drop, reaching its top speed of 108 km/h (67 mph), and then enters a cobra roll, followed by a vertical loop. The train then enters a small cave. Once exiting the cave, the train enters a corkscrew and another vertical loop before entering the mid-course brake run, descending another hill into another corkscrew and a turnaround. The train then tilts sideways as it enters a helix, which ends at this section's trim brakes.

The coaster train accelerates to its top speed of 240 km/h (149.1 mph) in 4.9 seconds using a hydraulic launch system. Riders experience up to 1.7 g-force during acceleration, and throughout the course of the ride may experience up to 4.8 g.

It then returns backwards through the covered section into the non-inverting loop. Returning forwards and accelerating riders to a top speed of 100 km/h (62 mph). It then travels up to a height of 46 m (150 ft) where it completes a heartline roll before exiting in a second non-inverting half-loop at 41 m (133 ft). It then enters the second twist. The ride can be compared to several of Maurer Söhne's X-Coasters.

The train's individual cars spin freely throughout the ride.

Once the train has reached the top of the lift, it turns right into a U-turn while traversing slightly downward.

The train rolls forward and backward to engage with the catch-car while the brake fins on the launch track retract in two stages. During the countdown, the voice repeats the safety instructions as five sets of red traffic lights above the train illuminate sequentially. When the lights turn green, the voice enthusiastically shouts "Go, Go, Go!", and the train is launched, accelerating rapidly from 130 km/h (0 to 80 mph). The train ascends the 62.5-metre (205 ft) top hat, turning 90 degrees to the left before

the crest. After cresting, it turns another 90 degrees to the left, descends to the base of the top hat, and continues over a airtime hill.

The car proceeds to swing around a few small helices before plunging back into the abandoned mine building, where it slows down. A spotlight turns on illuminating the canary cage seen earlier.

After it reaches the top, it drops 124 feet towards the ground and enters a zero-g roll, the first inversion, and then enters into a fly to lie element which makes riders face the sky. It then goes into a half pretzel loop and makes riders turn into a flying position again. Quickly, it turns into another pretzel loop, then enters a small tunnel at the bottom of the loop. It enters a small air-time element, then enters into the corkscrew. After the inversion, the train will enter a helix and comes to the inversion, an inline twist which is just before the trim brakes.

The train navigates a high banked turn over the covered section, dips beneath the chain lift hill, and enters a second hammerhead turn, this time to the left. The track then makes a right-hand uphill turn into the mid-course brake run.

It then makes a turn-around to the right followed by an 80 degree banked turn. After the turn, the train passes back under the track into a "Midway turn" which, being another 80-degree turn, faces the train towards the park's Celebration Plaza. Passing again back under the track, the coaster goes through the flat turnaround before making its way to the brake run.

The train makes a 180-degree right turn, passes over two small bunny hills, and begins its ascent up the 62-metre-tall (205 ft) chain lift hill. After cresting the top of the lift hill, the train drops 61 m (200 ft) at a 90-degree angle, reaching its maximum speed of 119 km/h (74 mph). After this drop, the train traverses a speed hill

followed by a climb into a larger airtime hill, which drops riders slightly to the right.

Each of the rides are Mack Rides' large park variation of their wild mouse roller coasters. After climbing the lift hill, the train makes a hairpin turn and drops down followed by a normal the ride track.

It then takes a sharp 180 degree turn before entering 2 back-to-back vertical loops. The train then ascends and turns 180 degrees into the mid-course brake run, which is also a small lift hill. At the top of the lift, it drops and enters 2 half-helix elements before entering its last 2 inversions, again done back to back.

The vehicles exit the hut by turning right, which leads into the third launch and then an airtime hill. Riders pass by an abandoned castle and make a left-handed dive into a tunnel.

The train takes a trundle round a corner, then travels gently downwards; the train picks up speed before hitting a banked corner. The train then travels through a loop, this is followed by a sharp corner. The train then rolls up and the brakes are only slightly applied. The train travels around more corners, followed by a small bump, then up and into the brake run at the end of the ride.

As the rider is totally immersed in the virtual reality world, one can even be tricked by giving hints on a wrong track direction and then e.g. have a giant creature grabbing the virtual cockpit and carrying it into a different direction (which turns out to be the actual direction of the rails). Also, the effect of physical track elements like block brakes can be utilized in the virtual reality experience for dramatic elements like crashing through a virtual barrier or building.

The train continues with a small right-handed turn, tracking towards the chain lift hill. Once at the top, the ride dives down into

a twisted drop. Riders experience a series of additional twists and turns before the ride takes an over-banked curve that brings riders close to flipping upside down above this section's entrance sign. The ride sharply swings right for several beats, then swings left for another few beats.

The 28 passenger train makes a wide "U" turn to the lift at the back of the ride. Rides ascend the 98.91 foot chain lift hill. Once at the top of the lift, riders drop towards the right reaching speeds of 50.20 MPH at the bottom of the drop. Upon exiting the drop, the train navigates a large vertical loop that runs parallel to the lift.

During two-train operation, the goal (for optimum rider throughput) is to have the second train ready to 'release' when the first train hits the midpoint of the track, which is the banked turn where the first train switches from "out" to "back".

This section features eight drops, seven camelback hills, a panoramic U-turn, and a diving spiral. The seven camelback hills represent each of Earnhardt's championships. From the first drop to the brake run, the camelback hills respectively measure 54 m (178 ft), 46 m (151 ft), 32 m (105 ft), 27 m (90 ft), 19 m (62 ft), 16 m (52 ft), and 15 m (48 ft) tall.

The riders traverse a banked left turn, followed by an upward helix to the left inside Wonder Mountain. Part of this helix is exposed, giving the riders a view of the surrounding area. Due to the nature of the train ride, the train picks up speed as it ascends the helix. The train then re-enters the mountain, traversing a downward double-helix to the right in complete darkness, with included special effects.

In the ride, visitors rode in a roller coaster car appearing as a boat, seating 10 passengers in 5 rows.

The train then banks right, makes a 12-metre (38 ft) dip, turns left and goes through a small 4.9-metre (16 ft) dip. The train then

makes one last 15-metre (49 ft) airtime drop before climbing up and into the trim brakes. There is approximately 26 seconds of airtime during the 2-minute, 15-second ride.

The train continues into a second elevated turnaround which continues with a double down. The train then navigates another camelback hill and raised turnaround before a series of hills split by a third turnaround. After a turn, trains enter this section's trim brakes.

After climbing 33 m (107 ft), the train turns left and begins a gradual drop at 18.5 degrees. The track tilts to the left as the train enters a double helix that features a highly banked turn to the left.

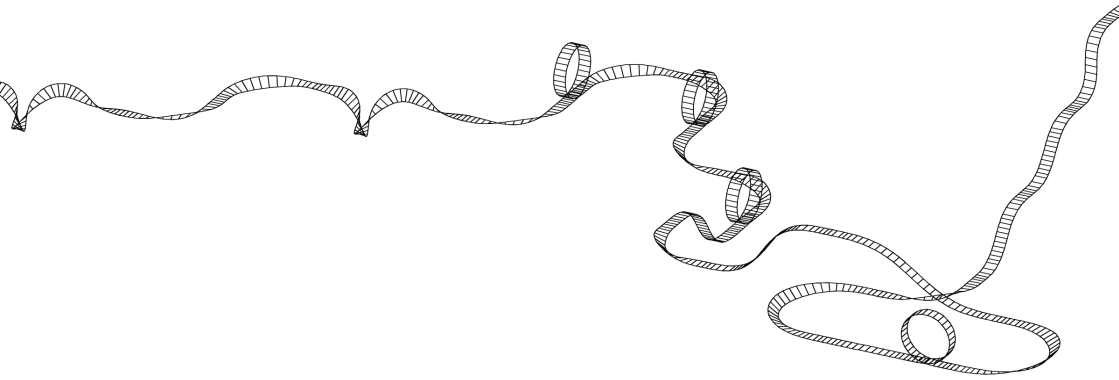
This section is 17 metres (56 ft) tall, and features a drop of 14 metres (47 ft). It has top speeds of 72 km/h (45 mph), and is 194 metres (635 ft) long. It has a maximum vertical angle of 52 degrees, and subjects riders to 4 Gs. This section has a single train with four cars. Each car has two rows that seat two riders, for a total of 16 riders per train.

A small prop area designed to resemble a mineshaft can be seen on the right as the train passes. A small left turn then leads the train into the first of two chain lift hills, along with a slow climb to 34 m (110 ft). Dramatic music is played over a speaker as a recorded announcement reminds guests to remain seated throughout the ride. After cresting the top, the train plunges 41 m (135 ft) at a 53-degree angle, entering a 38-metre-long (125 ft) underground tunnel and passing an on-ride camera.

The train enters a vertical loop before ascending a chain lift hill. When the train reaches the top, the lift disengages, and the train falls backward through the inversions and tunnel.

The train reaches the maximum speed of 117 km/h (73 mph) before entering a sharp turn and the first airtime hill. A 540-degree helix follows, in addition to another smaller airtime hill.

The circular nature of this section's loop, as well as its relatively small diameter of 7.6 metres (25 ft), meant that it could produce forces of approximately 12 g0 (120 m/s2).



Chain Lift Hill

The train continues into a right turn and two corkscrews which invert the riders twice. The track leads into a flat section followed by an s-curve drop and rise into the trim brakes, before turning right.

The train passes through a zero-g roll and a pair of banked turns overtop the Supersplash water coaster attraction before entering the second launch. The train navigates an airtime hill mid-launch before soaring up into a double-inverting dive loop (also known as a flying snake dive). Another airtime hill is traversed, followed by a turnaround and another airtime hill before popping up into the brake run.

Cresting the lift, it makes a small dip, a roughly 90-degree turn to the right, and drop down and into the two back-to-back vertical loops. Following the loops is a right turn into a double helix segment.

The ride layout is a standard the ride roller coaster layout.

This section's layout starts with a 90 degree turn into a short, 20 m (64 ft) Lift hill. after ascending the lift hill, the ride transitions into a curved drop and an mild hill, then turning into a series of banked turns overlooking the entrance to the park.

In total, it has 4 inversions.

The track travels along the terrain through various turns. The train goes down a drop then up another hill into a downward spiraling helix, before hitting the trim brakes.

The train continues and turns right. From there, it makes its way through a left turn and climbs the 30.6-metre (100.4 ft) chain lift hill. Upon reaching the top, the train drops 30 m (100 ft) to the right at 86.4 km/h (53.7 mph). Riders go through a right-handed banked turn after the drop.

The ride layout continues with a sidewinder and multiple turns. After passing through the mid-course brake run, riders spiral downward to the left, continuing to maneuver through the twisted ride structure. The train picks up speed as riders dive at the floor of the building, then passing through a Flatspin before arriving at the trim brakes.

The train then enters a double-up before making a slight right turn and going down a hill under the suspension bridge located at the south end of the park. The train then enters the large turnaround at the end of the boardwalk before entering a couple more airtime hills. the track then makes an unbanked right hand turn before going back under the structure of itself and the ride. The train then goes through one more bunny hill before hitting the trim brakes.

The train makes a small right turn to a 61-metre (200 ft) lift hill to begin its ascent. After reaching the top, it enters a 61-metre (200 ft) drop, in which the train reaches a top speed of 117 km/h

(73 mph).

Turning into the lift hill, riders pass by a projection of the Rookburgh city before the track and seats rotate riders into the prone flying position. The trains enter the outdoor portion and sends riders into the first launch. Immediately proceeding the launch, the trains navigate a left-hand cutback turn and a corkscrew inversion. Turning to the left, the layout twists over and under the launch before diving below the guest pathways, encircling the queue's descending stairs and entering the second launch below-ground. The launch takes the train through a darkened tunnel and up an incline, reaching the maximum above-ground height of the layout and sending riders into a spiral dive around the Hotel Charles Lindbergh.

The train makes a right-hand turn as it continues and climbs the initial chain lift hill.

The train executes a left-hand turn before ascending a 23 metres (75 ft) vertical chain lift hill. At the summit, it descends a beyond-vertical (~97°) first drop, leading into a twisted airtime hill, intertwining with Raven. The layout then enters this section's first inversion, a step-up under flip. Following the inversion, the train passes over a banked airtime hill that transitions from a right to a left bank before entering a right-hand turn. The train then navigates the second inversion, a zero-g stall, in which riders experience a brief period of weightlessness while the train is inverted. After exiting the element, the track banks to the right.

It then drops into two bunny-hops. In its minute and 24 second-long ride time, it reaches a top speed of 40 km/h (25 mph). It has 400 m (1,300 ft) of track.

The ride is considered a steel-tracked hybrid due to the track's use of tubular steel which sits on a wooden support structure.

Regardless of which path they take, riders are presented with

one of two battle scenes; rescue either Dragnor or Astradir. With three physical endings and two finales, a total combination of six possible endings are available on the ride.

The coaster goes to the left and down slightly. The coaster does a banked right turn up, and then travels through the top of the temple and then goes through the loop of MonteZOOMa: The Forbidden Fortress. The train then turns right and travels up a second lift hill. The track bends left, and the coaster goes through a series of bunny hills. As the track approaches the Timber Mountain Log Ride, it turns around. The train travels through some more bunny hills, then does a banked helix to the left.

Riders travel forward approximately 6 meters (20') before immediately descending a small, roughly 3–5 m (10-15') drop. They then ascend a small "bunny hill" which leads into another brief drop, this one curving to the right. The train then climbs up a small slope and enters into a pre-lift hill brake run, to reduce stress on the ride. After a very small drop down and a left-hand turn, riders begin to ascend the 67 m (220') lift hill.

The train descends a 44-metre (144 ft) drop, which banks to the left as it descends. The train then goes through an upward helix. Coming out of the helix, the train drops 31 m (102 ft) then turns right and rises up into the mid-course brake run. The train drops 15 m (48 ft) out of the brake run followed by another drop at 27 m (87 ft) toward the ravine.

The train then goes through the 24-metre-tall (78 ft) cobra roll, a roller coaster element which inverts riders twice. Riders then enter the mid-course brake run which is located next to the lift hill. The train drops out of the brake run to the right and enters a 270-degree helix followed by the two interlocking corkscrews. Riders go through a small dip then enter the trim brakes.

Following a series of turns, the vehicles approach the fourth

launch, initiating a double-right turn as Hagrid warns that the bikes are outside of his control. The vehicles then make a right turn and slow down on the mid-course brake run.

Riders then go through an over banked Horseshoe Curve element.

The ride then drops back off the quarry wall and enters a tunnel near its base. The tunnel features fog and lighting effects. After exiting the tunnel, the train enters brake run.

And as it turns and dips, you're elevated or (dipping) depending on where you sit, it's like being (out on) the wing of a plane. This section has two queues; the standby queue is for guests without passes and the Quick Queue is for guests who have a pass that allows them to bypass the queues for most of the major attractions at the park. The train continues and makes a downward right turn, immediately followed by an upward right turn, which leads to the 47-degree inclined chain lift hill. Once the train reaches the top of the 61-metre (200 ft) lift, it turns right into a holding brake, where it slowly moves over the first drop, stops for about four seconds, and then descends the 90-degree drop.

After leveling out, riders experience a slightly banked turn to the left into this section's third tunnel. As the train exits, the turn switches to the right and is followed by two ascending left turns that lead into the tunnel. A second brake run slows the train inside as it enters the second lift hill, which is completely enclosed. Large stone warrior statues in several rows with glowing eyes can be seen lining both sides during the ascent. Their arms move in sync rhythmically with clenched fists that are seemingly pounding down towards riders. At the top of the hill is a centrally mounted stone warrior with sharply lit eyes appearing to pour a vat of lava – a stationary lit object – on riders as they pass underneath. The train takes a short dip when exiting the tunnel,

and turns to the right into the trim brakes.

The ride continues with a mild 180° turn to the left, taking it straight into the lift hill. It takes approximately 23 seconds to ascend before the ride hits another 180° turnaround to the left, and plunges just over 80 feet before going into the inversion, a zero-gravity roll. It goes into the first high speed turnaround of many to the left, leading straight up an airtime hill that's followed by a "double down", then passing another bunny hill, and into a right hand turnaround. Two more small hills follow, and then goes towards the 115° over-banked turn to the right.

The following drop then sends the train speeding into a wooded area at a maximum speed of 41 mph. The following layout includes several airtime hills and banked turns, before ascending a short second lift hill.

Riders traverse a small straight drop before going into a curved drop to the right. After the drop, riders rise into a pretzel loop element and a left-hand dropping turn before a right-hand rising turn. It goes through a series of left- and right-hand turns before a 270-degree helix and an inline twist into the brakes.

As the car makes an U-turn, the spin lock mechanism on the car disengages, allowing the car to spin freely. The car spins out of control through three hairpin turns in pitch darkness before emerging at the gates of the giant furnace. The car enters the furnace illuminated by bright red lights, completing a wide turn, before exiting the furnace into a sudden dip and a "bunny hop" hill.

The train makes a slight right turn into a 19-metre (62 ft) inclined dive loop before going through a zero-gravity roll. Immediately after the roll, the train enters the first of two corkscrews. After exiting the corkscrew, the train then goes through a cobra roll. After an upward left turn, followed by a downward right turn, the train goes over an airtime hill before

entering the second corkscrew.

Fluffy, the three-headed dog from the film franchise, is seen on the left as riders pass by and make a turn to the left. After passing over a switchtrack, a fifth launch sends riders speeding into a tunnel with a group of Cornish Pixies that have taken over Arthur Weasley's broken-down flying Ford Anglia.

Like other suspended roller coasters, this section's trains are able to swing side to side.

Crossing the layout yet again, riders navigate a left-hand turnaround into a darkened tunnel with smoke and strobe effects. A few more banked airtime moments lead to a 270° counterclockwise helix, thus sending the train into the brake run.

As the train continues, riders make their way through a 180-degree left turn and begin climbing a 29-metre (95 ft) chain lift hill. At the top, the train turns left and drops 26.6-metre (87.3 ft), reaching 80.6 km/h (50.1 mph). The train speeds underneath the exit bridge followed by a right-handed banked turn.

The train then drops into the mine. After briefly leveling out at the bottom, it then climbs out.

The train then drops approximately 3 metres to the right, with a downwards left helix following. After exiting the helix, the train makes right-hand turn, before the track levels out on the brake run.

The train makes a left turn before going through an Inline twist. During this element, which is known as a keyhole, riders go through an airplane control tower, giving the illusion the train will hit the tower.

The ride enters a switch track, where the section of track the train sits on moves to the right while the train is on it. Next is a multi-pass launch, where the train is launched back and forth multiple times, building up speed. The train is first launched back-

wards at a slow speed over a small hill and towards a vertical spike.

The train turns 180 degrees to the right and ascends 68 metres (223 ft) up the lift hill, travelling at an angle of 47 degrees. It then turns 90 degrees to the right, halting at the 90-degree drop, where a portion of the train hangs slightly over.

The track then makes a small dip before navigating a 180° turn to the left. A second small dip is followed by another ascent before making a second 180° turn to the left.

The train seats 28 riders in an inverted roller coaster design.

The train continues into a left turn and a heartline roll inversion.

It turns 180 degrees right and exits the mountain, leading into a 18.3-metre (60 ft) chain lift hill. Once at the top, the train enters a shallow, right-hand drop, crossing over this section's first drop as it runs along the side of the mountain.

The ride enters the 20-metre (66 ft) tall vertical loop, where riders experience 4.3 G's.

After a series of small airtime hills and slight curves, the train cut through a short tunnel, followed by a helix up and to the left.

The 1,153 m (3,783 ft) roller coaster features 14 inversions, 1 zero-g roll, a fly-to-lie, 2 raven turns, and a half camelback twist.

Without enough speed to make it anywhere near the 33.0-metre (108.3 ft) peak, the train rolls back and is accelerated backwards by the launch, passing through the covered section and rolling partway up the dive loop. The train rolls forwards once again and is launched for the third time through the covered section at 95 km/h (59 mph), gaining enough speed to crest the twisted top hat, passing over top of the Monorail and into a "sling loop" (which is seen as a slightly extended Sidewinder inversion). The coaster navigates a camelback airtime hill and a right-hand overbooked helix (crossing over the Monorail once again), before entering the dive loop into the launch area once again.

The rollercoaster currently continues to go backwards after the New Year's Day event.

As the train exits the second horseshoe, it swoops down and begins a 270-degree helix to the right, which leads into this section's second inversion, an inline twist. The inline twist sees riders perform a full rotation around the track, starting from a position where they are facing downward.

Riders then pass by a semi-indoor scene featuring Hagrid, who says he will meet up with the Muggles at an abandoned hut at the edge of the Forbidden Forest. Once the bikes reach Hagrid's hut, a Blast-Ended Skrewt sprays smoke at the riders.

The train makes a left turn into the lift hill. After reaching the top of the 33.0-metre (108.3 ft) lift, the train drops back to the ground before making a sharp left over-banked turn. The train then enters a corkscrew followed by a slight upward left turn. The train goes back down slightly, also going through a set of trim brakes.

The car makes a 90° turn to the right and is carried to the top of a chain lift hill. The car then descends an 18 metre (60 ft) drop at an incline angle of 36°. The car then traverses several drops, banked turns and helixes. This section has top speeds of 64 km/h (40 mph) and is 450 metres (1,490 ft) long.

The break run consist of a block brake, a slow right-hand turn, a storage track and a straight with friction wheels to feed into the original break house.

The ride goes up a 30-metre (97 ft) lift, then drops 29 m (96 ft) at a 47-degree angle. After the first drop, the car goes up a hill and then makes a left 180-degree turn. The car drops back down another hill, goes up a small hill, and then up a larger hill, making another 180-degree turn. After the turn, there is another drop and then the track makes a right turn ("dog leg"), going through several bunny

hills before another left 180-degree turn.

The ride officially begins as trains plunge down the 62 m (205') first drop at an angle of about 60°. At the bottom of the drop, the ride reaches its top speed of 112.6 km/h (70 mph) as the trains briefly pass through a covered section of track, ascend the first large camelback airtime hill, and descend down another drop, this time veering to the left. They then climb a smaller slope and come down into a helix section, still veering left. The train then crests the second camelback airtime hill, followed by a drop veering to the right into another small helix; riders then briefly enter into the midcourse brake-run.

From here, the track makes an uphill right turn into the midcourse brake run. Off the midcourse brakes, the track drops into a the ride, and finish off with a figure-eight turn. The train then enters the trim brakes, and then makes a right turn, passing through the transfer track.

The train enters a short drop into a left-hand U-turn toward the 25-metre (82 ft) lift hill. It then descends the first drop, turns slightly left, and traverses two airtime hills. A larger hill leads into a wide banked left turn and a small drop. The train then turns slightly right and descends another large drop before entering a right-hand U-turn.

The train quickly navigates a back-to-back pair of Immelmann inversions. The train exits into a Roll over (two inversions).

It enters a small dip and into a short U-turn followed by a 21-metre-tall (70 ft) chain lift hill. When the train reaches the top of the hill, the train slopes down and into a 90-degree banked turn.

After the front gates open, the trains continue and head into its maximum vertical angle drop of 90-degrees. Descending the 27 m (90 ft) drop at its maximum speed of 81.0 km/h (50.3 mph), the trains enter a dive loop, head upwards into a left banked turn and

then a right-banked U-turn. Next, the trains enter the first linear synchronous motor (LSM) launch section where they stop momentarily before accelerating to 76 km/h (47 mph) in 3 seconds. The trains then enter a banked left horseshoe before continuing down left into the 29 m (95 ft) vertical loop.

The train makes a 180 degree turn leading to the 30-metre (100 ft) chain lift hill. Once at the top, the train goes through a predrop before making a sharp left hand turn leading into the first drop. Once the train is at the bottom of the first drop, it immediately goes through the first of two inversions, a vertical loop. The train then makes an upward right turn before making a left turn back down to the ground.

When the train reaches the bottom of the drop, it reaches its maximum speed of 104.9 km/h (65.2 mph) and enters a vertical loop, followed by a dive loop. After making a slight turn to the right, the train makes a banked right turn through the loop it passes through before. The train passes through a trim brake (to slow the train down) before entering a zero-gravity roll. The train then makes a banked turn to the left leading into the first of the interlocking corkscrews. After a banked turn to the right and a slight turn to the left, the train goes through the second corkscrew.

The train makes a left U-turn and ascends a 70-metre-tall (230 ft) lift hill. After cresting the top, the train drops 66 m (215 ft) at a 68-degree angle, reaching a maximum speed of 130 km/h (80 mph). The train then ascends a 58-metre-tall (189 ft) hill and dives down to the left, coasting over another large airtime hill. Afterwards, the ride enters an element known as a hammerhead turn, a tight U-turn, which veers to the right.

Dipping up and crossing over the Rio Grande Railway, riders plunge into a drop and navigate several mostly-low to the ground

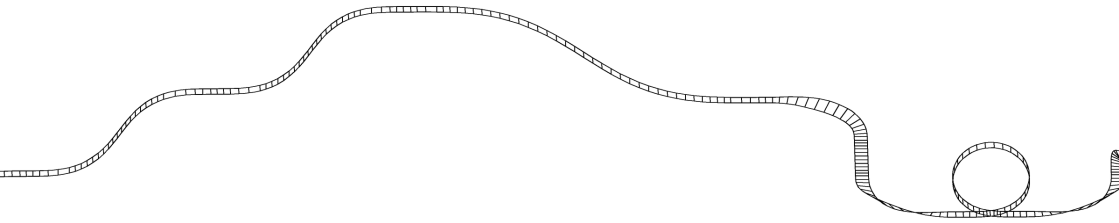
turns and helixes, interacting with thematic elements along the way. Shortly after the turn, riders hit the Brake run, where the train cars are righted.

As the train continues, it makes a small turn right followed by a small turn left.

The train makes a short dive and then banks left into a flat right turn. The train then maneuvers into a shallow drop, immediately followed by a corkscrew.

The train makes a highly banked right turn into a flat spin. The train makes a left turn extremely close to the ground, before entering a moment of airtime going into the brake run. After exiting the brake run, the train makes a right turn into a second, shorter set of brakes.

The third banked turn wraps in parallel to the first, but crosses over it and into the brake run. At this point riders will hear John Williams' famous score from Superman.



Chinese Tradition

In Chinese tradition, the number 8 signifies good luck and the number 4 signifies death, as the numbers are homonyms for characters representing luck and death, respectively. Entering tunnel 4, the train begins to climb a lift hill in the style of a traditional roller coaster. Once the train nears the top of the mountain the lift cable appears to break (next to Shaft 44, in reality the lift cable ends) causing the riders to go back down the hill in reverse and along a different track before entering another mine. Two animatronic bears, a mother and cub named Mother Lode and Nugget, are located inside the mine and the cub sets off some TNT while reaching for a hanging catch of fish left behind by one of the miners, causing the train to launch out of the mine on yet another track. The train then goes through a variety of twists and turns before arriving at the terminal brake run.

The ride proceeds into a small curve drop into a curve then a

drop into several camelbacks. After the camelbacks the train goes through a tower and navigates through Crazy Charlie's fireworks testing range before hitting an s-bend and entering Charlie Cherribaum's Gasoline and Fireworks Emporium, through the storage warehouse. Charlie panics, saying he sees smoke, but then dismisses it as a result of his paranoia as a spark ignites a box of firecrackers and a few tanks of gas. The fire catches Charlie by surprise, and he tells the volunteers (riders) to watch out for Big Bertha and to leave the building; as he says this, a large firecracker labeled "Big Bertha" falls into a position where it points directly at the train, and the fire ignites the fuse, which begins to throw sparks. As the fireworks start to go off, the room fills with smoke and a series of loud bangs are heard as the train accelerates backwards onto a different track, where it navigates a few more turns and passes underneath another fire tower before hitting the trim brakes.

The ride then enters a wide, highly banked helix to the right before perpendicularly crossing over the launch track at a height of 6 metres (20 ft). The ride then dips to the left and an on-ride photo is taken. The track then levels out. Upon entering the tunnel, another set of friction wheels are activated to launch the riders to the faster speed of 70 kilometres per hour (43 mph). The opposite end of the tunnel is covered in mist preventing riders from seeing the approaching track.

The train continues and immediately enters a dark tunnel. A siren and loud booming sound effect can be heard before the linear synchronous motor (LSM) launches the train straight up the 34-metre (110 ft) vertical launch hill at 66 km/h (41 mph). The launch is in two parts, with a short boost launch inside the tunnel and a longer, vertical launch straight up the top hat element. After cresting, it descends vertically down the other side of the top hat

element, accelerating to a maximum speed of 89 km/h (55 mph).

Riders enter a cobra roll. The train then rises uphill, makes a left turn under the lift hill, and enters the mid-course brake run. After the brakes, there is a small drop into a pair of interlocking corkscrews. Following the corkscrews, the train completes a 135-degree curve to the left that dives into a 270-degree curve along the ground, entering the trim brakes.

The ride makes two turns before riders climb the 17-metre (55 ft) lift hill. After the 15-metre (50 ft) drop, the ride turns and goes up a bunny hill. It is followed by many bunny hills, a tunnel, and even banked turns. Shortly after, the ride hits the brakes.

The train continues gently dipping to the left into a leisurely winding S-curve. Exiting the S-curve, the train enters the first linear synchronous motor launch, accelerating through a stone archway into the event building and cresting an airtime hill. This is followed by a dive into a series of high speed turns and a helix that guides the train onto a brake run, where a psychedelic visual of leaves with wind-blowing sounds are on display.

The train goes through a 30-metre (100 ft) tunnel next to the Wonderland Terminal, then curves upwards into a 50-metre-tall (164.0 ft) overbanked banked turn to the right. This is followed by another drop that enters a speed hill, which makes a high-speed turn slightly to the left at approximately 122 kilometres per hour (76 mph). The exit of the first high speed turn leads directly into a 56-metre-tall (184 ft) camelback, followed by a 44.8-metre-tall (147 ft), 115-degree hammerhead turn. Both of these elements are located above the guest parking lot and in front of the park's main entrance. After leaving the hammerhead turn, the train enters a second high-speed curve at approximately 96.5 kilometres per hour (60.0 mph).

One of the two helicopters is shot down, dropping riders into a

reverse freefall. Evading missiles, the car is launched backwards through a vertical loop and helix before reaching Maranello. The surviving helicopter appears in the town's plaza, and this section's spokeswoman proclaims the rider's deaths. A barrage of missiles are fired but ultimately the helicopter is blown up, the blast of which plunges riders sideways into Ferrari's underground factory.

The train enters a 30-metre (100 ft) vertical loop, then racing straight through a shallow pit towards onlookers, followed by an Immelmann loop. After the Immelmann, the train curves slightly right into a zero-g roll. The train continues through The Hollow, passing over walkways and heading towards the ride and the ride, then making a sharp left turn over Spring Creek. Due to the presence of Spring Creek, the train passes over several unconventional support structures wrapping under and around the train.

The park can adjust the degree of spin based on rider input, which will help them regulate the experience to average rider tolerance over time.

This section features a prototype seating arrangement to ensure an unobstructed view for every rider.

Each of the three towers featured a single floorless gondola seating eight abreast.

This section has a total of three trains, each containing seven cars. Riders are arranged four across in a single row for a total of 28 riders per train.

The train on this section uses a single lap bar to hold riders in the seat. This design works because the forces from the loop and helices press riders in their seats, and are a common feature with Schwarzkopf looping coasters.

On hot summer days, the park turns on this section's exterior waterfall, which falls from the exterior skull's eyes. It pours down

into the water below, wetting guests in this section's queue. While most of the park's coasters will operate in light rain, the ride remains open during thunderstorms, since it is enclosed.

At this point riders on the right hand side of the train experience a near miss with the water surrounding the helicopter. This is followed by a corkscrew, which passes through the inclined loop. Upon leaving the corkscrew, the train curves past a crashed fire engine with water and fire effects and proceeds through a keyhole: a partly destroyed church tower.

Partway through a drop, riders encounter a heartline roll, this section's first inversion.

It then goes through a left-handed curve. A smaller airtime hill leads to a 270-degree helix. Riders then go through a right turn and a left turn before hitting the brakes. The train slowly turns 90 degrees to the right, passing by the transfer track.

Trains then go through the double loop, the first of its kind in Vietnam. Then comes a double corkscrew, inverting the riders twice.

The turn continues with a short dip followed by an ascending 90-degree turn to the left. At the crest of the turn, the train descends again through a slightly banked turn to the right.

The train continually descends into a 38-metre (125 ft) tunnel, emerging in an upward right-banked turn. The train continues downward into a right turn, ascending a few hills before banking into the left turn.

The track leads into the main coaster section, with the train descending through a left-hand turnaround and an upwards clockwise helix. Entering the mountain, the train navigates another turnaround around the drop tower before entering the first gameplay section. The cars then rotate riders to face backwards during the next coaster section, consisting of an S-bend track leading

down to the second level of gameplay. During the next scene, the train is directed towards one of two physical endings; a drop track or a sliding track, both of which descend back to ground level. On the slide track, riders can be sent down facing either forwards or backwards.

Riders will reach a top speed of 40.2 kilometres per hour (25.0 mph). The background blends in with this section's Road Runner theming.

The train enters a straight section of track. The track then shifts to the right until it lines up with the launch track. At this point a door in front of the track opens. The train then begins to launch forward and travels part of the way up the first top hat.

It then immediately enters a 50-metre (165 ft) Immelmann loop before rising into the mid-course brake run. The train then drops 40 m (131 ft) and enters a dive loop, followed by a third inversion with a 270-degree zero-g roll. Riders experience an airtime hill and turn left before entering the trim brakes.

The turnaround starts in the second half of the ride. Diving off a straight section of track, riders descend a steep drop, making a left-hand turn, rising over an airtime hill, before making a right-hand turnaround underneath the turnaround in the first half. After rising over another airtime hill, the trains enter a 450-degree downward helix to the right, before rising over a hill and hitting the trim brakes.

The train makes an overbanked right turn and through a vertical loop followed by a sidewinder, the train travels through a double-up and into the trim brakes.

It then turns left, climbs an airtime hill, and then turns right. It then goes into 2 further back-to-back airtime hills, turns left, two more airtime hills, a slight left-hand turn, an airtime hill, a left-hand turn, and into the trim brakes. It runs for about 3 minutes,

including the lift hill.

The first drop is 65 m (213 ft) and is sloped at 89 degrees, at the base of which ride vehicles attain a maximum speed of 126 km/h (78 mph). During the initial drop, the seat assembly is rotated so that riders are positioned facing the ground. Similar to the ride, the seats are then rotated forward 360 degrees one to three times, simulating a front flip as the train descends the drop. The train then enters an inside raven turn, where the cars are rotated again halfway through the loop to create a "lie-to-fly" maneuver; however, unlike the ride, the seat assemblies rotate backward 360 degrees, simulating a backflip, on the top of the raven turn, before riders transition to a prone position, facing forward. After exiting the raven turn, the trains traverse through a zero-g roll.

The train makes a 180-degree right turn and climbs 37 m (120 ft) up a chain lift hill. The train turns 90 degrees to the left after reaching the top, then descends 47 m (155 ft) down the first drop.

Exiting the helix leads riders into a double inline twist that heads back toward the lake. The train curves again, dips, and rises up into the first brake run.

From there it turns right and climbs the first lift. After clearing the lift hill, the train turns right and descends into a trench while completing a U-turn. This is the "figure 8" area of the track. It rides through the "b" dips featuring 2 camel back humps. The train turns left then right and approaches the second lift hill.

Once the car is released from the top of the hill, it falls down towards the ground and enters a dive loop, an inverted top hat, and the seventh inversion, an immelmann loop. This section is approximately 2 minutes long.

Intricate rock work and foliage surrounds the track as trains pass through. The train turns to the left as it dives under itself and snaps back to the right, where the on-ride camera is located.

Riders continue before taking a sharp turn and up a 14.0 m (45.9 ft) chain lift into multiple 180° hairpin turns. Riders go up a small hill, take a sharp turn, and go down the biggest drop on the ride. Riders go up a hill and turn into two bunny hills, the second of which features this section's on-ride photo. Riders then turn into one more dip that leads into the brakes.

The vehicle does a sharp turn left onto the chain lift hill. The ride then reaches its highest point before a pre-drop, turning left and descending down the main drop. Riders ascend into a sweeping left hand turn, closely followed by a straight track and a second left hand turn. This section's train heads close to ground level, traversing a helix, followed by two inversions – a reverse sidewinder followed by a vertical loop. The ride ascends into a short straight track, before making a sharp turn to the right and entering the brakes.

The train turns 180 degrees to the left, then begins to ascend its 68-metre-tall (223 ft) chain lift hill. Upon reaching the crest of the hill, the train turns right and pauses for several seconds as it slightly hangs over the edge of the first drop. The train then suddenly releases, descending 65 m (214 ft) at a 90-degree angle and reaching a maximum speed of 121 km/h (75 mph).

The train whips into a corkscrew, and then two wide turns, first to the left, then the right, skirting around this section's vertical loop. Following another slight curve to the left, over the Coal Cracker flume ride, riders photos are taken and the train enters the trim brakes.

The train makes a right hand turn, entering the 40-metre (130 ft) lift hill. Upon reaching the top of the lift hill, the train enters the first element of the roller coaster, a dive drop. This element is similar to the dive drops found on the ride at Six Flags Great America and the ride at Thorpe Park, and it consists of the train

rotating 180 degrees into an inverted position before descending back to the ground. During this drop, the train reaches its maximum speed of 100 km/h (62 mph).

It then travels up to a height of 46 m (150 ft) where it completes a non-inverting half-loop, a heartline roll before exiting in a second non-inverting half-loop. It then enters a full non-inverting loop.

The train makes a left turn towards the 51-metre (167 ft) chain lift hill. After reaching the top, the sound of a banshee screaming is played as the train drops into a heavily banked right turn. After reaching the bottom of the drop, riders enter a dive loop immediately followed by a vertical loop around the lift hill.

It then dives down a 34 m (112 ft) drop back into the ravine, and rises up a third hill. At the top of the hill, the track makes a level right hand turn into the midcourse brakes. The track then takes a 31 m (103 ft) dive off the midcourse brakes, followed by a 22 m (72 ft) tall turnaround. Following this, the track makes a right turn, passes under the lift hill, and over a pair of smaller airtime hills with slight right turns, before making a 180 degree banked turnaround, leading into a bunny hop and the trim brakes.

The layout of the ride is a unique 90 degree curved out and back that follows the shore of a small lake. The train coasts down a small incline, around a curve, and heads up the lift hill to the top of the first drop. This drop, which is the largest in the ride, can provide for some air time depending on where riders are seated on the train. The second drop is considerably shorter but leads into the third drop, which contains a "head-chopper" moment at the bottom as the track ducks under some support structures for the return track, which is immediately to the rider's left at this point in the ride.

This $7 million GCI wooden roller coaster, which is identical to the ride at PowerPark, features an 24-metre (80 ft) drop with a top

speed of 77 km/h (48 mph), going through a course of 820 m (2,700 ft) of track. This section features 16 hills and multiple high banked turns at up to 67° angles.

The car goes into a banana roll, a corkscrew, and two airtime hills. The ride is slowed on a set of block brakes.

Water nozzles that spray mist are positioned under the pretzel-shaped helix, although these do not always run. After exiting the helix, the ride dips and rises once more, turns left and then right, before hitting the trim brakes.

The train completes a 180-degree right-hand turn, dipping through a smoke-filled tunnel with fire-effect lighting while sharply pivoting 180 degrees to the left.

After a 24-metre (78 ft) climb up its lift hill, the train descends 22 m (72 ft) at a 45-degree angle reaching a top speed of 64 km/h (40 mph). Riders then enter a series of two short hills which provide this section's maximum airtime followed by a larger, third hill that slows the train slightly. After the next drop, the train climbs into a 180-degree turn that sends riders over a short hill followed by three medium-sized hills on its way back.

The roller coaster is a boomerang ride wherein the train is taken backwards up a hill. At the top of the hill, the train is released and picks up speed.

The train moves forward and briefly stops. Mist blows in front of the train and thunder sounds as the train launches from 97 km/h (0 to 60 mph) in 3.5 seconds by linear synchronous motors. The train immediately enters a 43 m (140 ft) Immelmann loop. Following the Immelmann loop, the train then descends its first drop into a 38 m (125 ft) vertical loop.

A 180-degree right-handed turn follows, sending riders into the third inversion, a vertical loop. The train then makes a left-turning stall manoeuvre into a second corkscrew.

The instant that the train emerges from the tunnel it banks sharply to the right. This is followed by four quick, directional changing turns between rocks and past an animatronic sea lion. The track then elevates off the ground level to cross over the track a second time before entering the brake runs.

The train takes a sharp right turn as it continues and enters a dark tunnel. Upon reaching the end of the tunnel, the train ascends the 78-foot lift hill. After reaching the top of the lift hill, the train drops down a 72-foot initial descent before entering the first turnaround. Following a leftward turn, the train descends to ground level and traverses two smaller airtime hills before ascending into the second turnaround.

At the base of the first drop, the train reaches its maximum speed of 92 km/h (57 mph). Following the lift hill are two camelback hills. In both, riders experience "ejector air-time". The train then jumps a bunny hop and another camelback hill and a double uphill, before turning around for the second half of its course. After the turnaround, the train hits another bunny hop and another double uphill. After the double uphill, there is a series of six bunny hop hills.

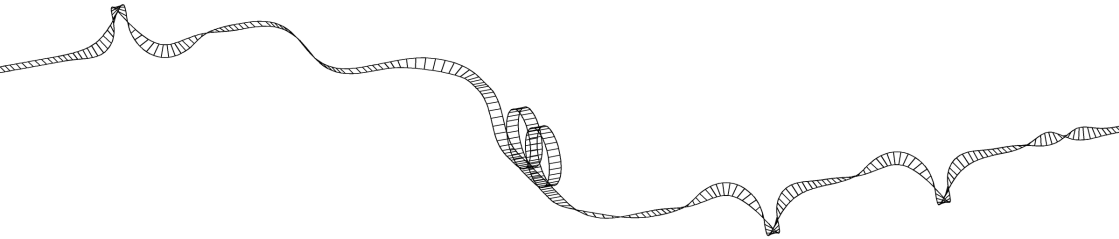
The layout and ride experience is the standard Vekoma the ride roller coaster design found at forty-three different amusement parks worldwide.

After going through the tower, the train goes through a near miss keyhole surrounded by a cloud of fog, before exiting the twist and making a tight right turn, passing by this section's corkscrews and turn, before making a slight left upward turn leading into the Main Brake. The train then makes a 180-degree left turn into the slowdown brake and the transfer track.

A projection mapping sequence appears where the ride is evidently displeased at trespassers in her temple and prepares to turn

the riders to stone. The train is suddenly launched backwards and blow past the lift hill. It then descends a curved drop and speed along a series of low-ground twists over the Romus et Rapidus before hitting the trim brakes.

After crossing under the blue track's zero-g roll, it then goes up a double up and turns left, where the train hits the trim brakes. Because this section is a racing coaster with one long continuous circuit, it may be considered a Möbius loop roller coaster.



Cavern Filled with Destroyed Bobsleds

Then, the turn leads into a vertical loop 32 metres (105 ft 0 in) high, which is the tallest loop on any launched roller coaster in Europe. One turn culminates in a mid-course brake run; and, exiting the brake run, the train completes a twisted horseshoe roll that partially passes over a lake. One airtime hill follows, and the ride continues with a heartline roll and trim brakes.

The train turns right and ascends the 15-metre (50 ft) chain lift hill. At the crest of the lift, the train descends briefly before climbing into a banked left turn leading to this section's main drop.

The train (the front part of the roller coaster looks like the locomotive of a steam train) continues, takes a 180 degree right turn, goes straight forward, and makes a 180 degree left turn. The train enters a right turn helix, followed by a tunnel. The train takes a 270 degree left turn. After going straight the train takes another 90 degree right turn drops into a short tunnel. The train goes up

the lift hill, out of the tunnel. There is a 270 degree left turn.

This is different from a traditional Möbius loop layout (such as Racer at Kennywood), which takes riders through only half of the ride's "complete" circuit.

The ride continues with a sharp left-hand turn, followed by a descending turn leading to the 63 m (206 ft) lift hill. As the train crests the hill, it slows down before descending its 91-degree drop, reaching its top speed of 122 km/h (76 mph). The train then climbs a hill and banks in the opposite direction making an outer-bank turn, returning downward in a left-hand turn and upwards to the right. Reaching the apex of the hill, the train navigates through a barrel roll downdrop, followed by an overbanked turn to the left.

This section features a 90-degree, 60-metre-tall (197 ft) vertical drop that exits into an Immelmann loop. After the Immelmann, riders swoop down to the ground then climb a curve up to the block brake, before going over a second near vertical dive dropping into a tunnel. Upon exiting the tunnel, riders make an overbanked turn into a water splash feature which exits into a bunny hill then into a helix before entering the trim brakes.

Once the train is loaded and secured, the floor is retracted and the train continues. A U-turn to the right leads to the 46-metre-tall (150 ft) chain lift hill. Once riders reach the top, they go through a small pre-drop before dropping 43 m (141 ft) to the right. The train then reaches a top speed of 101 km/h (63 mph) and enters the 39-metre-tall (128 ft) vertical loop followed by a turn to the left into the 29-metre-tall (96 ft) dive loop.

After a mid-course brake run, the train descends into a left-hand drop under the brake run, followed by a corkscrews, after which the on-ride photo is taken. The train then travels through an inclined helix to the left before entering the second corkscrew, a right-hand helix, and an S-turn and into the trim brakes.

The train dives down to the right transitioning into a brief straight section of track. Riders then enter the first of two flat spins which rotates the train 360 degrees to the left. The track straightens briefly again before turning to the right and entering a short dip before taking riders into the second flat spin.

The roller coaster takes passengers on a test flight of the Hippogriff over the Forbidden Forest and Hagrid's Hut. Riders are instructed to bow to the Hippogriff before the ride progresses up the lift hill.

The track then drops to near ground level and enters a vertical loop.

It then goes through a small tunnel and enters the vertical loop. Riders are then transitioned back to a prone position after entering the following "Lie-to-Fly" element, which travels over water. Riders then experience a double inline twist element, and a helix that turns to the left.

Guests then "plunge" down the past-vertical drop at 97 degrees, and ride through: a vertical loop at 40 miles per hour, speed hill, cutback overbank turn, heartline roll, mid-course brake run, overbank helix, an additional brake run, then.

Ride operators and promoters engaged in incredible levels of exaggeration regarding the speed and dimensions of the ride.

The sudden brake right out of the turn causes the cars to swing briefly even after the train has stopped moving forward.

On-ride photos and the ride merchandise can be purchased here.

This section has an oval-shaped twister layout with multiple levels. It is modeled after the ride, being a near mirror image, though with an extra double down element and a maximum height of 30 m (100 ft) compared with Cyclone's maximum height of 23 m (75 ft). Its layout features multiple banked turns, eleven

drops, and nine moments of air time.

A sidewinder is similar to an Immelmann loop however it features a half loop followed by a half corkscrew (rather than an inline twist).

Modern looping roller coasters use teardrop-shaped loops to greatly reduce these g-forces.

Although it is not beyond vertical, many sources note the 90 degree drop on the ride. The coaster also has a total of two inversions—both dive loops. It is also noteworthy for it sometimes banking toward the outside of turns. Guests ride the coaster in individual four-rider cars containing two rows of two.

Diabolical's Cliffhanger is 46 m (150 ft) in height, reaches a maximum speed of 97 km/h (60 mph), and a track length of 762 m (2,501 ft). This section features a 95-degree (a beyond vertical drop), as well as a second 23-metre (75 ft), near-vertical drop.

After a brief straightaway, riders experience a weightlessness maneuver in a zero-g stall inversion. This inversion travels over Kennywood's lagoon, part of the park's National Historic Landmark designation.

After ascending the lift, the train goes through a slight right-hand drop into the pretzel loop. After the pretzel loop, you do a slight left-hand drop and an inline twist into a right-hand drop. After the drop, the train goes through a zero-g roll and into the midcourse brake run. After the midcourse, the train goes through a right turn and a drop towards a pond, with fountains going off as it skims the pond. After, the train goes into a right-hand turn into an inline twist, into a left turn, and into the brakes.

This section operates with single car trains.

At the same time, two bellows actuators fill up with air. Meanwhile, near the back of the train, two drive tires that are holding the train in place must retract.

The sound of creaking wood is heard before the horizontal track section jolts, then free-falls vertically (together with the train and floor section) to a lower level. Here several statues light up, before the train moves backwards out of the building and descends a short drop into an underground tunnel. The train completes a backwards helix in the tunnel, then re-emerges outside and stops.

During normal operation, the track is inspected every two hours.

The ride currently operates with 4 trains (2 on each side).

It measures around 140 by 30 metres (459 by 98 feet), and the highest point of the track is 21 metres (69 feet). The track length is 982.40 metres (3,223.1 feet), which makes this ride the second longest scenic railway in the world. The track is wooden and the running rails are in a trough with walkways either side. The ride is often called a side-friction coaster, which is misleading as the train makes no contact with the side walls of the track and the running wheels are flanged like those of railway vehicles.

Both are veteran attractions from the opening day, and have shared the same plot of land for almost a century.

According to the Gerstlauer website, the manufacturer has two different models of its spinning roller coaster collection, as well as one custom option for a park to fit its desires. The five Pandemoniums make up both of the two, the 400 and the 420. Each model has different statistics, so the Pandemoniums range in height from 16 m (52 ft) to 16 m (54 ft) with the same speed of 50 km/h (31 mph).

The ride is featured in an area of its own, with guests having to cross a bridge over the Pangea attraction in order to access the queue. Around the attraction itself, there are a few comic strip-esque signs around the attraction, showing the ride story. The queue itself is very lightly themed.

Once the bobsled car has disengaged from the chain, riders

pass through a dark tunnel, where the Abominable Snowman's roars echo. The trains emerge into a cavern filled with destroyed bobsleds, skyway buckets, some climbing equipment, and a crate stamped with "Wells Expedition" may also be seen. Both tracks take a left-hand turn, then split off before they can crash into the first animatronic. The cars swirl around the mountain, dipping under tunnels and by the waterfalls. Each track then passes another animatronic of the Snowman.

The route is divided into two areas, a dark ride section, in which the ride starts and ends, and the main part is an outdoor route.

This section has one train, which has 10 cars with 2 seats on each car catering 650 riders per hour.

Blordak urges guests to seek shelter inside one of the vaults. After entering the vault, the walls come crashing down and Nagini appears in an attempt to attack guests. Bellatrix reappears, this time with Lord Voldemort, who confronts guests and questions them on their knowledge of Harry Potter's whereabouts. He blames Bellatrix for the vault infiltration, who again decides to torture guests sending them to a lava-filled chamber deep in the vaults.

This section is designed for families and children, featuring a minimum height requirement of 97 cm (38 in). The two-across, ten-car trains feature an individual T-shaped lap bar for each rider.

This section's trains are themed to the character widely known in the Batman comics. Each train has six cars, each seating four people with two across in two rows, for a total capacity of 24 riders per train. As the train continues, it enters several small 'bunny hills' and turns, reminiscent of pre-lift elements on the ride at Six Flags Magic Mountain.

The beginning of this section features two large drops, the sec-

ond of which is a 'headchopper' where the train dives under some of the support structure of the ride. There are also other drops and a bunnyhop sequence which gives very good air-time both at the front and back of the train. In total there are nine drops. A typical ride time is 3 minutes 20 seconds, but can vary depending on the styles of the particular brakesperson.

The train rotates 360 degrees in an inline twist and passes by the two keyhole towers. The track veers slightly left as the train enters the mid-course brake run. This is followed by another hill that dips low to the ground and a 360-degree downward helix.

Once the train ascends the 48-metre-tall (156 ft) lift it goes through a small pre-drop, before dropping 45 m (146 ft) on a banked left turn. It then enters the 38-metre-tall (124 ft) vertical loop which wraps around the lift hill. Two dive loops follow, the second of which also wraps around the lift hill. An inclined loop is followed by two right turns that lead up and into the mid-course brake run. After dropping out of the brake run, the train immediately enters a corkscrew to the left.

The train drops a bit and curves and makes the 66 ft. drop, entering the double loops. Afterward, the train turns right into the double corkscrews.

The train then enters a zero-g roll followed by the cobra roll, a first for inverted roller coasters, which inverts riders twice. After the cobra roll, the train makes a slight right turn followed by an upward left-hand helix and enters the mid-course brake run.

The exit from the brake run led into a pair of interlocking corkscrews.

After going up the 40-metre (130 ft) tall lift hill, the train goes down a 87° drop. This leads into a dive loop and then an airtime hill with 180° stall. The train next goes up into a turnaround, which drops down into a zero-G roll.

The 97° drop makes the ride the steepest dark ride rollercoaster in the world, and the steepest roller coaster in Germany. It is the first combination of a Euro-Fighter and a Launched roller coaster.

The cart quickly drops beneath the grinder and whips around a hairpin curve. As the car slows to a stop, a crow caws loudly from somewhere above and a small lever labeled "mine gas" cranks into the on position. The car ascends up a vertical wall of track at a 90-degree angle.

The train drops a short length into a dip that leads into the tallest hill of the ride. After cresting the second hill, the train drops 165 feet and reaches a maximum speed of 117 km/h (73 mph). A climb to the left follows, and the train enters the first of several steeply banked turns. It then rises to the right, snaps to the left, and then back to the right once more. This is followed by a right-hand turn, another snap-left-then-right sequence, and a double-up climb.

This section has two tracked sections and three floating sections: Block brakes are placed after almost every element of the tracked sections, to ensure a small minimum distance between individual boats.

This section's layout is vastly similar to those of Vekoma's 335m Junior Coaster layout. After making a right hand turn, the train is pushed up the lift hill by booster wheels, all the way to the 48.6 foot height.

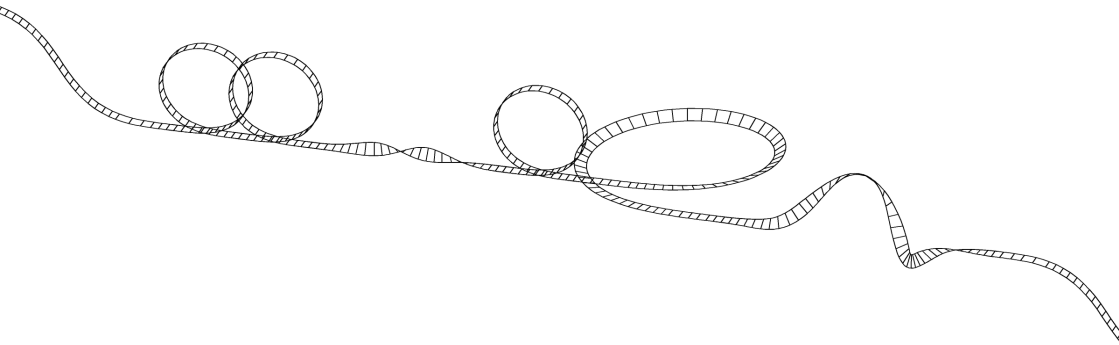
This way a capacity of 1690 guests per hour can be ensured, despite each boat only holding 8 guests. The total time it takes to ride the coaster is around 5 minutes and 35 seconds in the front boat, and around 6 minutes in the second boat, with the park's own website citing 5 minutes and 50 seconds.

The ride' theme is based on the several volunteer-based fire departments surrounding the Great Smoky Mountains National

Park in the 1940s.

The train makes 2 circuits per ride cycle. This section's track and supports are painted blue and there is a large amount of marine styled theming, which reflects the aquatic theme of The Octonauts. This section's minimum height restriction is 90 cm and unaccompanied by an adult, it is 110 cm.

In a figure 8 track layout, the track crosses over itself, forming an "8" shape. The double figure eight is a variation where the track contains two separate figure-eight shapes. Some examples include the Cobra at Six Flags Discovery Kingdom and the Harley Quinn Crazy Train at Six Flags Great Adventure.



Corkscrew

This section is located in the Underland Zone area of the park, which is themed around a mining town where ogres live. The entrance and exit of the coaster is decorated to simulate a jewel mine, and there are depictions of Lorry, one of the theme park's mascots, going on a mining trail with ogres in the inside walls of the waiting area.

Riders enter the ride from the Streets of New York section of Movie Park Germany. The outside of the building is themed to a 1950s American garage with the storyline taking place in the 1960s. The queue line and pre-shows inform guests that they are touring the ride.

The train consists of five cars. Each car has two rows, each row seating two riders, for a total of 20 riders per train. The trains also feature buzz bars.

A series of magnetic trim brakes engage and slow the train as

needed, followed by a turn to the right through a heavily wooded area. Veering left, the track enters a second tunnel that spans 82 m (269 ft). A quarter of the tunnel is underground, and the train exits above ground gaining speed and veering right, which leads into another hard right turn on a slight incline. The train descends downhill, rises uphill, and then passes over a set of trim brakes before entering the second lift hill.

These brakes lead into a left turn with reverse or outward banking. It then drops under the block brakes and into the first inversion – a Dive Loop. The inversion begins with half of an inline twist which transitions into half of a vertical loop. Upon exiting the inversion the track inclines into a second set of block brakes.

There are five trains, although only three are currently in operation. Each train is made up of three cars, each of which has five 2-person bench seats, allowing 30 riders per train. There are manual lap-bars and grab rails for each seat. The brakesperson rides between the first and second cars.

This section's single seven-car (28-passenger) train runs along a 200 m (656 ft) U-shaped track, incorporating two 56-metre (185 ft) vertical spikes. The forward spike incorporates a twisted spiral, and the rearward spike provides a straight freefall. The 20 m (65 ft) train, propelled by linear induction motors (LIMs,) is accelerated in less than four seconds to 109 km/h (68 mph) toward the forward tower before dropping back down through the covered section house and up the rearward tower.

The five Premier LIM catapult coasters share a similar layout and have the same technical specifications.

Once the trains are loaded, they move forward to a holding area, where a lap bar check is made and so the train may wait its turn to start the circuit. When the rocket is released, the red strobe lights surrounding the train flash and it makes a right turn, be-

fore climbing the first lift hill. At the top, two strobe lights flash on and the rockets enter a tunnel of flashing blue lights to signify the transfer of power to the trains.

The car enters the trim brakes.

This section has one of the highest capacities of any ride in the park.

Throughout the course of the ride, there are multiple "foot choppers" (support beams that draw near to the rider's feet, giving the illusion that the rider's legs are going to be chopped off, hence the name "foot chopper"). This section is 395 m (1,295 ft) long.

As with recent SeaWorld installations, the attraction endeavors to educate while entertaining and to promote the park's core message of conservation and preservation.

This section has the same layout as Insane at Gröna Lund in Sweden.

This section's single seven-car (28-passenger) train runs along a 200 m (656 ft) U-shaped track, incorporating two 57-metre (186 ft) vertical towers. The forward tower incorporates a twisted spiral and the rearward towering provides a straight freefall. The 20 m (65 ft) train, propelled by linear induction motors (LIMs), is accelerated in less than four seconds to 110 km/h (70 mph) toward the forward tower before dropping back down through the covered section house and up the rearward tower.

As virtual reality allows for several modifications and extensions of the actual track layout, the size of the virtual reality track can be much larger than the real one. This of course means that speeds can be much faster and heights much taller, as these aspects also grow with the increased dimensions.

After a right turn, the train goes through a block tunnel and slows down. It stops at the second launch and the countdown begins. Riders are then launched through a series of GO! rings

as the train passes the on-ride camera, which takes photos of the guests. The train makes its way through an airtime hill and a left turn.

It uses GCI's Millennium Flyer trains. The ride continues with a slow turn into the 147ft lift hill. At the top, it traverses a 270° right hand turn before heading down the 131ft triple down drop. It then goes around a large left hand turn, before dipping down under the previous piece of track.

The lift takes riders up to the top of Wonder Mountain. Upon cresting the lift, the train slowly turns to the right adding to the suspense of the upcoming drop. After passing beneath Alpen-Fury's double-inverting top hat, the train then quickly dives off the side of the mountain, passing under the track of the ride and AlpenFury's zero-g roll. The train then sweeps to the right and climbs up the "fan-turn" element of the ride that flies over guests on the midway.

This section is 44.6 m (146.3 ft) in height, reaches a maximum speed of 99.9 km/h (62.1 mph), and has a track length of 216.0 m (708.7 ft). This section features two vehicles, each seating 20 riders in 5 rows of 4 riders each.

In the background, the sounds of a car engine revving and a pit crew performing maintenance create an immersive atmosphere.

This section's superstructure is entirely timber, predominantly fir and pine, and is clad in steel sheets.

Off the midcourse brake run, the track makes a near-miss with this section's first drop as it drops to the right, crossing under the exit from the first loop, and into a second vertical loop which wraps around the zero g roll. Leaving the second loop, the track passes goes through a 270 degree counterclockwise upward helix, before entering a set of interlocking corkscrews.

This section's storyline takes place in a fantastical kingdom

whose day and night cycle is perpetuated by the mystical dragon Dragnor and owl Astradir. When evil forces capture and imprison the two creatures in the Dolomite Mountains, a lynx known as Queen Azdra implores riders to free them in order to restore peace to the realm. Guests are informed of the story through an animatronic of Queen Azdra, and supplied with 3D glasses for the ride.

The track is 1,285 m (4,216 ft), and this section's top speed is over 104 km/h (65 mph).

Graffiti litters the sheet metal walls on both sides. After reaching the top, the car drops down a short hill outside the ride building. A sign advertises "Burnt out bridge ahead!", and the car swerves across the trestle.

Riders reach a top speed of 40.2 kilometers per hour (25.0 mph) on the forty-second ride.

As the trains crest the lift, the galaxy swirls up and vanishes. The music climaxes as the trains emerge into the main part of the ride, the inside of the dome. Trains make a U-turn before ascending one more small lift hill. A countdown begins as the vehicles crest the hill and the rockets are sent into a high-speed ride through the dome. This section consists of many turns and small dips with the illusion of speed given by the fact that the track is barely visible to the riders.

Bellatrix Lestrange appears, after being notified of an impostor infiltration, and decides to torture the guests by cutting off the track and tilting it down. This sends the cart speeding toward the vaults as stalagmites can be seen to give an impression of being deep underground in the vaults. Bill Weasley casts Arresto Momentum to stop the cart just before Harry, Ron, Hermione, Griphook and Bogrod emerge from the Thief's Downfall, which has washed away the Polyjuice Potion's effect off Hermione. Their

cart drops them from a high altitude as security trolls climb up, pushing Bill and Blordak off their cart.

During the brake run, speakers alongside the track play audio of Emmaline excitedly announcing the experiment's success to her father.

The brakes then lower while the train rolls back, allowing it to hook onto the 2.1-metre-long (7 ft) catch car. Once connected, the motor engages to launch the train. The catch car passes over a set of magnetic brakes, and the motor must work harder to attain proper launch speed if the train is not fully loaded with passengers.

The ride subjects riders to a maximum of 4.3 Gs. This section has three trains. Each train has four cars, each car featuring two two-person rows, (excluding the train with the Ensō car, which seats only two). This makes for a total of 16 riders per train (14 on the Ensō train).

After climbing into the new Morgan trains, the operator pushes a lever to release the still used manual braking system. The trains roll down a short inclined curve toward the lift hill. Once at the top, riders are given a quick view of the Long Island Sound before plunging down an abrupt dip which gives the train a rush of momentum. The train takes a quick left and proceeds down the main drop of about 80 ft.

The track then sweeps back down over the grass and above the river, swooping left and right at 89 km/h (55 mph). The train dives into a downward helix, which swings only a few feet above the surface of the water and wraps around this section's first drop. The train swoops left and right a few times more until it charges into the brake run. The immense swinging force as the train enters the brake run causes the cars to swing even after the train stops moving.

This section uses two 28-passenger trains, each with seven cars carrying four passengers in two rows of two seats.

It exits the tunnel at the beginning of the ascent, where a skeleton wearing a safari outfit is seen perched to the right with a spear through the chest and a snake wrapped around its neck. Riders pass under several wooden beams, each featuring a tiki-themed face mounted at the center. After reaching the top, the train makes a short dip before descending quickly down a winding 270-degree turn to the right.

It then traverses a triple down which goes through the covered section for a second time. It goes through a 270° left hand turn and two airtime hills going over the break run. The ride then continues with a long right hand turn.

This system allows the ride to achieve its relatively high throughput of 1920 riders per hour.

The brakeman rode between the first and second cars on the bogie and operated the brakes with a large lever.

The coaster doesn't have a mid-course brake run like the similar rides at Kings Island and Kings Dominion; instead this section has multiple rings that the train goes through. After the rings, riders spiral downward and to the left, and after more twists and turns they pass through a corkscrew before arriving at this section's trim brakes.

A window effect above riders shows the storm going on outside the shaft. Another flash of lightning strikes the top of the simulated tower protruding above, which collapses onto the window.

The humming sound of a hornet is played twice during the ascent – once at the beginning and a second time after a ride safety announcement. After reaching the top, the train enters its largest drop of 98 m (320 ft) at an 81-degree angle, accelerating up to 153 km/h (95 mph).

This section's entrance and queue is located directly underneath the covered section. The ride continues with a left-hand turn into the lift that ascends 40 m (131 ft). At the top, the inverted train dips slightly before turning left ninety degrees as it drops 36 m (119 ft) down the first hill. From the bottom the train immediately enters a vertical loop.

This section's story and themes are introduced in the queue as guests enter.

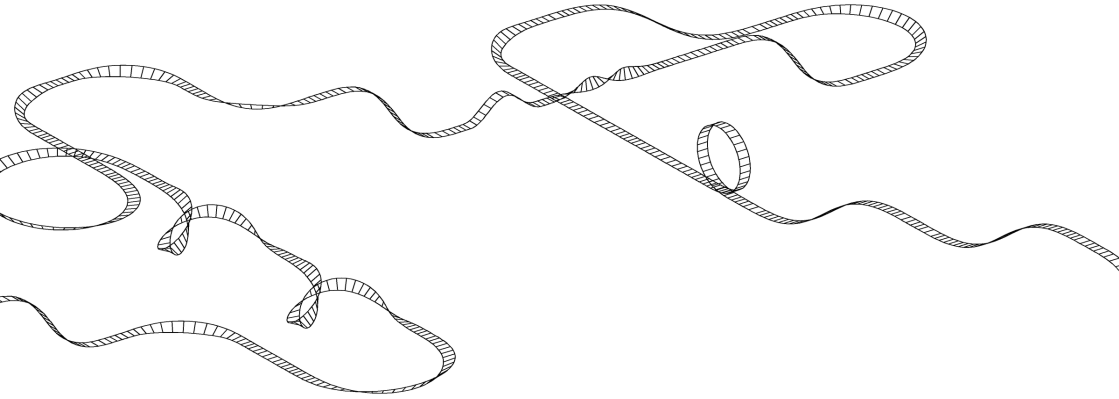
This section has 350 metres (1,160 ft) of track and the cars reach a top speed of around 72 km/h (45 mph). It has a maximum height of 18 metres (59 ft). This section has three trains consisting of seven cars each. Each car can hold two riders, for a total of 14 passengers per train. A maximum of only two trains can be operated at any given time.

This section has near-miss points with three attractions: the Monorail, Dry Gulch Railroad, and the ride. As the train continues, it travels through the switch track, and onto the launch track.

The ride entrance is located in the Ghost Town section of Knott's Berry Farm. This section's layout passes through three of the park's themed areas: Ghost Town, Fiesta Village, and Indian Trails. It makes a 90 degree right turn before beginning to climb the 45-metre (146 ft) lift hill. Once at the top, the train goes through a pre-drop before entering the 33-metre (109 ft) downward right drop. Once at the bottom, the train enters a 32-metre (105 ft) vertical loop. After exiting the loop, the train makes a banked left turn leading into a cobra roll.

Shortly after, the Guardians of the Galaxy arrive in their ship as Eson prepares to throw the Cosmic Generator towards Terra, who disregards Quill's music from the Guardians' ship as "noise." The Guardians then blast the Cosmic Generator from the Celestial's hands, opening a jump point which proceeds to suck every-

thing in, taking them back in time to the Big Bang until eventually fading into complete darkness. Suddenly, the evacuation shuttles are launched backwards in an explosion, and it is at this point that the vehicles begin spinning on their axes to achieve a sensation of "space travel." While the story develops around them, one of the six songs randomly selected plays.



Distance and Soon Becomes Visible

Once the train is loaded and secured, the steel floor is retracted and the gate in front of the train opens. The train immediately begins to climb the 40-metre (132 ft) chain lift hill. At the top, the train drops down a small straight section before finishing the drop with a sharp turn to the right.

The train goes through a tunnel, where the passengers wait until the countdown starts. After the countdown, the trains, which exist out of 2 across in 2 rows for a total of 24 riders per train, are accelerated from 90 km/h (0 - 55.9 mph) in 2.8 seconds. It first climbs into a Sea Serpent Roll, which is a 2 inversion element. The train plunges into numerous turns and helixes and the 3rd inversion, a corkscrew. The train then passes in through a mid course brake run, makes a few turns then travels into the trim brakes.

With the retirement of the ride, the ride is the only coaster installation of any kind with this feature.

The ride sits in the center of Lake Compounce. The ride utilizes its design to twist around itself. The train turns slightly right, then left, before climbing the 85-foot lift hill and dropping 78 feet. The lift hill is adjacent to the park entrance.

A security troll pushes the guests' cart further, past the Thief's Downfall and stops in front of more security trolls. One of them is hit by a mace but hangs on to the guests' cart and sends them falling. Bill casts Arresto Momentum again to stop the cart from falling, followed by Aguamenti to stop the Ukrainian Ironbelly dragon from breathing fire on the guests (the audience gets sprayed with water at this point) and Wingardium Leviosa, sending guests' through the vaults, in order to help them escape. Meanwhile, Harry, Ron and Hermione are attempting to escape the vaults on the Ukrainian Ironbelly.

On some ride signs, the name is stylized as "ThuNderaTion", a reference to TNT to coincide with this section's mining theme.

The first car passes an initial encounter with Billy the Puppet, who laughs as it moves through. The second car is briefly halted while Billy delivers a short monologue outlining the experience ahead; this pause also serves to conceal a block section, ensuring safe spacing between vehicles. The car then advances toward two dimly lit, swinging pendulum blades that appear to close in before a sudden, near-vertical hidden drop. This creates the illusion of plunging directly into a pit of strobe-lit spikes, which are narrowly avoided.

It then travels up to a height of 46 m (150 ft) and twists into a slow heartline roll at the top. Upon exiting the roll, riders twist down into a full non-inverting loop.

This section features three cars, each of which seats eight riders.

The vehicle enters a brake run, during which air blasts are directed at riders to simulate the firing of syringes from crossbows

positioned to appear as though they narrowly miss riders' heads. Following another surprise drop, the car turns left into a bathroom scene, where it executes a heartline roll above a screaming, heavily injured victim in a pool of blood—an explicit reference to a trap featured in the first Saw film. At this point, water is sprayed at riders to simulate blood. Exiting the warehouse section, the car enters another brake run before proceeding toward the 30-metre (100 ft) vertical lift hill. Prior to engaging the lift, the car stops in front of two video screens and a further digital countdown timer.

The piston suddenly discharged a blast of air from beneath the water, creating a loud noise and splash designed to startle the exiting riders.

Only four riders could ride the Olentangy the ride at once.

As the evacuation shuttles travel through multiple jumpholes along the way, the Guardians of the Galaxy engage in combat with Eson around the Terrans as they also cross multiple jump points. The Guardians then lead a fleet of Nova Corp Starblasters to Eson, who manage to contain the Celestial and recover the Cosmic Generator. As the Guardians of the Galaxy travel far from the scene, they lead the Terrans to a series of jump points that take them to the Milky Way, then around the Moon before locking the coordinates for EPCOT and sending them back to Terra. As they arrive, Peter Quill welcomes the Terrans to the Guardians of the Galaxy.

To further torture guests into revealing Harry Potter's whereabouts, Voldemort decides to administer "a little dose of pain to help you remember" and conjures a ball of fire. The Ukrainian Ironbelly crashes in from the opposite side, and Harry casts a counterspell, while the dragon spits fire towards Voldemort, who shields himself and Bellatrix and both later furiously retreat. Hermione casts a spell to hook the cart to the dragon in order to get guests safely out of the vaults and Ron casts Confringo to

create an escape route. The cart is then launched through the escape route and Harry orders Hermione to cut the cart loose, away from peril.

The splash is created by a series of water jets that shoot up when the train passes, creating the splashdown effect. The splashdown is followed by another set of camelback hills. Following these hills, the train passes through the mid-course brake run. Then, after making a banked downward left turn, the train passes over the hill.

This section features ten trains with two rows of two riders. This section is 400.0 m (1,312.3 ft) long and reaches a top speed of 56 km/h (35 mph). The track layout consists of hairpin turns followed by a pair of steep drops.

The ride is a Euro-Fighter by German company Gerstlauer. The roller coaster stands 30 m (100 ft) above the pier with some parts of the roller coaster standing over the Gulf of Mexico. This section features 4 inversions.

The ride covers 13 acres (5.3 ha); it runs parallel to the shoreline of Sandusky Bay, then travels to an island located inside the park. There are two tunnels, three overbanked turns and three hills.

This led into the mid-course brake run slowing the train before descending a hill into a second corkscrew and a turn-around.

The car goes slowly through the covered section, loads riders, and then flips the car straight up, into the lift hill, which is partially very slow, but soon speeds up as the car is almost at the top.

The lava tube entrance of the attraction features a photo-opportunity allowing guests to sit on a throne and be crowned as a the ride champion.

The signage provides details on stingrays found in the Texas Gulf and how everyone is connected to the ocean. In addition, 5% of the sales of the ride merchandise is donated to HRI. As the

train continues it curves to the right to engage the 96-foot lift hill, another right turn off the lift leads to an atypical straight first drop of 100 feet into a small valley.

The park can adjust the degree of spin based on rider input, which will help them regulate the experience to average rider tolerance over time.

The trains still have the blue and purple cars with flames but with a tint of grey-brown to make the ride look damaged by the supernatural forest.

This gives riders the opportunity to experience the ride in both directions moving forwards and backwards.

Full size figures of astronauts are suspended upside-down beneath the ship's engines to give the illusion of zero-gravity. As trains climb up the lift hill, projections of the Sun, Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto, stars, comets, meteors, asteroids and the Milky Way can be seen when looking past the lift hill bay's open ceiling. It is possible to make out trains climbing the lift hill on the other side and also rockets passing through on the side, as well as the trains of the PeopleMover passing between the two tracks. Once at the top, the trains then make a small and quick dip before plunging into numerous twists and turns as they travel around the mountain in near-complete darkness, including the coaster tracks' steepest drop of 39 degrees. The ride continues with the trains passing through a red and orange swirling wormhole, before hitting the trim brakes.

While the car is on the vertical spike at the back end of this section's layout, the water flows into the splashdown pool in approximately six seconds raising the level. The car then travels down the spike and enters the splashdown pool, which is now filled with water. This serves to slow the car down before it re-enters the

turntable to unload riders.

This uncontrolled spinning resulted in a slightly different ride experience every time.

The train slows as it reaches the top producing a moment of weightlessness as the train begins its descent. The dive is followed by a turn to the right and a heartline roll, in which riders enter at speeds up to 97 km/h (60 mph).

This section comes to an end as the train is slowed by magnetic brakes.

On the ride, the seat rotation is controlled by the position of two rails on the outside of the track, resulting in a "choreographed" rotation that is identical on every ride.

The train exits the brake run with a left turn into a dive loop. A left turn leads the train into the trim brakes.

All versions of the ride share a similar concept. The ride begins as a slow-moving dark ride through a medieval castle populated with scenes featuring knights, wizards, and dragons, often using animatronics. The trains then exit the building, ascend a lift hill, and complete a family-friendly coaster circuit with gentle turns and drops. The rides do not feature inversions and are designed to be accessible to younger guests.

The train zooms through the Sector 3 Transport Tunnel, the first two tunnels. Inside, there are red lights and fog effects.

After a left turn, the train enters the midcourse brake run. The track makes a 180° turn to the right and enters two consecutive corkscrews. It then traverses a 540° helix, featuring a 110 foot long tunnel, at the entrance of which an on-ride camera is mounted, before.

The layout consists of several "segments" separated by brakes. The first segment comprises a swooping drop and climb into the second segment, which is a series of S-shaped turns. The third

segment is a heavily banked figure-8; the fourth is a series of hills that form a semicircle; the fifth is a pair of small hills; and the sixth is a helix into the seventh segment; a long loop, which leads to a massive hill and then into the brake run. For most of the ride, the cart is spinning around rapidly.

The train then enters the brake run, slowing. During previous events, different launch announcements have been used:.

…There's nothing above you and nothing below you.

After going around a low-to-the-ground curve, the train enters a 30-metre-tall (100 ft) double down followed by a 21-metre-tall (70 ft) double up. This is followed by a left turn into a "twist and turn" element, which is then followed by a "wave turn" that drops to the right into a small airtime hill known as a high-speed float. This section's two inversions are heartline rolls, where the rotation axis occurs at approximately chest level. The train then enters the brake run. A single ride cycle is completed in approximately 1 minute and 27 seconds.

After a few seconds, the holding brake is released and the trains roll down a drop into a strobe tunnel of pulsating blue lights, which flash at a gradually increasing frequency as the train progresses down the tunnel. A repetitive warping sound signifies an energy charge as the riders roll towards a blue orb. At the far end of the tunnel, a field of stars appears as the blue strobe lights turn off. The train then makes a 180 degree turnaround and climbs the lift hill. During the turn, the on-ride photo is taken.

Riders are required to wear safety goggles to protect from airborne particles such as dust. For safety reasons, this section has a height restriction (140-195 cm) in place.

As the train moves into position, the sound of a heartbeat is played. After a brief pause, the train rolls backwards slightly, the brakes on the launch track retract, and a pre-recorded voice says

"Now, get ready! Here we go!", and the train is launched from 0-72 mph in 2 seconds.

Riders enter the queue area by passing through a shipping container with a rusty the ride sign on it. A short outdoor path leads riders towards a stack of containers. Riders enter the containers through a series of black rubber flaps, before emerging into a warehouse.

The train enters the tunnel, where a matching scent to the one selected by the host can be smelled by passengers. The train then traverses into a vertical loop and goes up, ascending the chain lift hill. When the train reaches the top, the lift disengages, and the train falls backward through the inversions and tunnel.

Normally this section operates for two laps.

The layout has since been reproduced by Chinese manufacturers. The train first ascends a 117 ft chain lift hill and drops 111 ft to the right. The train then immediately enters the first inversion, a right vertical loop, before slightly banking right then straight again.

The train makes a slow U-turn followed by a slight left turn onto the launch track. Once clear, the train is launched to a speed of 80km/h (49.7 mph). The layout is woven between the surrounding buildings and structures of Klugheim, and through tunnels. The riders aboard the train narrowly miss walls of faux rock faces and waterfalls. During the course of the ride, the train crosses over or under its own track 116 times at 58 intersecting track points, more than any other launched roller coaster in the world. The second launch section is in the lowest point in the canyon and it is taken without pause, accelerating the train to its top speed of 117 km/h (72.7 mph).

The track crosses under the ride and enters the mid-course brake run, before descending into a downward helix. The train

continues through an angled decline through Nickelodeon Streak into a tunnel, where an on-ride photo is taken, and a short ascent into the trim brakes.

This momentum sends the train going into the mouth of a dragon. As they enter the tunnel, a roaring sound-effect plays and the eyes of the dragon light up. The tunnel wraps around the turn off the lift hill's upper middle, and sends the riders out into this section's left leg. The trains dip down out of the tunnel into a short straightaway, and then climb up into a turnaround at the far end of the left leg. The trains head back toward the center lift hill turn portion. After a few dips, the trains reach another turn around, directly under the dragon tunnel.

Pre-ride videos are played throughout this section's line queue featuring actor Renny Krupinski, who plays a character named the Lord of Darkness. This section features seven single-car trains which each accommodate sixteen passengers in two rows of eight, allowing for a theoretical capacity of 1,820 riders per hour.

It enters a block brake before more full helix elements. It banks to the right where the on-ride photo is taken. After the photo is taken, the ride enters the brakes.

The layout crosses over and under itself seventeen times. This ride also features two 24-passenger Millennium Flyer trains for better cornering.

Backward-facing riders do not see this transition. In the next scene, riders—who are surrounded by a lake inside the mountain—battle Wet Ones and Crazy Crabs. The scene starts above the surface of the water but later moves downwards. As the scene ends, the train passes two waterfalls; this transition is absent for backward-facing riders. Riders then arrive in an underground city named Draconian City, where they battle against Draconians and Archers.

Goofy is pictured on billboards throughout the ride teaching guests the step-by-step process of flying a plane:.

Once guests have boarded the bobsled, riders enter a dark tunnel at the base of the mountain where strange growls are heard. The bobsleds then climb an 80 foot lift hill up towards the South part of the mountain. Near the top of the lift the bobsleds pass by a sheet of ice on each side where the Abominable Snowman simultaneously climbing the mountain is angered at the sight of intruders.

The ride started by a slow ascent of 22 metres (72 feet), powered by a 75 kilowatts (101 horsepower) lift motor.

Mack Rides sells a modified version of this section's layout, which excludes the opening dark ride section. A rather egregious example is Battle for the ride at Quancheng Euro Park, which also lifts the name from the ride. Another notable example is the ride at Dreamworld, which adds a swing launch system and a spiral rollback in addition to this section's original layout.

The other Vekoma Hurricane is located at Walygator Parc as the ride which only travels riders forwards and not backwards.

The former has much thinner track rails than the tubular steel of the latter.

The ride queue is themed to resemble an interpretation center dedicated to the study of sea monsters. It features a pre-show that talks about various sea monsters around the world including Champ, Ponik and the ride.

This section is located on the side of a mountain, and the coaster weaves around it throughout its layout.

After sitting down in one of two sets of benches which face each other, the exit doors close and the lights dim. A crystal ball illuminates, showing Houdini's head inside the ball, (not in the New England version.) He denounces Margery and other frauds who

claim to bring Houdini back to the world of the living, since he claims he never left. In order to show you his power, he locks you in place with a set of magical keys (projected onto mirrors) and tells you he will show you his greatest illusion to date. The room begins to rotate, and the benches swing. At the climax of the ride, the room and benches stop so that riders should theoretically be upside down.

Now in darkness, the car stops at the peak of the hill. A lit fuse is heard in the distance and soon becomes visible, snaking along both sides of the car.

The ride follows the surrounding terrain with a track length of 990 m (3,250 ft). Its maximum height is 21 m (70 ft), but because of the track layout and the natural ravines, the maximum drop is 27 m (90 ft).

The track then returns to approximately two thirds of its original height before entering the first set of block brakes. These brakes lead into a left turn with reverse or outward banking. It then drops under the block brakes and into the first inversion – a Dive Loop. The inversion begins with half of an inline twist which transitions into half of a vertical loop. Upon exiting the inversion the track makes an incline and into a second set of block brakes.

Ride operators clap and chant "Eejanaika, Eejanaika" as the train continues. As the train makes a 180-degree turn onto the lift hill, the cars are rotated 90 degrees backward before rotating back 45 degrees shortly before entering the hill. After ascending 76 m (249 ft), the train enters a pre-drop. During this lift, riders are facing backwards.

For the track layout, the train continues and immediately enters a tunnel. After a few drops and turns in the tunnel, the train emerges at the base of the lift hill. Upon reaching the top of 21 m (70 ft) lift hill, the train drops 20 m (65 ft), reaching a top speed of

89 km/h (55 mph).

The XT 150 has the option of sending riders around a second cycle at the operator's discretion.

Once a color is selected, the host announces the flavor, beginning this section's cycle.

Riders are then confronted by more Draconians and Archers before entering a gold-covered area where King Adelsten's crown is found. Soon after, Ormaar arrives and breathes fire toward the ground. After a few seconds, the drop track releases the train into a 9-metre (30 ft) free fall, escaping from Ormaar. As the train leaves the drop track, the trains are rotated to their original positions and all of the riders' scores are displayed on both sides of the track.

Once it lost momentum the train then returns down passing through the covered section again. It then proceeds back up the drop and partially into the heartline roll before once again losing momentum and returning to the ground. The train is stopped near the top of the lift hill.

The train dips into a tunnel before lifting and leveling out into the trim brakes. Every so often, a launched train will fail to make it over the apex of the tower. At this time, the emergency brakes will deploy and the car will enter a slowed descent back into the launch tunnel.

Ride operators dressed in themed miner clothing with hard hats and fluorescent jackets.

The ride begins in the Big Grizzly Mountain Mining Company building. This is home to the mining company's sawmill and one of the mining shafts, where the sounds of a snoring bear can be heard.

The ride continues into the dark ride section of the coaster, passing through several scenes training them for their mission

before blasting into hyperspace on a show screen. Exiting the building, the ride makes a 90° turn into the first of two tire propelled launches, enabling riders to reach their max speed of 55.0 km/h (34.2 mph), as well as letting the cars spin freely throughout the layout. The train navigates a curved airtime hill, a valley below the ride entrance, and a turnaround before hitting the second tunneled tire propelled launch. The coaster navigates a short section of simple airtime hills and curves before hitting the brakes, and re-entering the building.

Riders are batched into rows of 4.

The ride is a 4th Dimension coaster, where riders sit in horizontally spinning cars on either side of the track and freely flip throughout the ride.

It makes a slow turn to the right. A quick drop follows, before the train rounds a series of left turns into the first inversion, an inline twist.

This section can accommodate a maximum of 950 riders per hour when using two vehicles. After riders are loaded, the turntable rotates, and the vehicle lines up with the track. Once the track is locked in place, the vehicle accelerates backward over a small hill before traveling through a straight section of track. This straight section of track goes through this section's splashdown pool. The water in the pool is low enough at this point in the ride that the vehicle can travel over it.

The complete ride is about 20-30 seconds.

This section consists of a wooden-tracked railway with flat steel rails supported by a wooden structure. The ride occupies a space approximately 180-metre-long (580 ft) and 37-metre-wide (120 ft). The track is in a trough, and as such this scenic railway is often incorrectly referred to as a side-friction coaster.

This effect distracts riders from noticing an automatic track

switch in front of them. As the shadow moves away, the train rolls forward out of the mountain, going past the on-ride camera and plunging down the main 24-metre (80 ft) drop. It enters a 250° turn and speeds back up through another cave in the mountain, where the roars of the yeti are heard once more.

The track length is 740 m (2,427 ft).

This section features two PTC trains, one yellow and one orange, each capable of accommodating 24 passengers.

The train enters the second phase of the layout. It snakes slightly to the left before passing through two corkscrew elements: the first inverts riders over the airtime hill, while the second passes over a pathway leading to the ride entrance. On-ride photographs are captured as the train levels out after the second corkscrew, at which point the train's speed decreases rapidly. The ride continues through four consecutive clockwise heartline rolls, passing just 3 m (10 ft) above the pathway.

Riders approach Hollywood Dream within the Hollywood area of Universal Studios Japan. Admission is through four entrance options: general admittance, holders of Universal Express Passes, single riders, and "child switch". Child switch is designed for children with adults, where one adult takes the ride while the other looks after a child, both then switching roles.

As the shuttles depart, Terrans begin making their way out of the Nova ship to enter the jump point in their evacuation shuttles.

Riders enter the queue for the ride from the Hollywood section of Universal Studios Japan. As part of this section's theme, queue features an 26-metre-long (85 ft) wall which is able to recognise the movements of up 132 hands on 66 guests without them physically having to touch the wall. The wall is powered by 22 3D cameras and 13 projectors connected to an array of 16 quad core computers. This section features separate queues for single riders

and holders of Universal Express Passes, as well as offering "child switch" where one parent can experience the ride while the other parent looks after their child before switching their roles.

The overall layout is a standard paperclip layout with two out-and-back style runs (at Kings Dominion this extended into the courtyard adjacent to Grizzly). The ride is also a mirror image of the ride at Kentucky Kingdom.

Riders sit in an enclosed car. The car climbs vertically inside a hollow steel tower, then spirals back down around the outside of the same tower. Each vehicle has a single rubber tire with a hydraulic clutch braking system that governs the speed of the vehicle as it descends the tower. The rubber tire engages a center rail that begins halfway through the first spiral.

The car continues part of the way up a vertical spike before losing momentum and reversing direction. It then travels forward over the pool and launches with a much faster forward acceleration into a second vertical spike, and then comes back down the spike before entering a second backward launch. While the car is on the vertical spike at the back end of this section's layout, water flows from the side reservoirs into the splashdown pool, raising the level in about six seconds.

At that point, pictures are automatically taken of the riders (who can buy souvenir pictures at the beach hut themed stall after the ride). The train then speeds through the covered section at its top speed, and then goes through a cobra roll and a vertical loop. It then climbs to the top of a second hill.

The ride then flies over the covered section and enters into a last inline twist before a mid-course brake run. Following this, the train makes a banked left-hand turn (which despite being directly after a brake run, can generate strong forces for riders on the rear-right of the train) into the brakes.

The train exits the tunnel and travels along a slight decline.

This section has a track length of 361 metres (1,184 ft), a height of 22 metres (72 ft), and achieves a maximum speed of 70 kilometres per hour (43 mph). This section has three inversions: a vertical loop, a cutback and a heartline roll. The coaster layout for the ride is particularly close to two other Euro-Fighter (model 320+) roller coasters: Rage at Adventure Island amusement park and the ride at Canobie Lake Park.

At this point, the machine explodes, flooding the screen with wine. The cars then move forwards then back again, and on the screen the professor dives through the wine and then the ride launches to 135.0 kilometres per hour (83.9 mph) in 3.5 seconds up a small hill before navigating a series of tight turns and transitions through the vineyard and a tunnel. The ride then dives into an inline twist before dropping and twisting through a turn very close to the lake before rising up and going into the brake run where the monkey is hanging on the ceiling holding some grapes. Then the riders are let off the ride and into the shop where they can purchase an onride photo or a video of their ride.

Coyote can be seen standing on top of some Acme crates.

The second lift pulls riders upwards for a few seconds, then releases, sending riders backwards. The train encounters the loop first this time, only to then go through the cobra roll once again, which leads riders back through the covered section and partially up the first lift section again.

The train slows to a stop approximately ten hours and thirteen seconds after it began.

Sources

Thunderhawk (Michigan's Adventure), Copperhead Strike, Wabash Cannonball, Shambhala, Griffon, Thunder Road, Superman El Último Escape, The Flash: Vertical Velocity (Six Flags Discovery Kingdom), Superman: Krypton Coaster, Skull Mountain, Superman Escape, Behemoth, Super Manège, The Joker (Six Flags México), Manta (SeaWorld Orlando), VelociCoaster, Formuła (Energylandia), Woodland Run, The Monster (Walygator Parc), Leviathan (Canada's Wonderland), Good Gravy!, Twisted Colossus, Ravine Flyer II, Woodstock's Air Rail, Demon, Mako, Arkansas Twister, Loch Ness Monster, Texas Tornado, High Roller (Stratosphere), Takabisha, American Eagle, Abyss, Mission Ferrari, Dæmonen, Wiener Looping, Voltron Nevera, Ice Breaker, Wave Breaker: The Rescue Coaster, Wrath of Rakshasa, Mumbo Jumbo, Steel Taipan, Goliath (Six Flags Great America), Nitro (Six Flags Great Adventure), Rampage, Goliath (Six Flags Magic Mountain), Expedition Everest, Hyperion, Booster Bike, Teststrecke, Dragon Khan, Formule X, Goliath (La Ronde), Toutatis, Full Throttle, Panic Coaster – Back Daaan, Steel Force, Alpina Blitz, Texas Stingray, OzIris, Wildfire (Kolmården

Wildlife Park), White Lightning, Candymonium, DarKoaster: Escape The Storm, Pyrenees, Rapterra, Hagrid's Magical Creatures Motorbike Adventure, Flight of Fear, Coney Island Cyclone, Coast Rider, Silver Comet, The Smiler, Maverick, The Wave (Drayton Manor), Diving Machine G5, Rattlesnake, Mighty Canadian Minebuster, Dahlonega Mine Train, The Oath of Kärnan, Big Dipper (Luna Park Sydney, 2021), Hair Raiser, Cornball Express, The Joker (Six Flags Discovery Kingdom), Zeus, ArieForce One, Slinky Dog Dash, Raik, Fire Runner, Rewind Racers, River King Mine Train, Road Runner Rollercoaster, Batman: The Ride (S&S Free Spin), The Riddler Mindbender, X-Coaster, Pipeline: The Surf Coaster, Taiga, Wildcat (Lake Compounce), Mr. Freeze, El Toro (Six Flags Great Adventure), Thunder Striker, Millennium Force, Pulsar, Big Dipper (Camden Park), Iron Rattler, Batman: The Dark Knight, Harpy (Xishuangbanna Theme Park), Tigris, Kenny's Forest Flyer, Iron Shark, Phoenix, Siren's Curse, Santa Monica West Coaster, Magnum XL-200, Roller Coaster (Lagoon), The Exterminator, Mystery Mine, Wild Mouse (Lagoon), Afterburn, Verbolten, New Mexico Rattler, Iron Dragon, The Legend, Kong, Mystic, Tiki-Waka, TMNT Shellraiser, Abyssus, Viper (Six Flags Magic Mountain), Wakala, Chupacabra, Bulldog Coaster (Brean Leisure Park), Orion, Galactica, Eagle's Pursuit, Vampire, Nemesis Reborn, Stampida, Vortex (Carowinds), Battlestar Galactica: Human vs. Cylon, Gold Striker, Sea Viper, The Flying Cobras, Electric Eel, Monimo Rush, Looping Star (Nagashima Spa Land), Sandy's Blasting Bronco, Twister (Knoebels Amusement Resort), Storm Runner, Parrot Coaster, Fahrenheit, Silver Bullet (Frontier City), Nickelodeon Slime Streak, Lech Coaster, Wicked, Skyrush, Shuttle Loop (Nagashima Spa Land), Rollo Coaster, Turbulence, Tornado (Parque de Atracciones de Madrid), Hurler, Flying Aces, Icon, Mystic Timbers, Banshee, Garuda Glide, Firebird, Dynamite (Freizeitpark Plohn), Raging Bull, Tranan, Storm Coaster, Backlot Stunt Coaster, SheiKra, Untamed (Canobie Lake Park), Fury 325, Giant Dipper (Santa Cruz Beach Boardwalk), Dragon Slayer, Flyvende Ørn, Goliath (Six Flags Over Georgia), Thunderhead, Wonder Woman Golden Lasso Coaster, Steel Eel, Shock Wave (Six Flags Over Texas), Cannibal, The Beast, Whizzer, Yukon Striker, Toxic Garden, Steel Dragon 2000, Matterhorn Bobsleds, Gemini, Poltergeist, Tatsu,

Blue Hawk, Wild Eagle, Goliath (Walibi Holland), Cedar Creek Mine Ride, Kondaa, Cheetah Hunt, Wicker Man, Patriot (Worlds of Fun), Penguin Trek, HangTime, Phoenix Rising, Tonnerre 2 Zeus, Thunderbird (Holiday World), Iron Gwazi, Flight Deck (California's Great America), Medusa (Six Flags Discovery Kingdom), Project Zero, Sky Rocket, Pégase Express, Rougarou, Eurostar, Adrenaline Peak, Cheetah (Wild Adventures), Cobra's Curse, Diamondback (Kings Island), Dragon's Run, Oblivion, Valkyria, Thunderbolt Express, Viper (Six Flags Darien Lake), The Joker (S&S Worldwide), F.L.Y., Escape of Novgorod, Batman & Robin: The Chiller, Twister III: Storm Chaser, Apocalypse: The Ride, Comet (Waldameer), The Monster (Adventureland), Junker, GateKeeper, Expedition GeForce, Flug der Dämonen, Invertigo, Stardust Racers, Oblivion: The Black Hole, New Texas Giant, Rita, Tennessee Tornado, Batman: The Ride, Hollywood Dream – The Ride, Pantherian, Wildfire (Silver Dollar City), X-Flight, Corkscrew (Cedar Point), Giant Digger, Woodstock Express (Carowinds), Runaway Mine Train (Six Flags Over Texas), Great Bear, Xcelerator, Avalanche (Blackpool Pleasure Beach), Hyperia, Raptor (Cedar Point), Kentucky Flyer, El Loco, Krake, Mahuka, Green Lantern Coaster, Figure 8 roller coaster, Mayan, Steel Curtain, Alpengeist, Speed – The Ride, Orkanen, Tantrum, Thirteen, Wild Thing (Valleyfair), DC Rivals HyperCoaster, Montu, Diabolik Invertigo, Merlin's Mayhem, Medusa (Six Flags Great Adventure), Tempesto, Schweizer Bobbahn (Europa-Park), Kirnu, Saw – The Ride, Star Trek: Operation Enterprise, Carolina Cyclone, Stunt Pilot, Batgirl Batarang, Thunder Run (Canada's Wonderland), Dragon Coaster (Playland), Wild Mouse (Luna Park Sydney), Scenic Railway (Dreamland Margate), Centrifugal railway, Big Bad Wolf: The Wolf's Revenge, Space Fantasy – The Ride, Thunderbolt (Kennywood), Impulse, Olympia Looping, Shuttle Loop, Colossus (Thorpe Park), Infusion, Space Shuttle, Time Traveler, Zoomerang (Lake Compounce), SooperDooperLooper, Timber Wolf, Wonder Mountain's Guardian, Blue Fire, Jumbo Jet (Morey's Piers), Big Apple Coaster, Canyon Blaster (Adventuredome), Goofy's Sky School, The Swarm, Giant Inverted Boomerang, Eejanaika, Nemesis Inferno, The Joker's Jinx, Fairly Odd Coaster, Tron Lightcycle Power Run, Soarin' Eagle, Talon, Raptor (Gardaland), Kraken, Outlaw Run,

Roller Coaster (Dai Nam Van Hien), Corkscrew (Silverwood), Mine Blower, Titan (Six Flags Over Texas), Sky Scrapper, Roller Coaster (Great Yarmouth Pleasure Beach), Wilderness Run (Carowinds), Ednör – L'Attaque, RailBlazer, Jetline, The Big One, Gotham City Gauntlet: Escape from Arkham Asylum, Superman: Ultimate Flight, The Bat (Kings Island; opened 1993), Boomerang (Mid America Adventure), Muntanya Russa (Tibidabo Amusement Park), Silver Bullet (Knott's Berry Farm), Viper (Six Flags Great America), The Riddler's Revenge, Apollo's Chariot, Revolution (Blackpool Pleasure Beach), Fønix, Trailblazer, Red Force, The Ride to Happiness, Katun, Steel Vengeance, Moonsault Scramble, Ride of Steel, Herky and Timmy's Racing Coaster, Space Mountain (Disneyland), Dragon Fyre, Eurofighter (Zoosafari Fasanolandia), Dominator, Alpenexpress Enzian, Werewolf Gorge, Furius Baco, Tower of Terror II, Comet (Hersheypark), Pepsi Orange Streak, Big Bad Wolf, Superman: Ultimate Flight (Six Flags Discovery Kingdom), GhostRider, The Great Scenic Railway, Falcons Flight, Maxx Force, InvadR, Decepticoaster, Formula Rossa, Stealth, The Flying Dinosaur, Lego Technic Test Track, Virtual reality roller coaster, Tornado (Adventureland), Vonkaputous, Flip Flap Railway, Dragon (Adventureland), The Dark Knight Coaster, Freedom Flyer, Kumali, Hoosier Hurricane, Lightning Rod, Drakon, Primordial, Jaguar!, Scream, Batwing, Adventure Express, Crystal Wing, Hydra the Revenge, Vortex (Canada's Wonderland), Octonauts Rollercoaster Adventure, Magic Flyer, Viper (Six Flags Great Adventure), The Ultimate, Gold Rush (Slagharen), Astro Storm, Nitro (Imagicaa), Storm Chaser (Paultons Park), Big Grizzly Mountain Runaway Mine Cars, FireChaser Express, Jet Rescue, Kingdom Coaster, Zumanjaro: Drop of Doom, Road Runner Express (Six Flags Magic Mountain), Valravn, Ricochet (Carowinds), The Gold Coaster, The Boss, American Thunder, Blue Streak (Cedar Point), Wipeout, Shivering Timbers, Zydeco Scream, Thunderation, Dive Coaster (Chimelong Paradise), Flight of the Hippogriff, SpongeBob SquarePants Rock Bottom Plunge, El Loco (Adventuredome), Dr. Diabolical's Cliffhanger, Acrobat, Pandemonium, Harry Potter and the Escape from Gringotts, Kumba, Jersey Devil Coaster, Poseidon, Silver Streak (Canada's Wonderland), Van Helsing's Factory, Blue Flyer, Steel

Venom (Valleyfair), Green Lantern: First Flight, The Flash: Vertical Velocity (Six Flags Great America), Wood Coaster (Mountain Flyer), Aquaman: Power Wave, The Barnstormer, Guardians of the Galaxy: Cosmic Rewind, Xpress: Platform 13, Orient Express, Raptor Attack, Loop the Loop (Olentangy Park), Apple Zapple (Kings Dominion), The Incredible Hulk Coaster, Ultra Twister (Nagashima Spa Land), Space Mountain (Magic Kingdom), Bat (Lagoon), The Dragon, Jolly Rancher Remix, Taron, Corkscrew (Alton Towers), The Jester, Houdini's Great Escape, Objectif Mars, Toboggan, Falcon (Duinrell).